Canonical GHZ configurations - cluster P2

```
species: electron
polarization cluster: P2
source cluster configurations: 97
canonical configurations collected: 56

canonical display criterion: exactly 2 retained final-state components
retained-component cutoff: fraction >= 2.0%
normalized amplitude display: bar length = |A_tilde|, bar color/text = phase

cluster retained-component count distribution:
  2 components: 56 configurations
  4 components: 7 configurations
  5 components: 8 configurations
  6 components: 21 configurations
  7 components: 4 configurations
  8 components: 1 configurations

CSV index:
  Output\GradientPhaseSpaceScan\electron\Data\GHZ\dghz\polarization_cluster_02\combined\
  canonical_dghz_polarization_cluster_02_configurations_electron.csv

Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.
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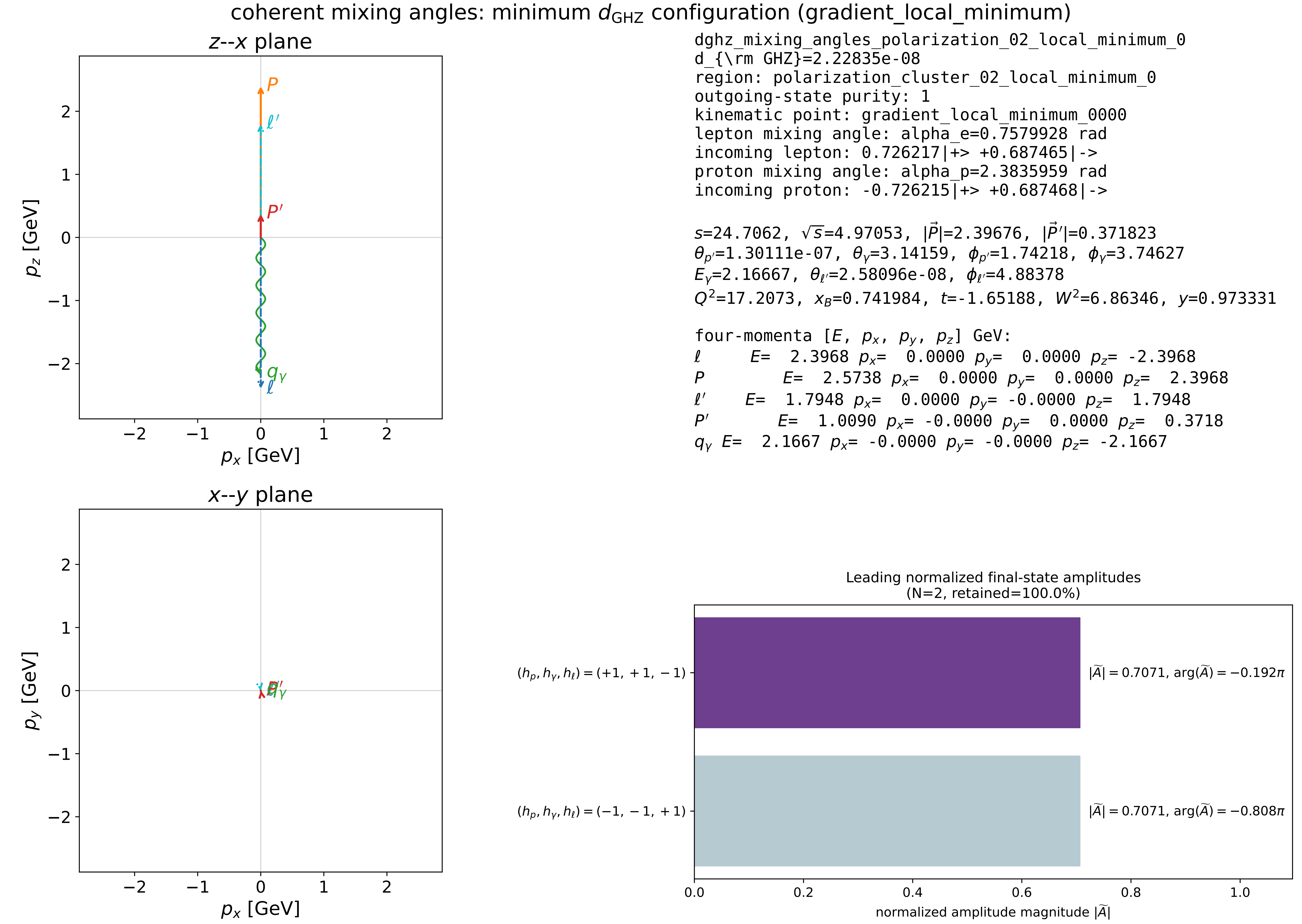
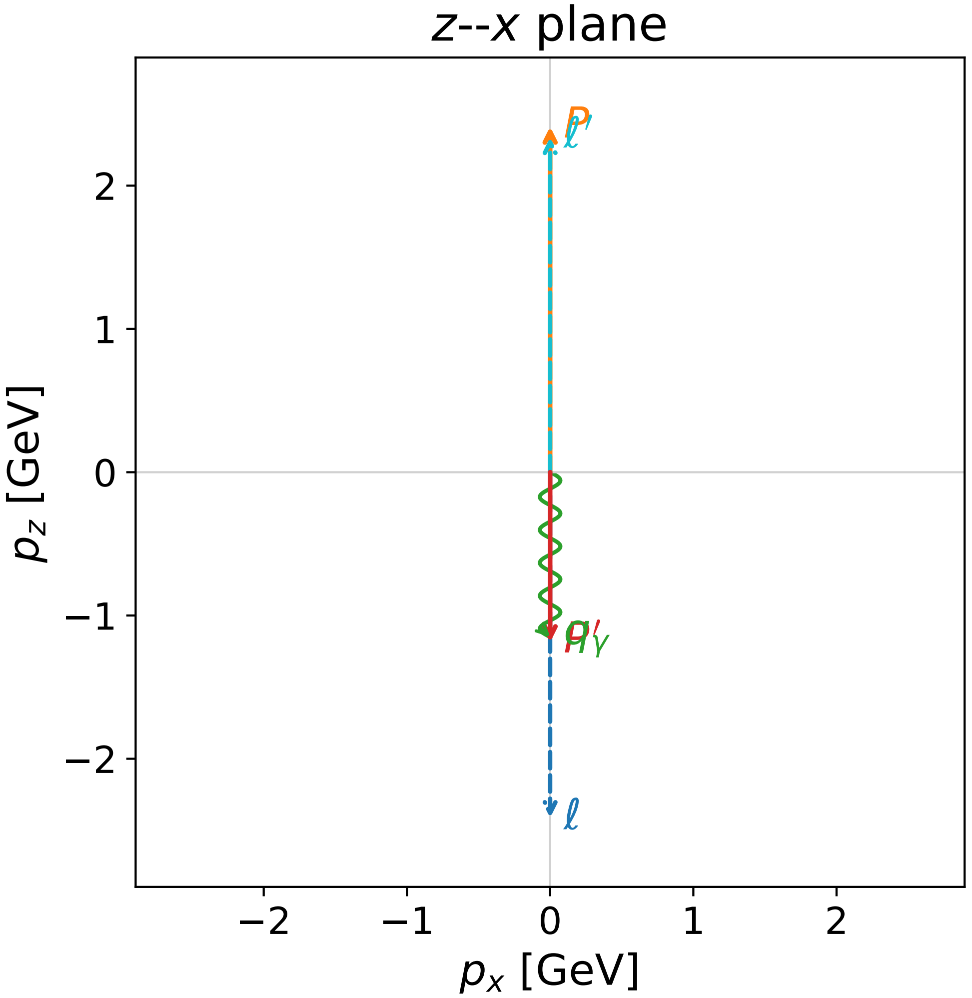

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
dghz_mixing_angles_polarization_02_local_minimum_0
d_{\rm GHZ}=2.22835e-08
region: polarization_cluster_02_local_minimum_0
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0000
lepton mixing angle: alpha_e=0.7579928 rad
incoming lepton: 0.726217|+> +0.687465|->
proton mixing angle: alpha_p=2.3835959 rad
incoming proton: -0.726215|+> +0.687468|->
s=24.7062, $\sqrt{s}$=4.97053, $|\vec{P}|$=2.39676, $|\vec{P}'|$=0.371823
$\theta_{p'}$=1.30111e-07, θ_γ=3.14159, $\phi_{p'}$=1.74218, ϕ_γ=3.74627
E_γ=2.16667, $\theta_{\ell'}$=2.58096e-08, $\phi_{\ell'}$=4.88378
Q^2=17.2073, x_B=0.741984, t=-1.65188, W^2=6.86346, y=0.973331
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3968 p_x= 0.0000 p_y= 0.0000 p_z= -2.3968
P E= 2.5738 p_x= 0.0000 p_y= 0.0000 p_z= 2.3968
ℓ' E= 1.7948 p_x= 0.0000 p_y= -0.0000 p_z= 1.7948
P' E= 1.0090 p_x= -0.0000 p_y= 0.0000 p_z= 0.3718
q_γ E= 2.1667 p_x= -0.0000 p_y= -0.0000 p_z= -2.1667
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.192\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.808\pi$
normalized amplitude magnitude $|\widetilde{A}|$

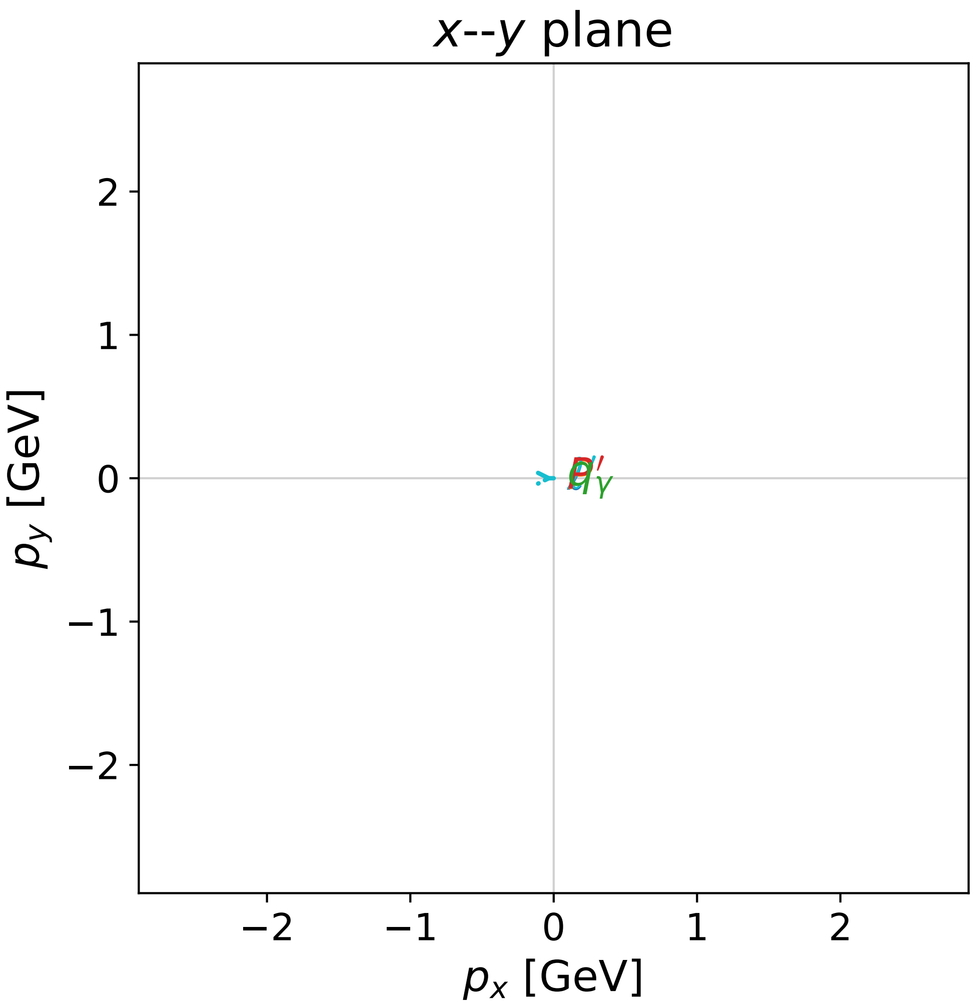
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

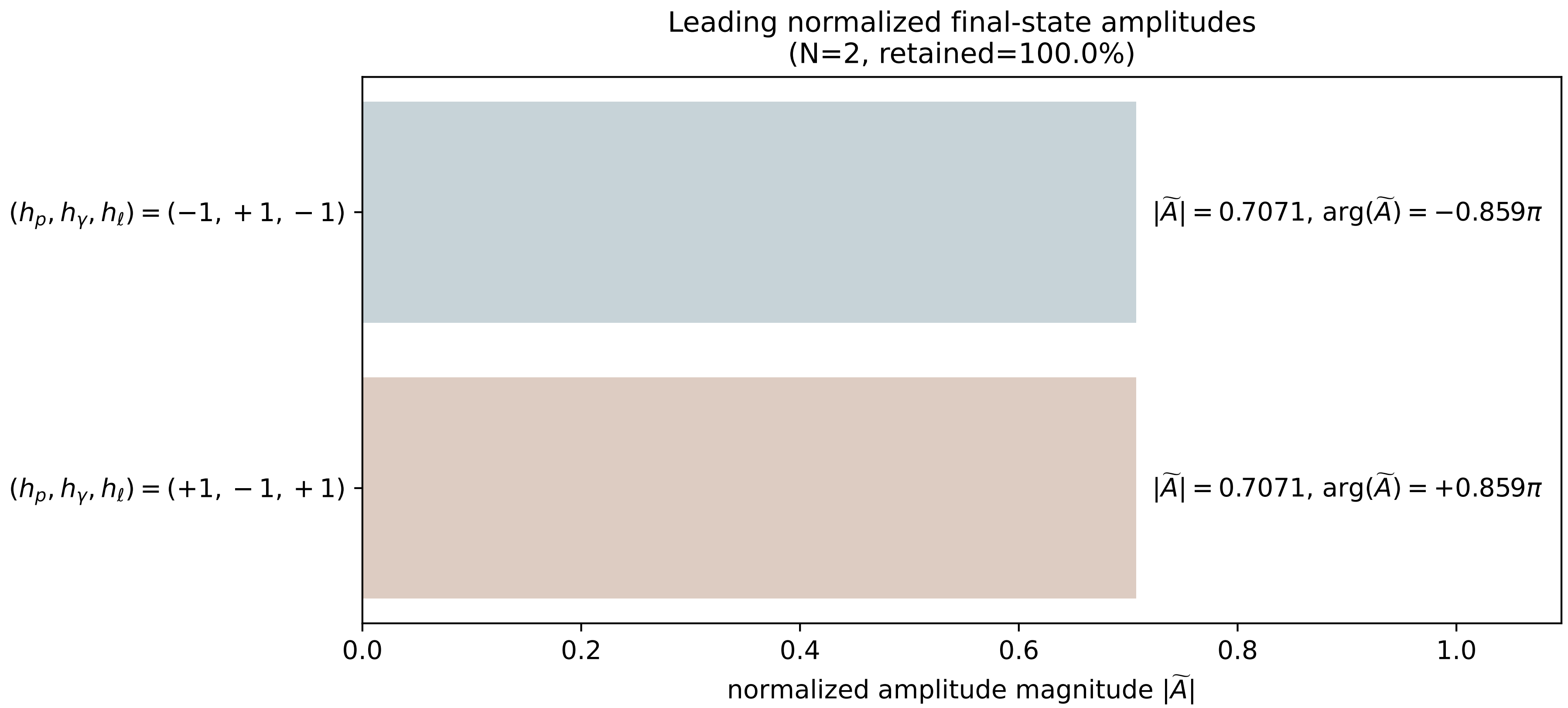



```
dghz_mixing_angles_polarization_02_local_minimum_3
d_{\rm GHZ}=2.32433e-08
region: polarization_cluster_02_local_minimum_3
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0003
lepton mixing angle: alpha_e=0.82685667 rad
incoming lepton: 0.677192|+> +0.735806|->
proton mixing angle: alpha_p=2.3147292 rad
incoming proton: -0.677187|+> +0.735811|->
```

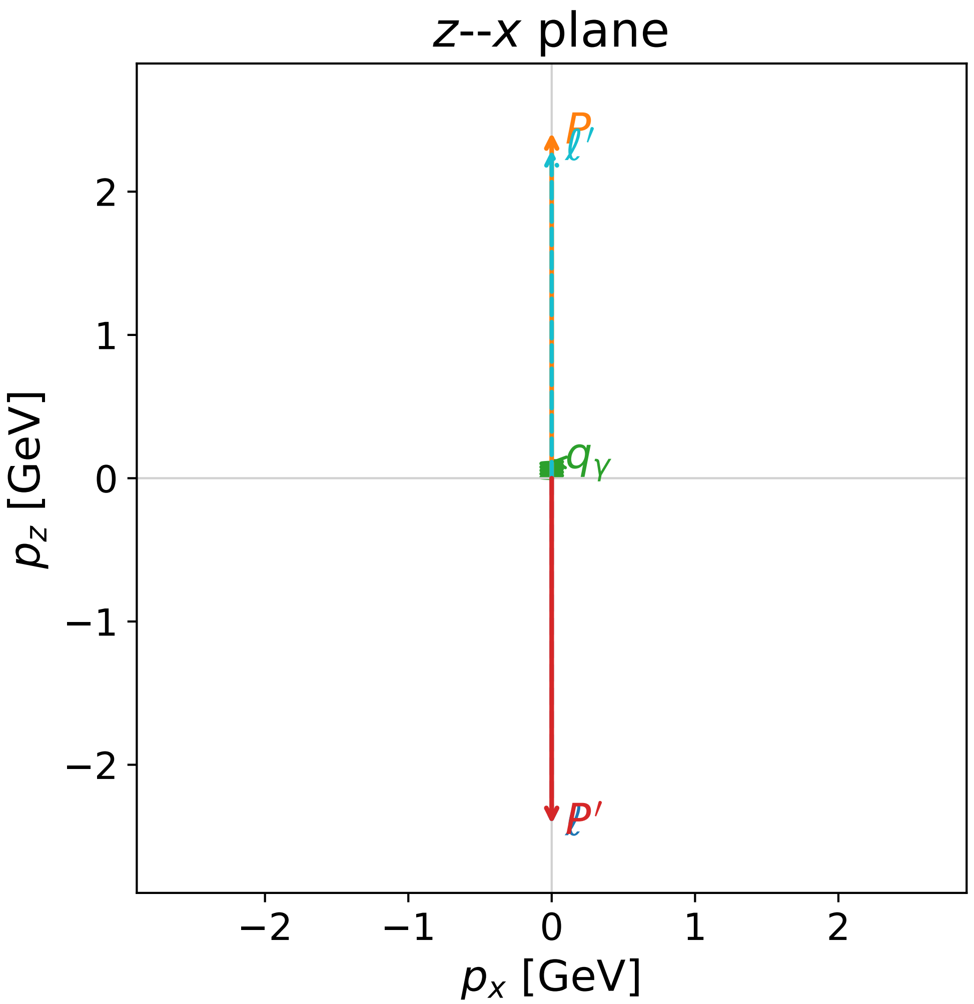
s=24.9996, $\sqrt{s}$=4.99996, $|\vec{P}|$=2.412, $|\vec{P}'|$=1.18707
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=2.98106, ϕ_γ=2.69995
E_γ=1.14997, $\theta_{\ell'}$=0, $\phi_{\ell'}$=5.98435
Q^2=22.5478, x_B=0.96783, t=-11.7976, W^2=1.62932, y=0.965899

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.3370 p_x= 0.0000 p_y= -0.0000 p_z= 2.3370
P' E= 1.5129 p_x= -0.0000 p_y= 0.0000 p_z= -1.1871
q_γ E= 1.1500 p_x= -0.0000 p_y= 0.0000 p_z= -1.1500





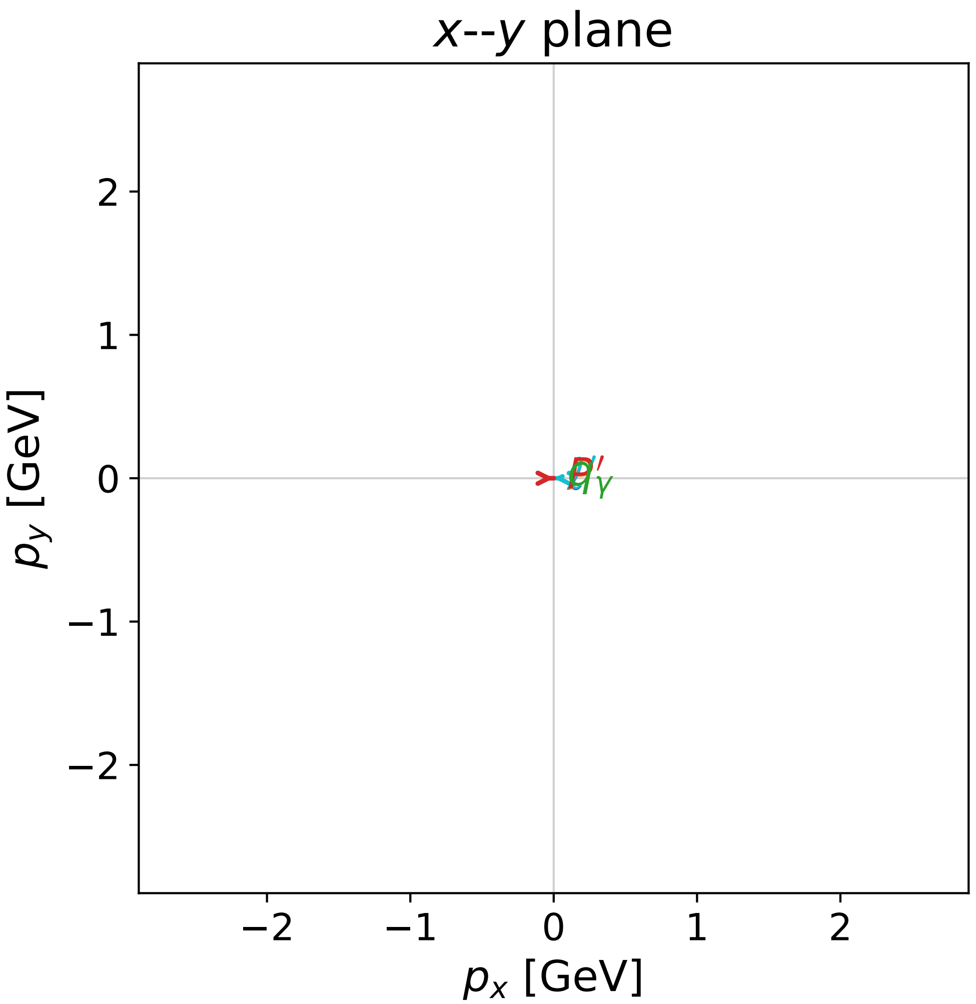
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

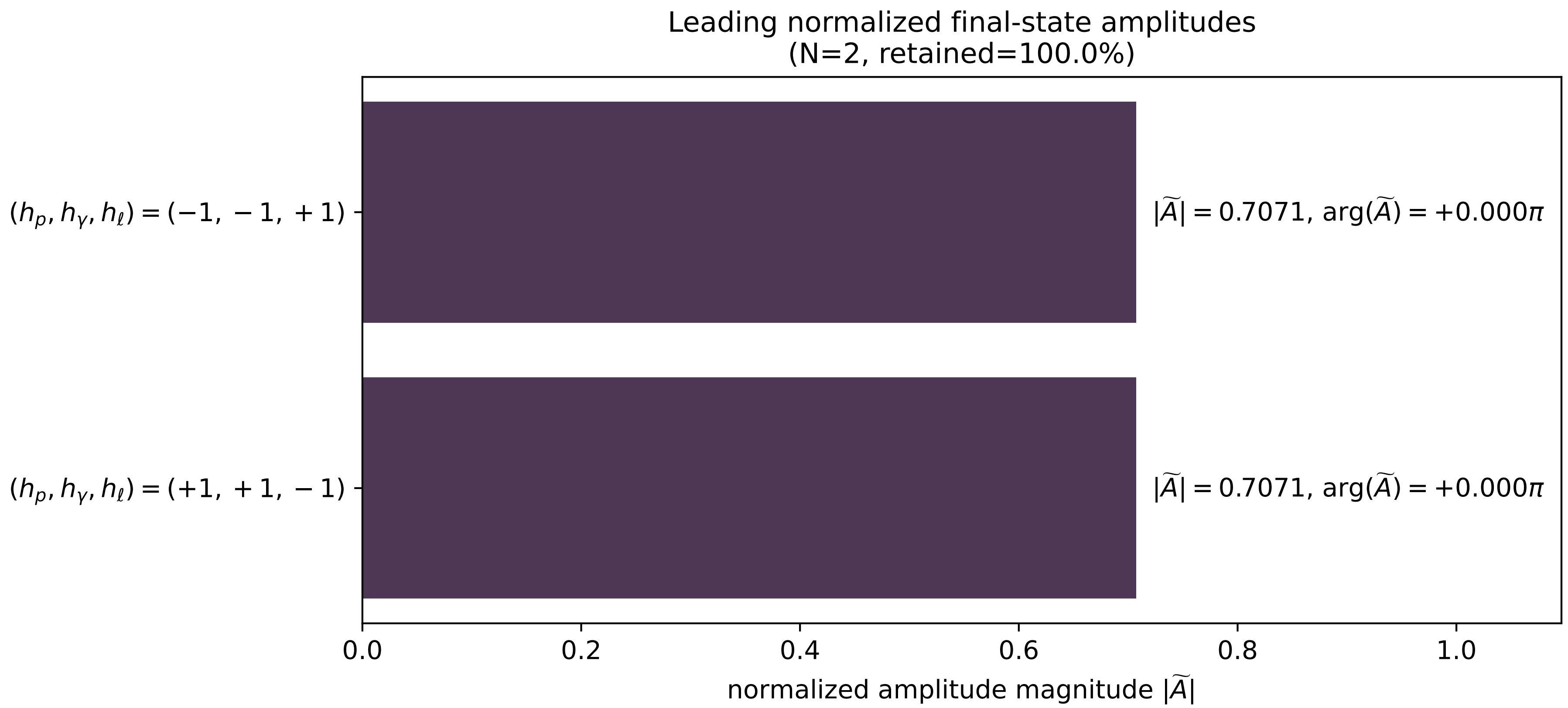


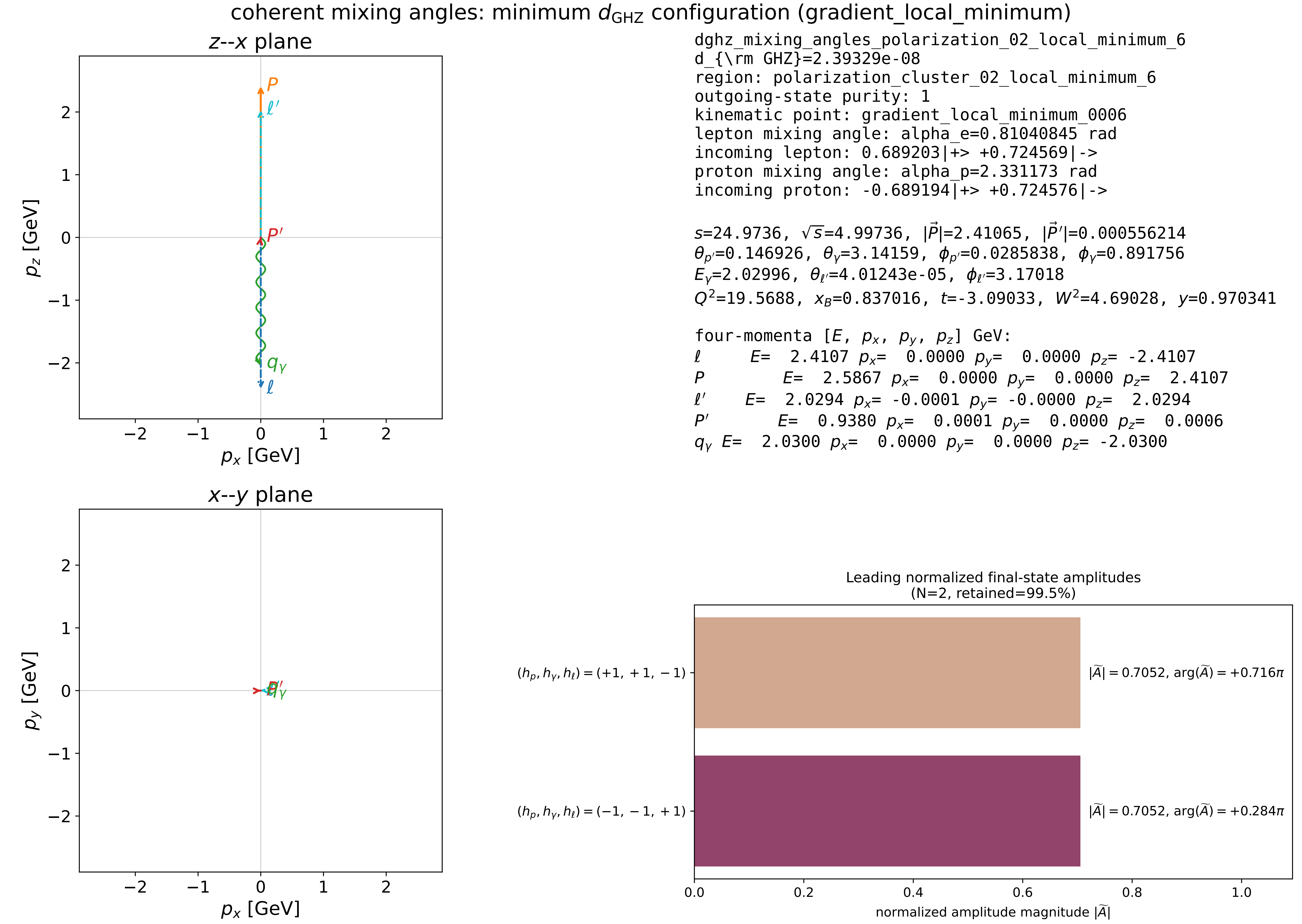
dghz_mixing_angles_polarization_02_local_minimum_4
d_{\rm GHZ}=2.36714e-08
region: polarization_cluster_02_local_minimum_4
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0004
lepton mixing angle: alpha_e=0.81232036 rad
incoming lepton: 0.687816|+> +0.725885|->
proton mixing angle: alpha_p=2.3831047 rad
incoming proton: -0.725877|+> +0.687825|->

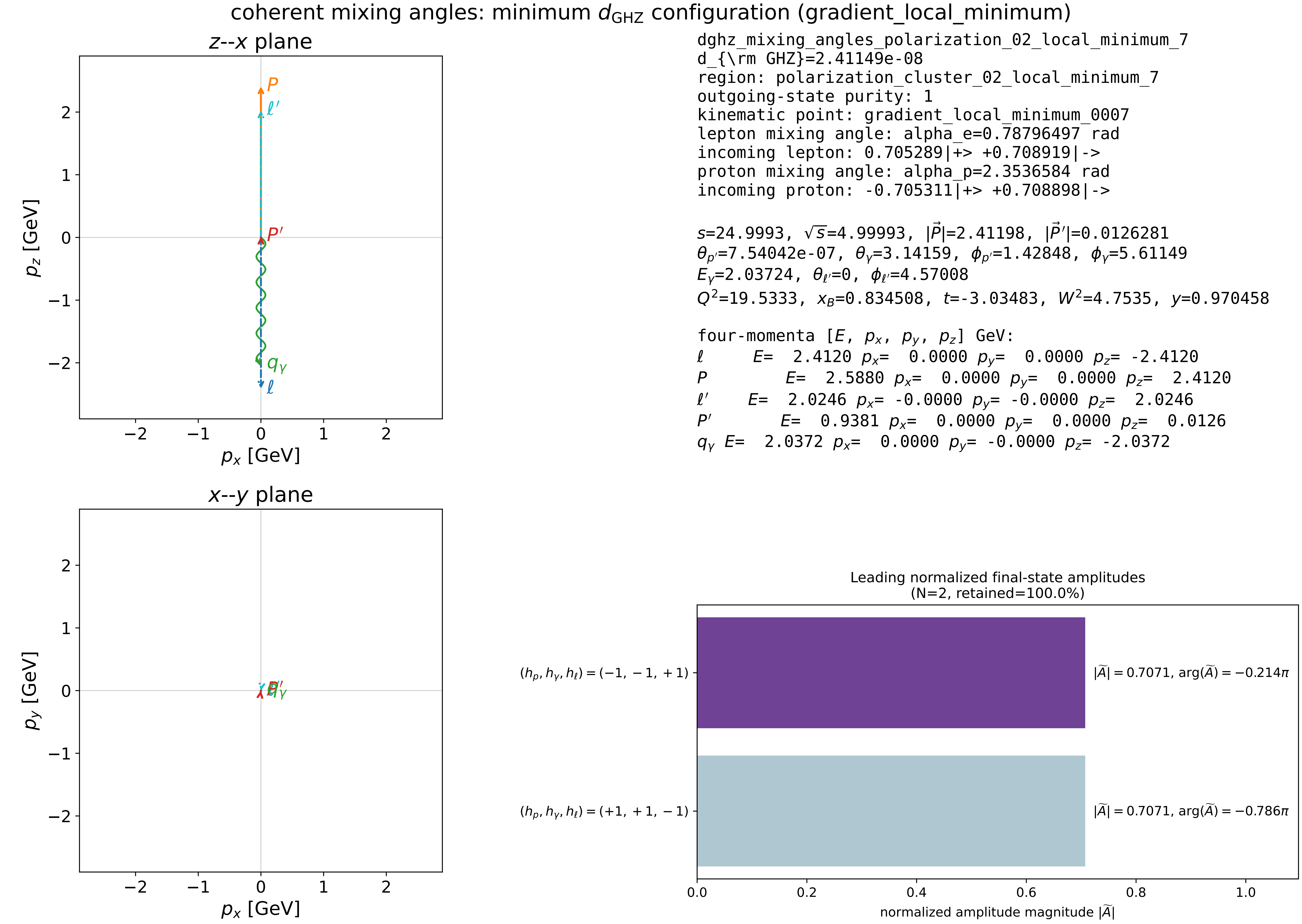
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=2.41202
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=0, ϕ_γ=0.437635
E_γ=0.120596, $\theta_{\ell'}$=0, $\phi_{\ell'}$=3.14159
Q^2=22.1078, x_B=0.948273, t=-23.2713, W^2=2.0858, y=0.966566

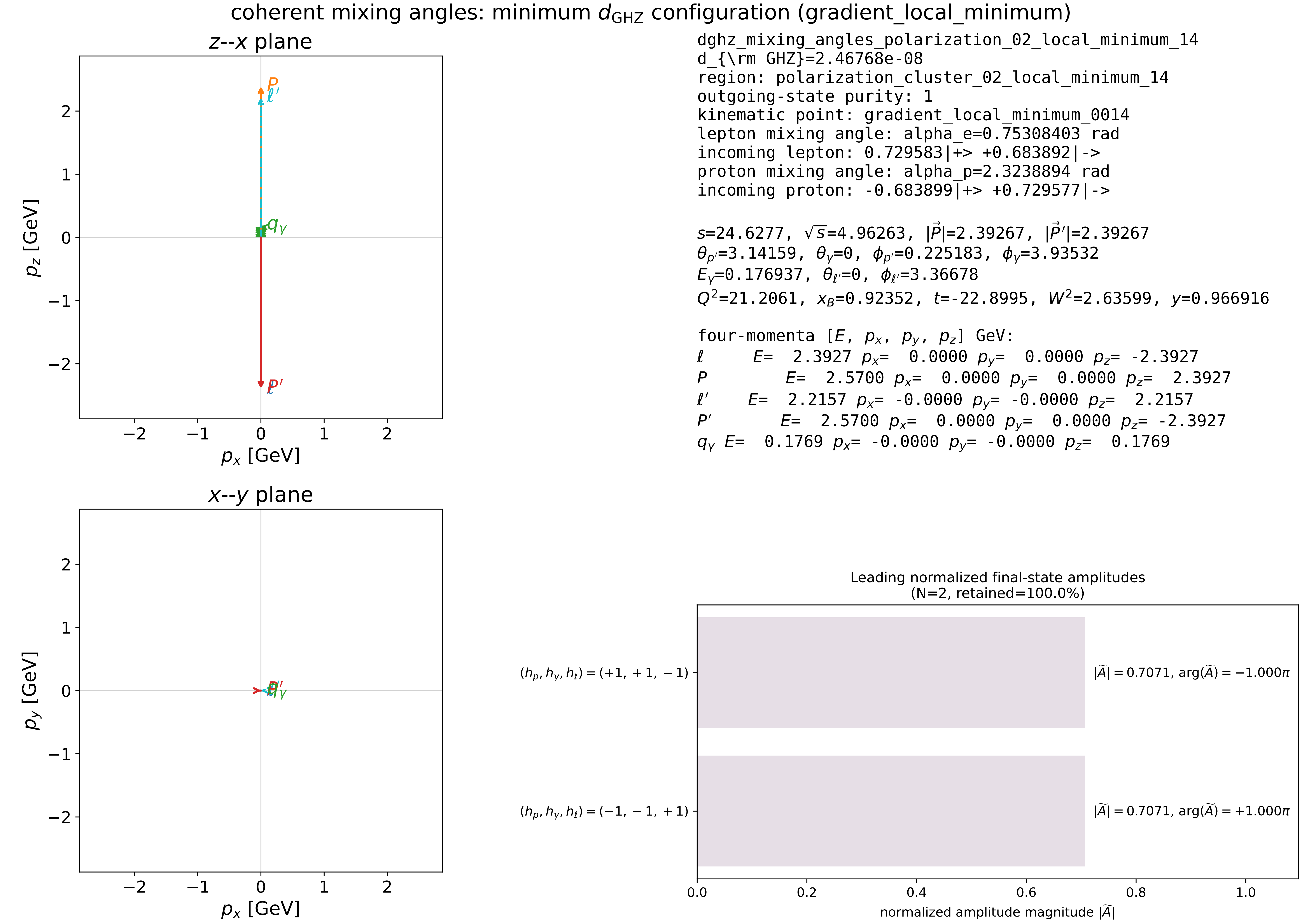
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.2914 p_x= -0.0000 p_y= -0.0000 p_z= 2.2914
P' E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
q_γ E= 0.1206 p_x= 0.0000 p_y= 0.0000 p_z= 0.1206



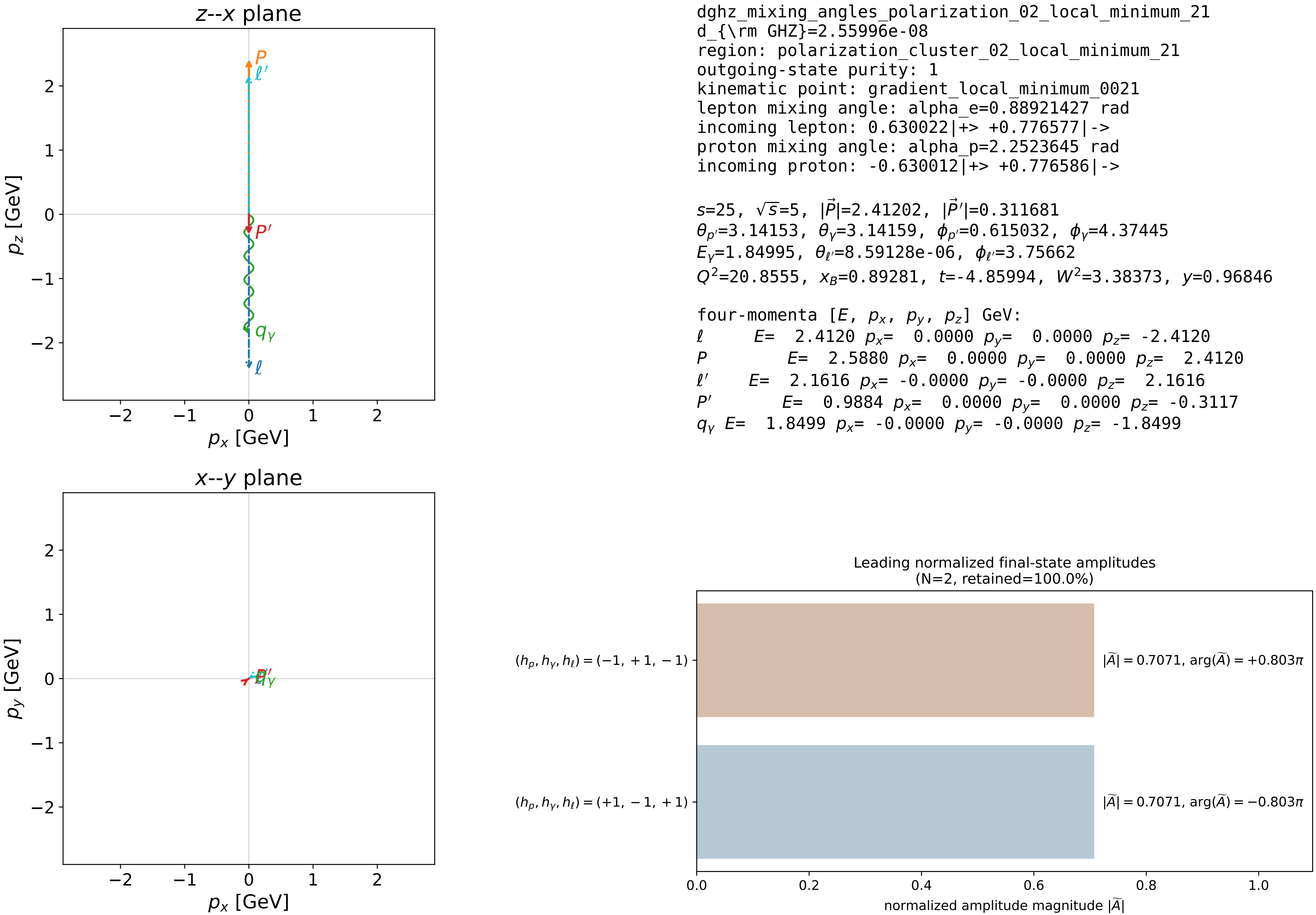



coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_6
d_{\rm GHZ}=2.39329e-08
region: polarization_cluster_02_local_minimum_6
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0006
lepton mixing angle: alpha_e=0.81040845 rad
incoming lepton: 0.689203|+> +0.724569|->
proton mixing angle: alpha_p=2.331173 rad
incoming proton: -0.689194|+> +0.724576|->
s=24.9736, $\sqrt{s}$=4.99736, $|\vec{P}|$=2.41065, $|\vec{P}'|$=0.000556214
$\theta_{p'}$=0.146926, θ_γ=3.14159, $\phi_{p'}$=0.0285838, ϕ_γ=0.891756
E_γ=2.02996, $\theta_{\ell'}$=4.01243e-05, $\phi_{\ell'}$=3.17018
Q^2=19.5688, x_B=0.837016, t=-3.09033, W^2=4.69028, y=0.970341
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4107 p_x= 0.0000 p_y= 0.0000 p_z= -2.4107
P E= 2.5867 p_x= 0.0000 p_y= 0.0000 p_z= 2.4107
ℓ' E= 2.0294 p_x= -0.0001 p_y= -0.0000 p_z= 2.0294
P' E= 0.9380 p_x= 0.0001 p_y= 0.0000 p_z= 0.0006
q_γ E= 2.0300 p_x= 0.0000 p_y= 0.0000 p_z= -2.0300
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=99.5%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7052$, $\arg(\widetilde{A}) = +0.716\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7052$, $\arg(\widetilde{A}) = +0.284\pi$
normalized amplitude magnitude $|\widetilde{A}|$

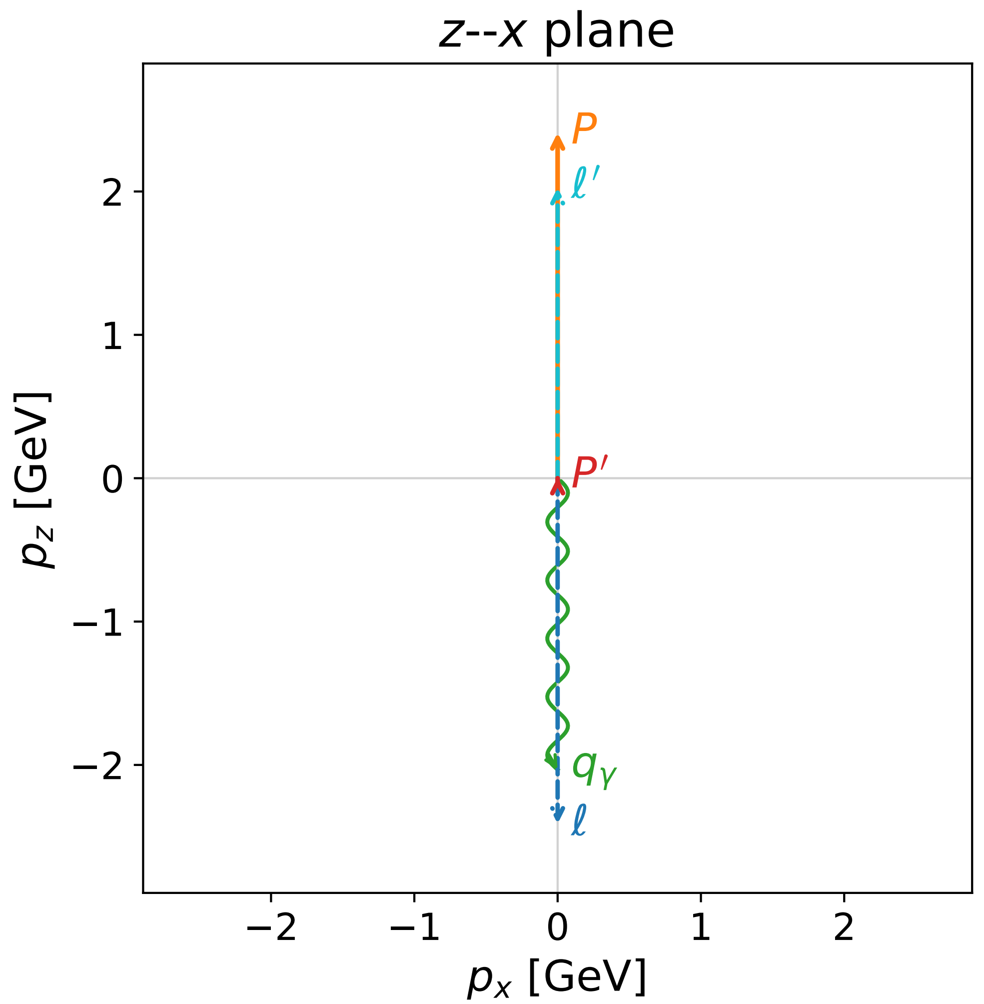

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_7
d_{\rm GHZ}=2.41149e-08
region: polarization_cluster_02_local_minimum_7
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0007
lepton mixing angle: alpha_e=0.78796497 rad
incoming lepton: 0.705289|+> +0.708919|->
proton mixing angle: alpha_p=2.3536584 rad
incoming proton: -0.705311|+> +0.708898|->
s=24.9993, $\sqrt{s}$=4.99993, $|\vec{P}|$=2.41198, $|\vec{P}'|$=0.0126281
$\theta_{p'}$=7.54042e-07, θ_γ=3.14159, $\phi_{p'}$=1.42848, ϕ_γ=5.61149
E_γ=2.03724, $\theta_{\ell'}$=0, $\phi_{\ell'}$=4.57008
Q^2=19.5333, x_B=0.834508, t=-3.03483, W^2=4.7535, y=0.970458
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.0246 p_x= -0.0000 p_y= -0.0000 p_z= 2.0246
P' E= 0.9381 p_x= 0.0000 p_y= 0.0000 p_z= 0.0126
q_γ E= 2.0372 p_x= 0.0000 p_y= -0.0000 p_z= -2.0372
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.214\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.786\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
q_γ
P'
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_14
d_{\rm GHZ}=2.46768e-08
region: polarization_cluster_02_local_minimum_14
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0014
lepton mixing angle: alpha_e=0.75308403 rad
incoming lepton: 0.729583|+> +0.683892|->
proton mixing angle: alpha_p=2.3238894 rad
incoming proton: -0.683899|+> +0.729577|->
s=24.6277, $\sqrt{s}$=4.96263, $|\vec{P}|$=2.39267, $|\vec{P}'|$=2.39267
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=0.225183, ϕ_γ=3.93532
E_γ=0.176937, $\theta_{\ell'}$=0, $\phi_{\ell'}$=3.36678
Q^2=21.2061, x_B=0.92352, t=-22.8995, W^2=2.63599, y=0.966916
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3927 p_x= 0.0000 p_y= 0.0000 p_z= -2.3927
P E= 2.5700 p_x= 0.0000 p_y= 0.0000 p_z= 2.3927
ℓ' E= 2.2157 p_x= -0.0000 p_y= -0.0000 p_z= 2.2157
P' E= 2.5700 p_x= 0.0000 p_y= 0.0000 p_z= -2.3927
q_γ E= 0.1769 p_x= -0.0000 p_y= -0.0000 p_z= 0.1769
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)



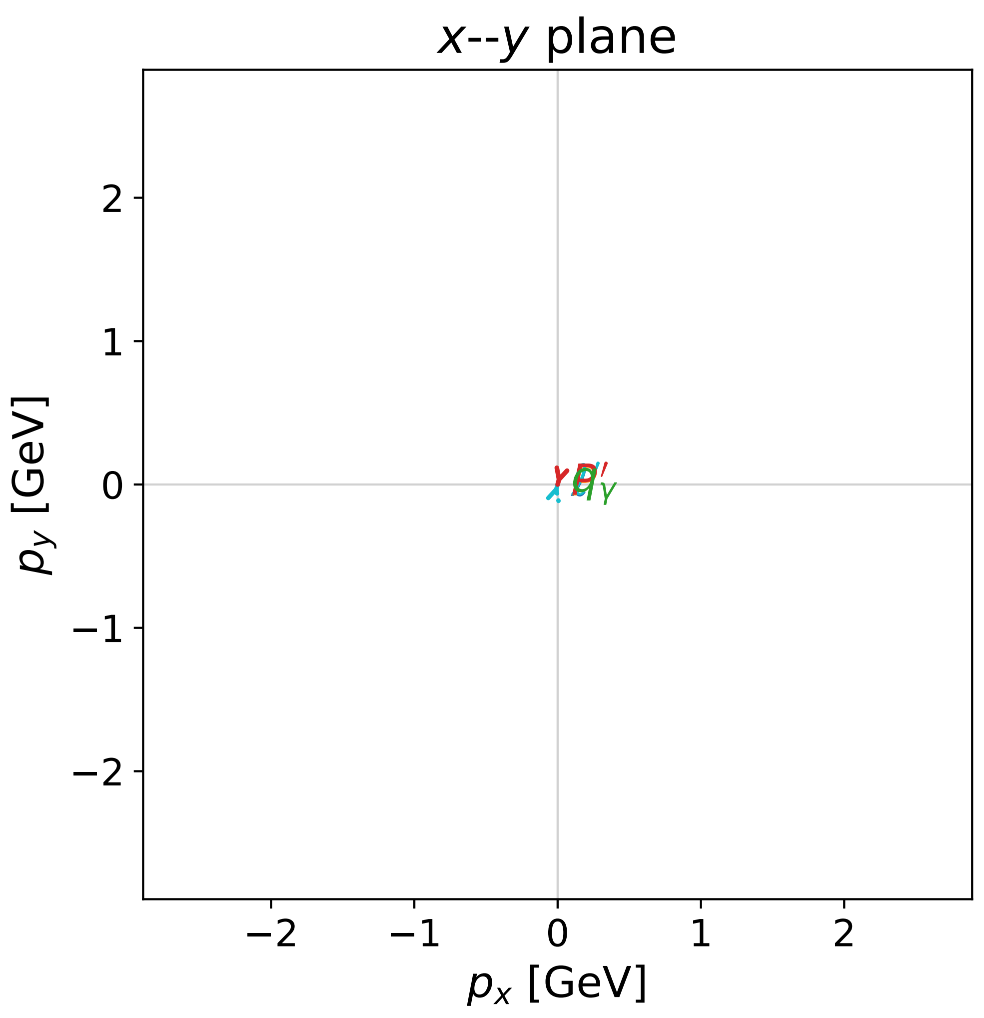
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

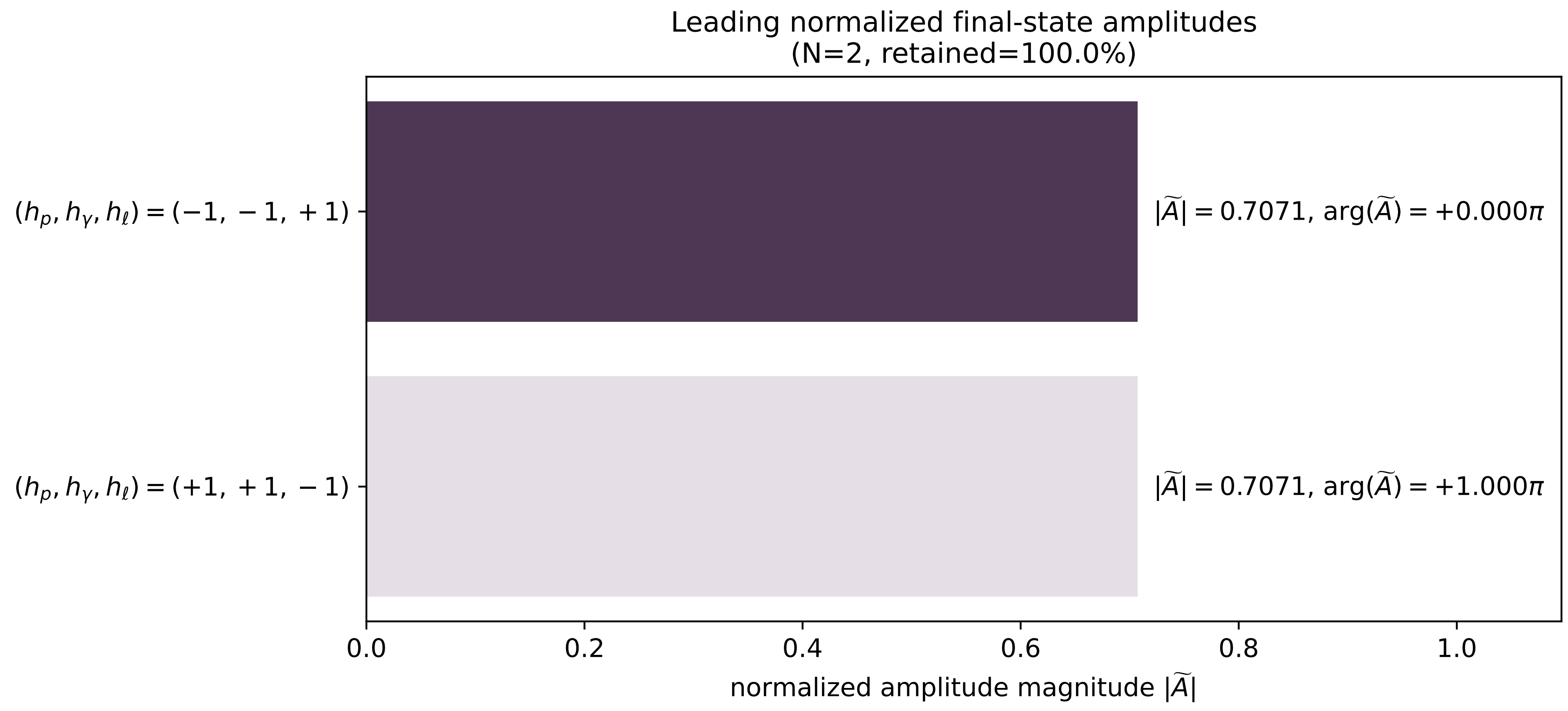


dghz_mixing_angles_polarization_02_local_minimum_26
d_{\rm GHZ}=2.59712e-08
region: polarization_cluster_02_local_minimum_26
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0026
lepton mixing angle: alpha_e=0.77423168 rad
incoming lepton: 0.714958|+> +0.699167|->
proton mixing angle: alpha_p=2.3673692 rad
incoming proton: -0.714964|+> +0.699161|->

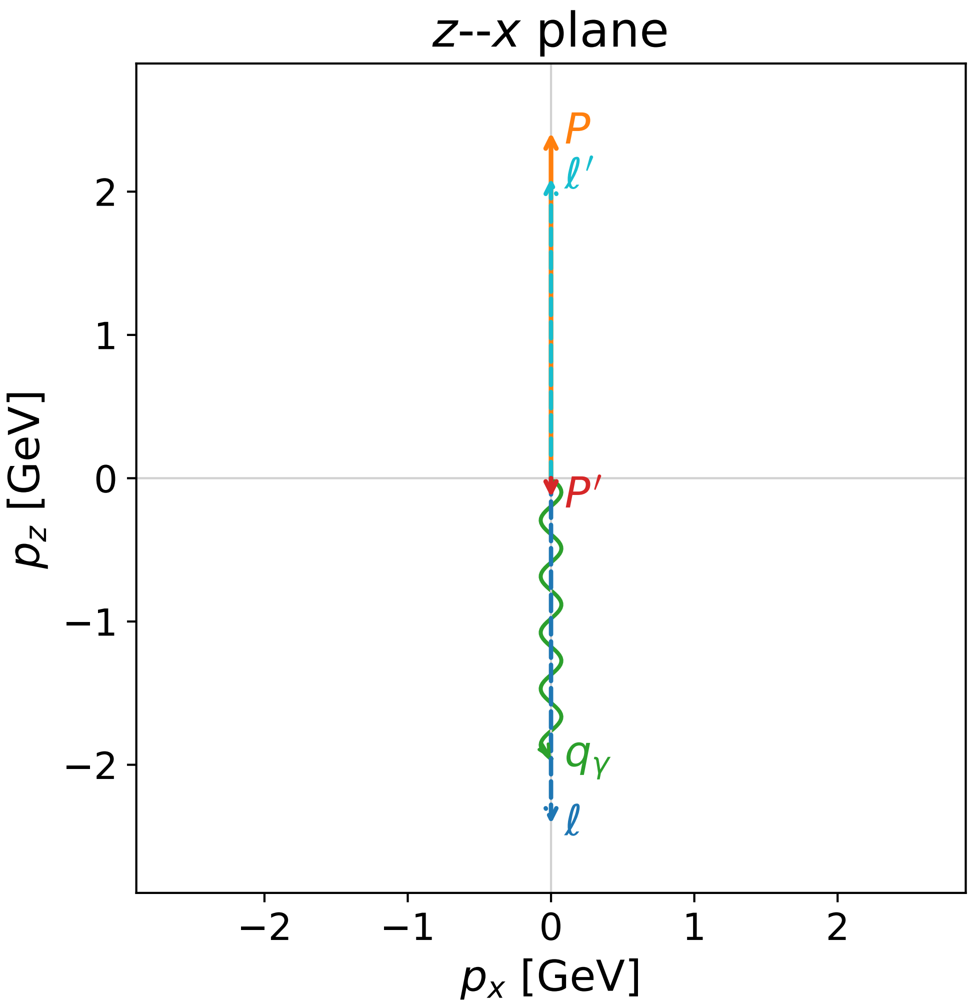
s=24.9729, $\sqrt{s}$=4.99729, $|\vec{P}|$=2.41061, $|\vec{P}'|$=0.00645061
$\theta_{p'}$=0.00979874, θ_γ=3.14159, $\phi_{p'}$=4.43638, ϕ_γ=6.38506e-05
E_γ=2.03286, $\theta_{\ell'}$=3.11916e-05, $\phi_{\ell'}$=1.29479
Q^2=19.5395, x_B=0.835755, t=-3.06193, W^2=4.7198, y=0.970383

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4106 p_x= 0.0000 p_y= 0.0000 p_z= -2.4106
P E= 2.5867 p_x= 0.0000 p_y= 0.0000 p_z= 2.4106
ℓ' E= 2.0264 p_x= 0.0000 p_y= 0.0001 p_z= 2.0264
P' E= 0.9380 p_x= -0.0000 p_y= -0.0001 p_z= 0.0065
q_γ E= 2.0329 p_x= 0.0000 p_y= 0.0000 p_z= -2.0329





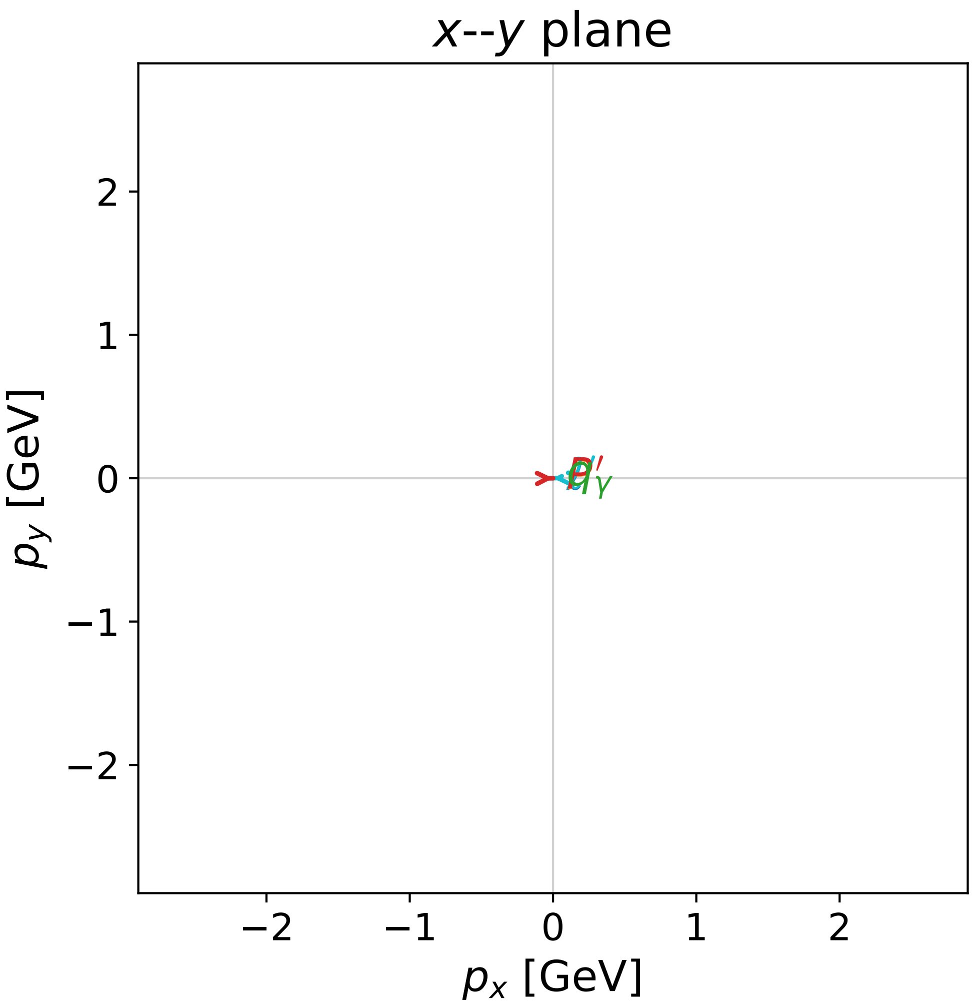
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

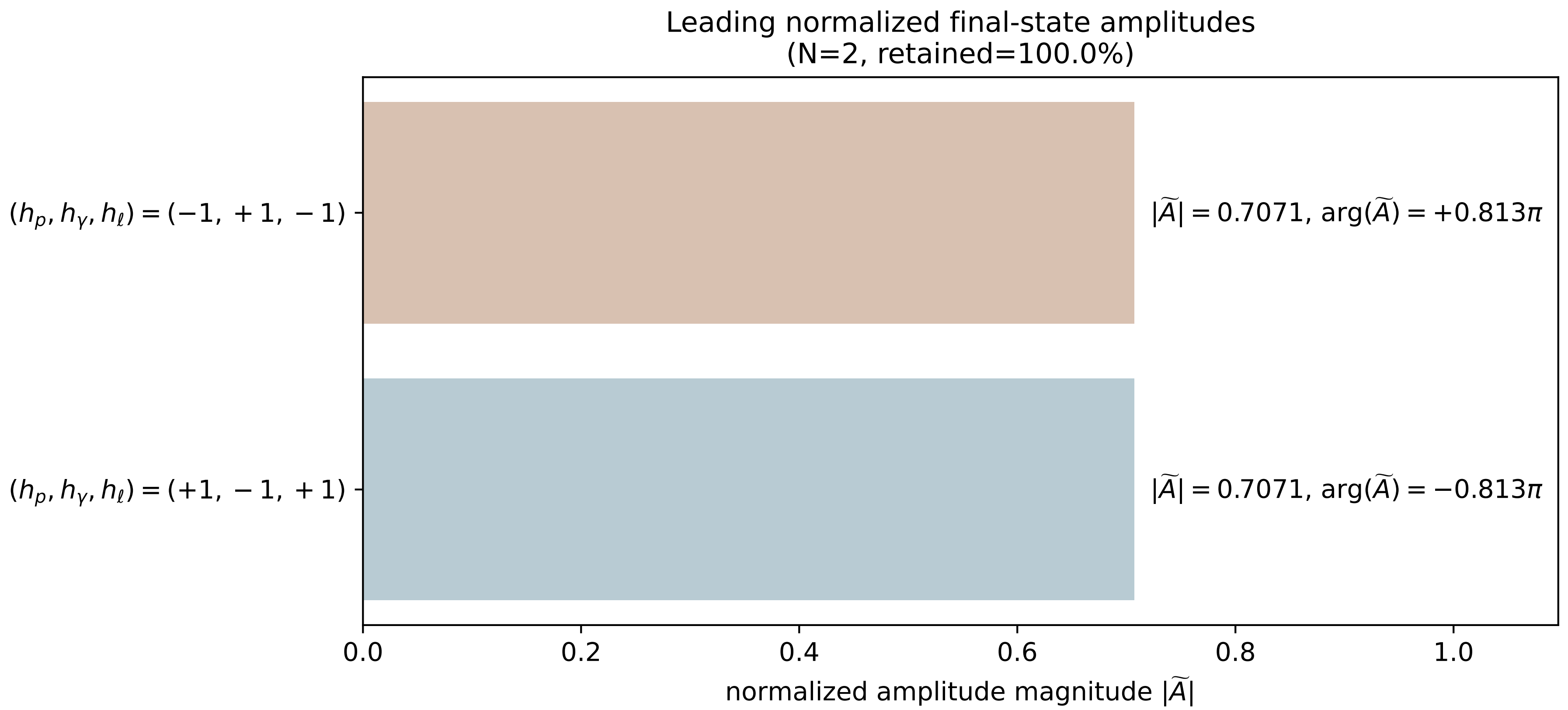


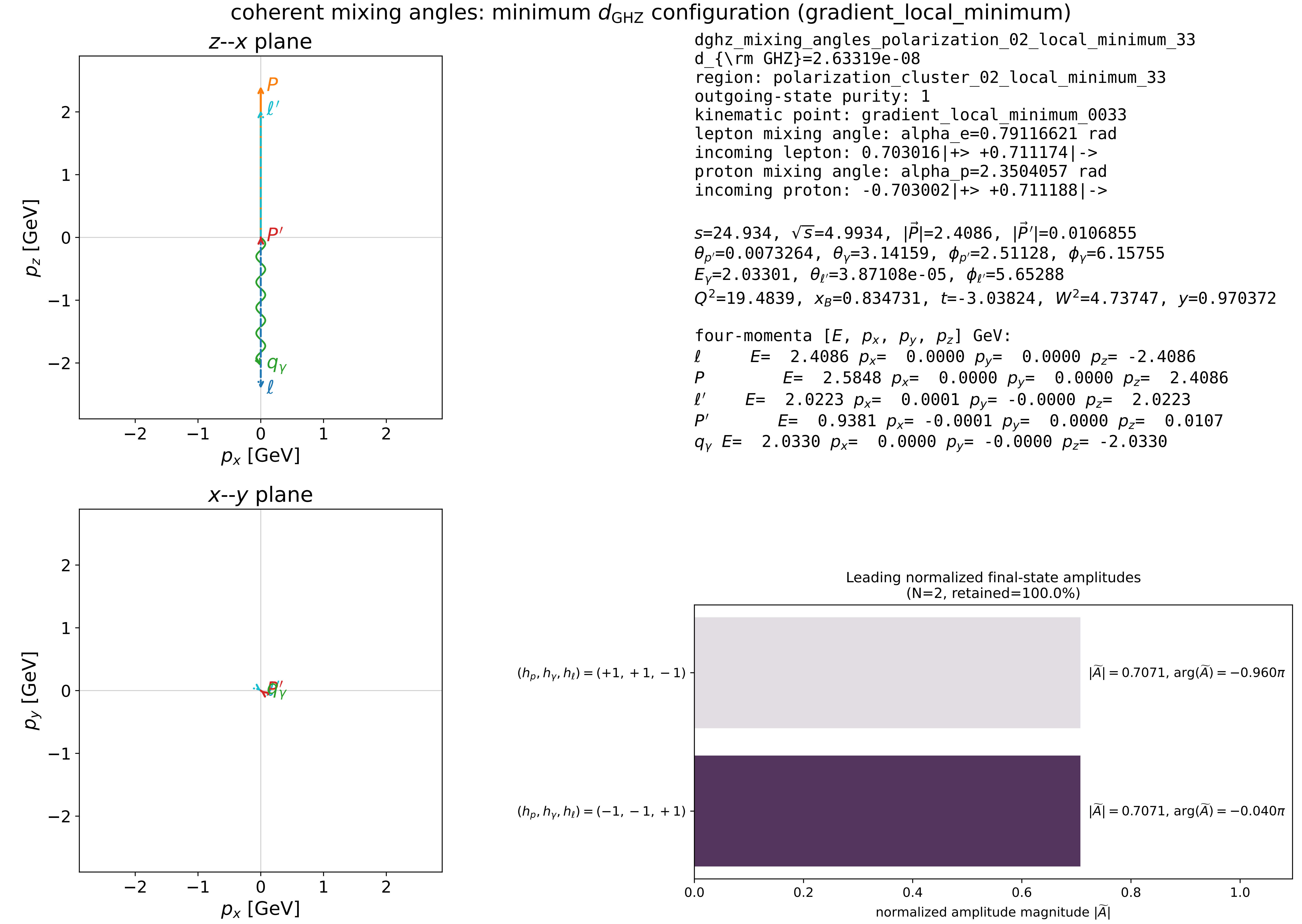
dghz_mixing_angles_polarization_02_local_minimum_31
d_{\rm GHZ}=2.62572e-08
region: polarization_cluster_02_local_minimum_31
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0031
lepton mixing angle: alpha_e=0.72294087 rad
incoming lepton: 0.749863|+> +0.661593|->
proton mixing angle: alpha_p=2.4186407 rad
incoming proton: -0.749856|+> +0.661601|->

s=24.9982, $\sqrt{s}$=4.99982, $|\vec{P}|$=2.41192, $|\vec{P}'|$=0.133473
$\theta_{p'}$=3.14093, θ_γ=3.14159, $\phi_{p'}$=0.0144773, ϕ_γ=3.74298
E_γ=1.95945, $\theta_{\ell'}$=4.19601e-05, $\phi_{\ell'}$=3.15607
Q^2=20.1919, x_B=0.863574, t=-3.78797, W^2=4.06974, y=0.969459

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4119 p_x= 0.0000 p_y= 0.0000 p_z= -2.4119
P E= 2.5879 p_x= 0.0000 p_y= 0.0000 p_z= 2.4119
ℓ' E= 2.0929 p_x= -0.0001 p_y= -0.0000 p_z= 2.0929
P' E= 0.9474 p_x= 0.0001 p_y= 0.0000 p_z= -0.1335
q_γ E= 1.9595 p_x= -0.0000 p_y= -0.0000 p_z= -1.9595

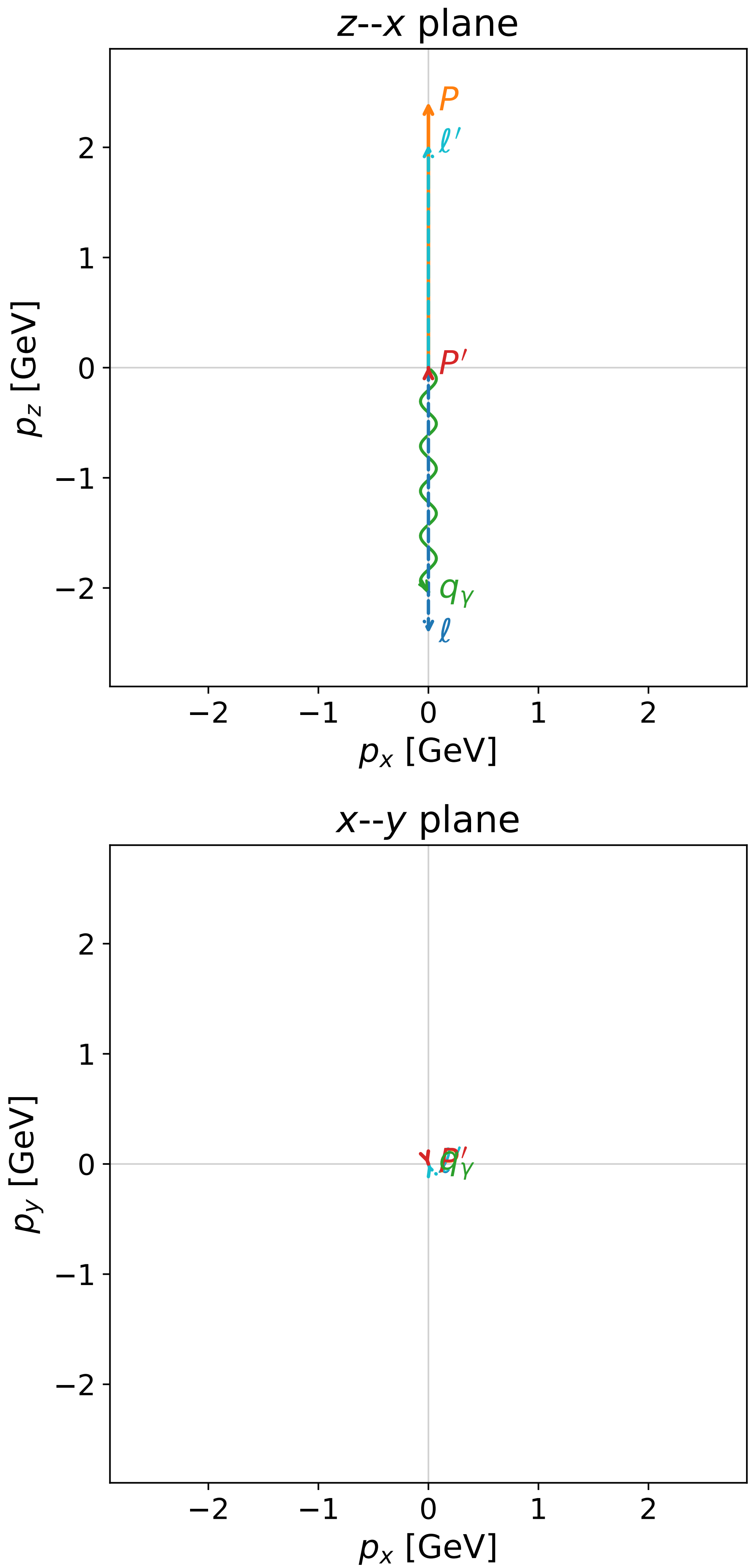






coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_33
d_{\rm GHZ}=2.63319e-08
region: polarization_cluster_02_local_minimum_33
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0033
lepton mixing angle: alpha_e=0.79116621 rad
incoming lepton: 0.703016|+> +0.711174|->
proton mixing angle: alpha_p=2.3504057 rad
incoming proton: -0.703002|+> +0.711188|->
s=24.934, $\sqrt{s}$=4.9934, $|\vec{P}|$=2.4086, $|\vec{P}'|$=0.0106855
$\theta_{p'}$=0.0073264, θ_γ=3.14159, $\phi_{p'}$=2.51128, ϕ_γ=6.15755
E_γ=2.03301, $\theta_{\ell'}$=3.87108e-05, $\phi_{\ell'}$=5.65288
Q^2=19.4839, x_B=0.834731, t=-3.03824, W^2=4.73747, y=0.970372
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4086 p_x= 0.0000 p_y= 0.0000 p_z= -2.4086
P E= 2.5848 p_x= 0.0000 p_y= 0.0000 p_z= 2.4086
ℓ' E= 2.0223 p_x= 0.0001 p_y= -0.0000 p_z= 2.0223
P' E= 0.9381 p_x= -0.0001 p_y= 0.0000 p_z= 0.0107
q_γ E= 2.0330 p_x= 0.0000 p_y= -0.0000 p_z= -2.0330
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.960\pi$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.040\pi$
normalized amplitude magnitude $|\widetilde{A}|$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

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dghz_mixing_angles_polarization_02_local_minimum_34
d_{\rm GHZ}=2.63343e-08
region: polarization_cluster_02_local_minimum_34
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0034
lepton mixing angle: alpha_e=0.76007598 rad
incoming lepton: 0.724784|+> +0.688977|->
proton mixing angle: alpha_p=2.3815115 rad
incoming proton: -0.72478|+> +0.68898|->
```
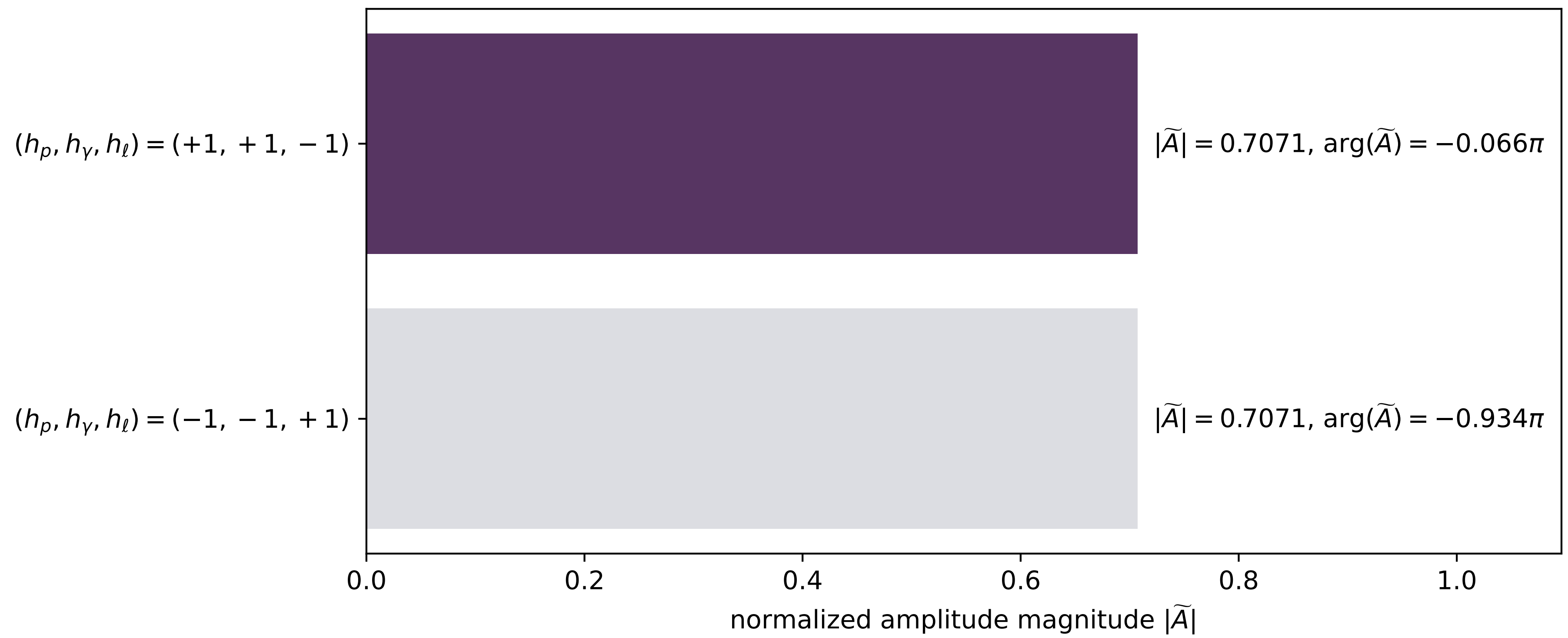
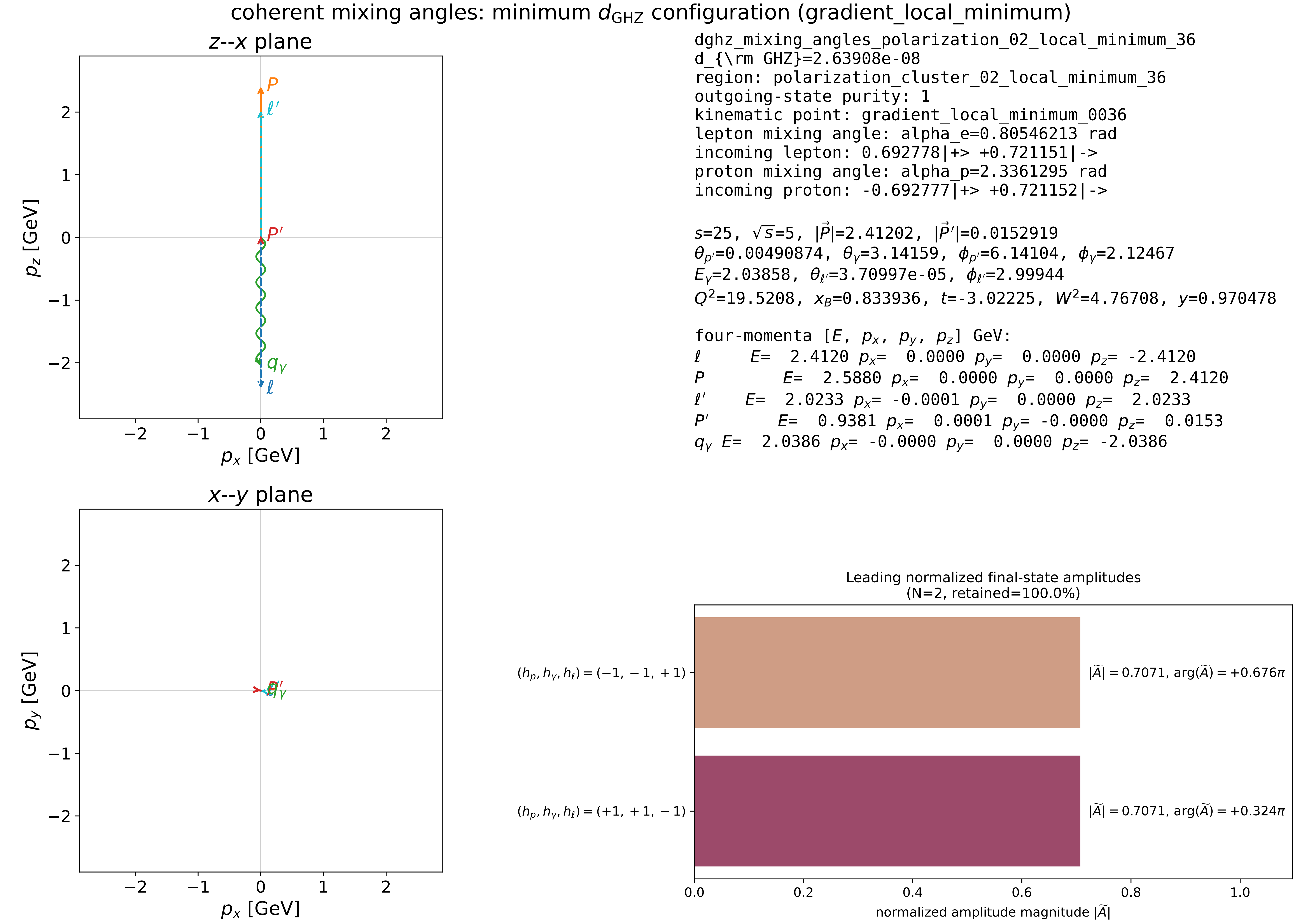

s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.0121954
$\theta_{p'}$=0.00245437, θ_γ=3.14159, $\phi_{p'}$=5.04048, ϕ_γ=3.34859
E_γ=2.03706, $\theta_{\ell'}$=1.47823e-05, $\phi_{\ell'}$=1.89889
Q^2=19.536, x_B=0.834603, t=-3.03695, W^2=4.75137, y=0.970455

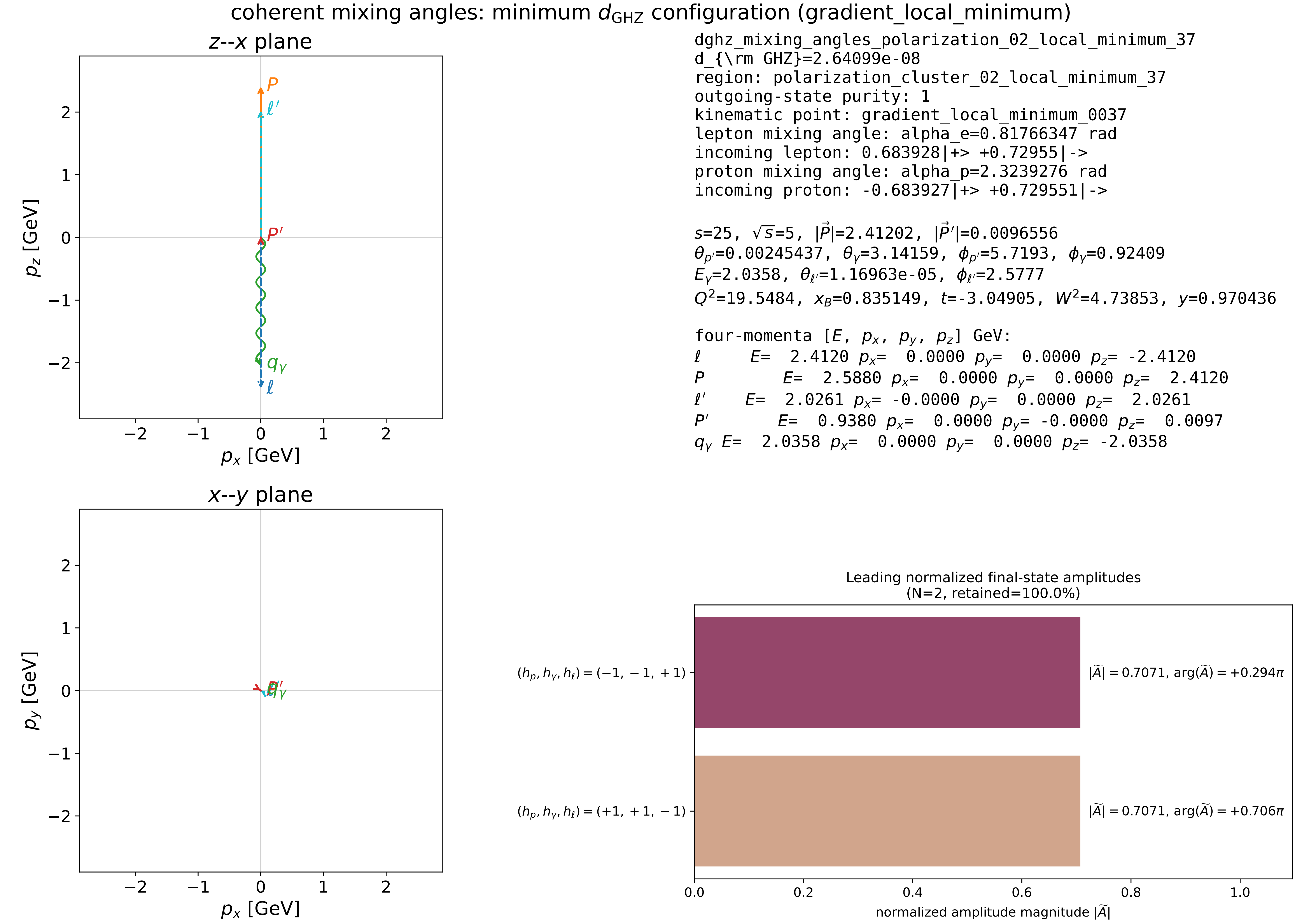
four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 2.4120	p_x= 0.0000	p_y= 0.0000	p_z= -2.4120
P	E= 2.5880	p_x= 0.0000	p_y= 0.0000	p_z= 2.4120
ℓ'	E= 2.0249	p_x= -0.0000	p_y= 0.0000	p_z= 2.0249
P'	E= 0.9381	p_x= 0.0000	p_y= -0.0000	p_z= 0.0122
q_γ	E= 2.0371	p_x= -0.0000	p_y= -0.0000	p_z= -2.0371

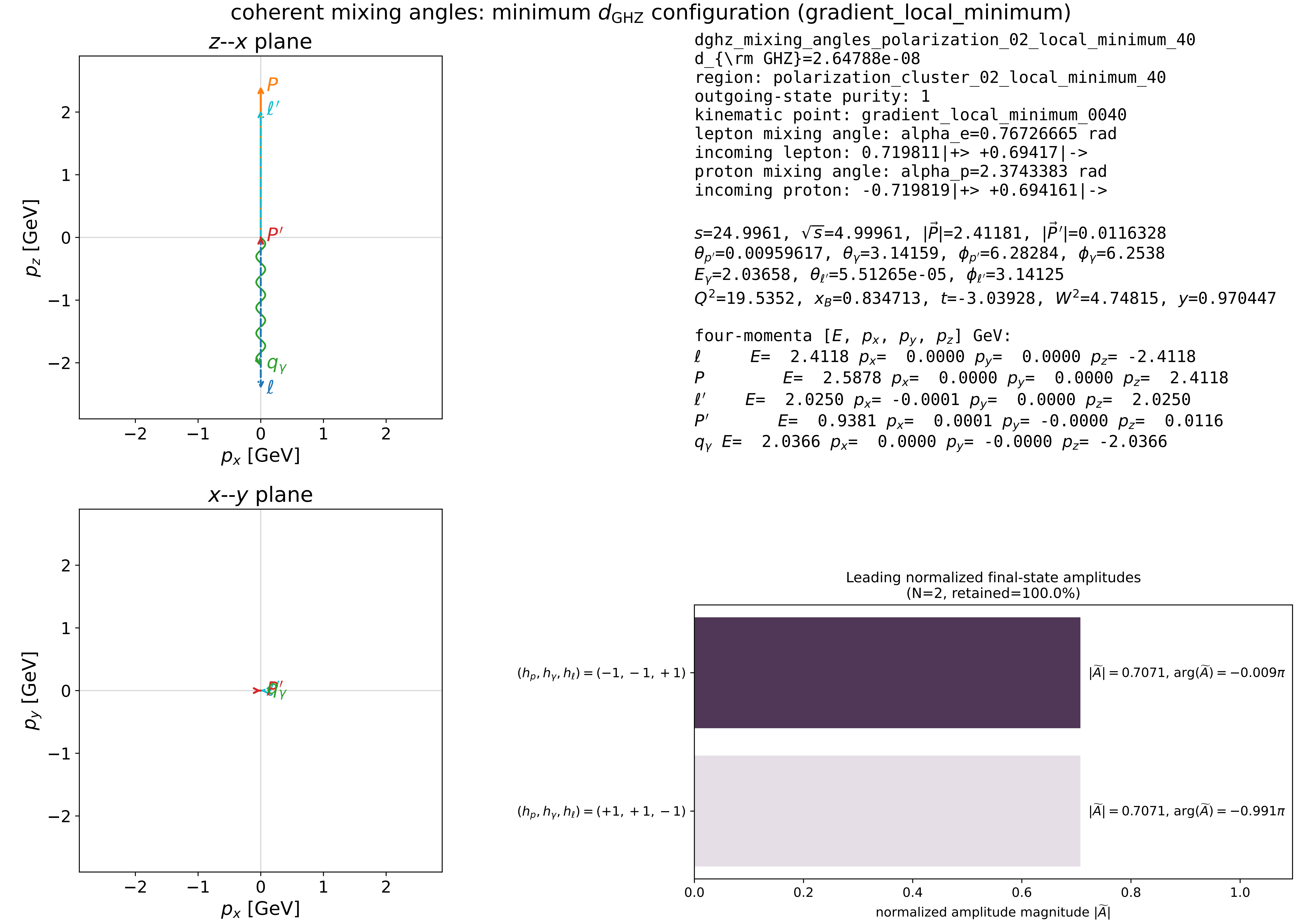
Leading normalized final-state amplitudes
(N=2, retained=100.0%)

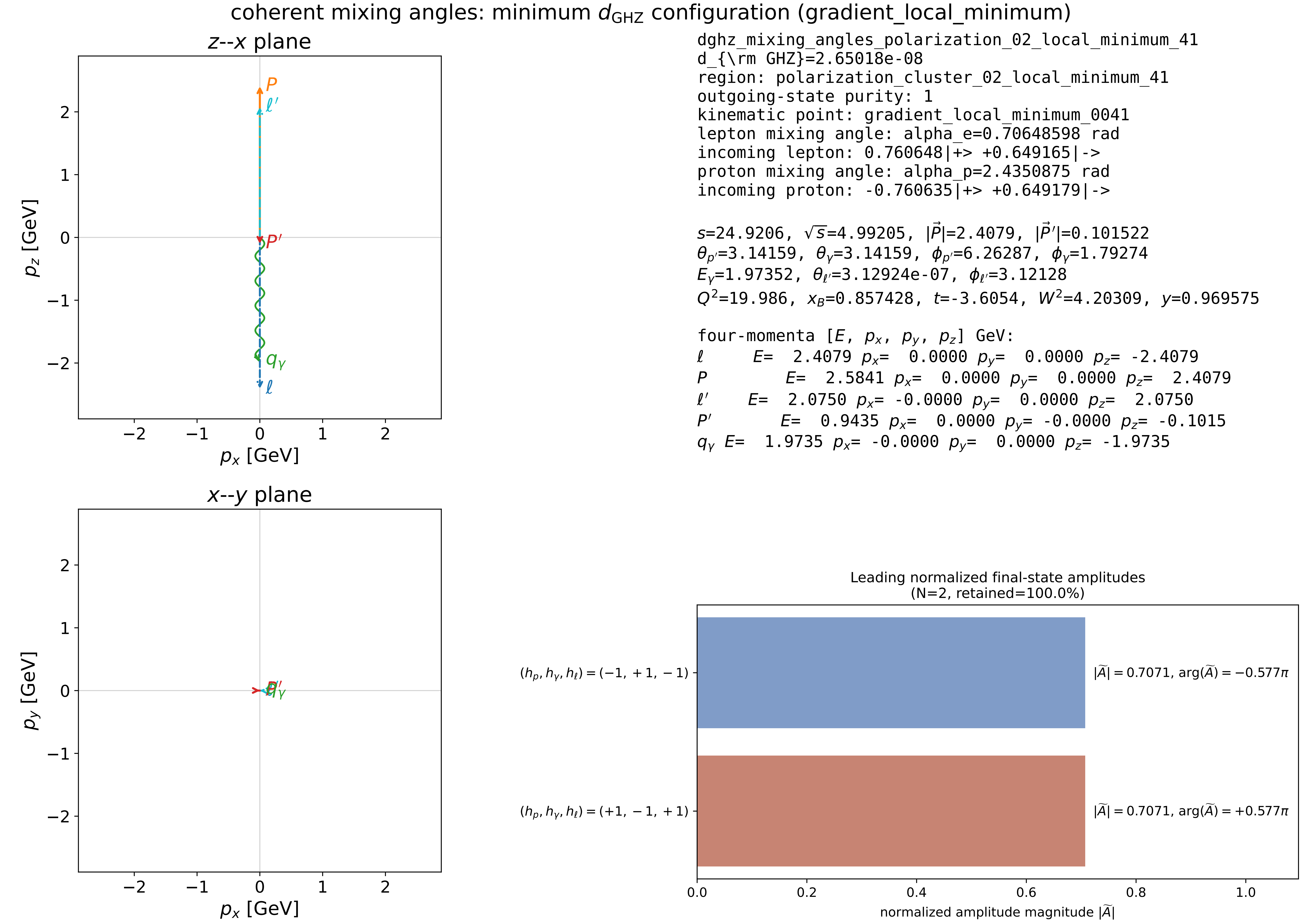


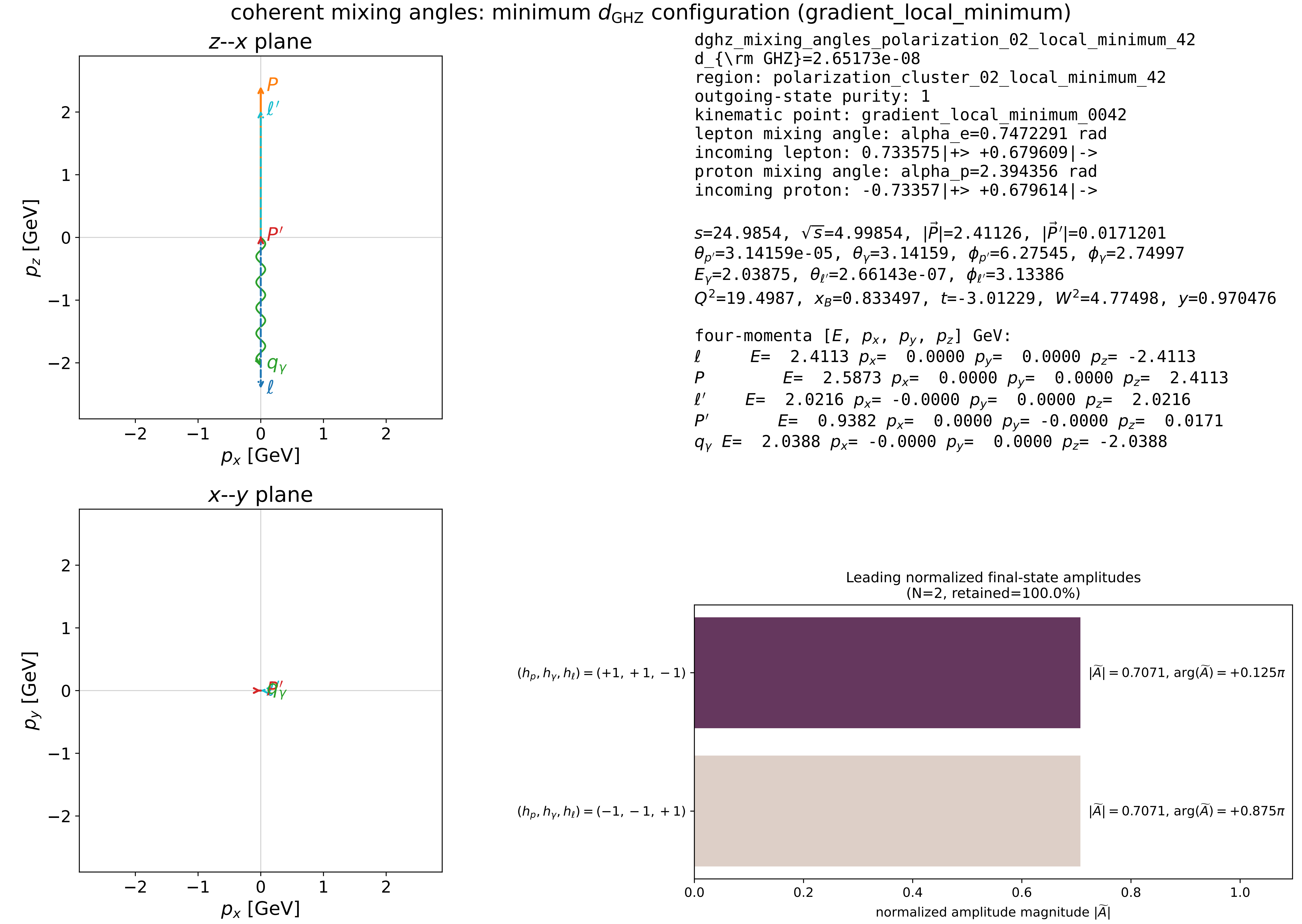

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
dghz_mixing_angles_polarization_02_local_minimum_36
d_{\rm GHZ}=2.63908e-08
region: polarization_cluster_02_local_minimum_36
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0036
lepton mixing angle: alpha_e=0.80546213 rad
incoming lepton: 0.692778|+> +0.721151|->
proton mixing angle: alpha_p=2.3361295 rad
incoming proton: -0.692777|+> +0.721152|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.0152919
$\theta_{p'}$=0.00490874, θ_γ=3.14159, $\phi_{p'}$=6.14104, ϕ_γ=2.12467
E_γ=2.03858, $\theta_{\ell'}$=3.70997e-05, $\phi_{\ell'}$=2.99944
Q^2=19.5208, x_B=0.833936, t=-3.02225, W^2=4.76708, y=0.970478
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.0233 p_x= -0.0001 p_y= 0.0000 p_z= 2.0233
P' E= 0.9381 p_x= 0.0001 p_y= -0.0000 p_z= 0.0153
q_γ E= 2.0386 p_x= -0.0000 p_y= 0.0000 p_z= -2.0386
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.676\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.324\pi$
normalized amplitude magnitude $|\widetilde{A}|$

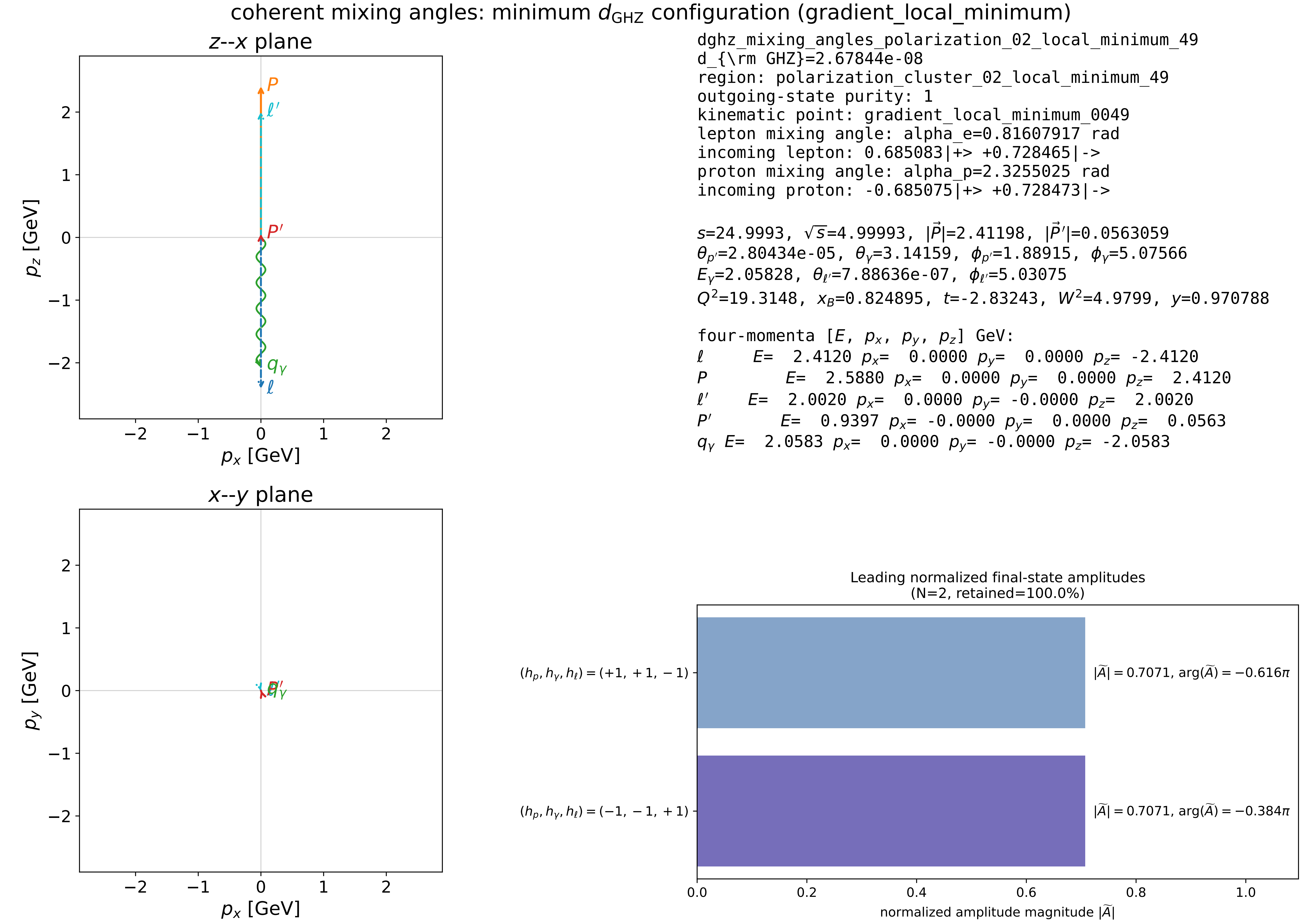


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_37
d_{\rm GHZ}=2.64099e-08
region: polarization_cluster_02_local_minimum_37
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0037
lepton mixing angle: alpha_e=0.81766347 rad
incoming lepton: 0.683928|+> +0.72955|->
proton mixing angle: alpha_p=2.3239276 rad
incoming proton: -0.683927|+> +0.729551|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.0096556
$\theta_{p'}$=0.00245437, θ_γ=3.14159, $\phi_{p'}$=5.7193, ϕ_γ=0.92409
E_γ=2.0358, $\theta_{\ell'}$=1.16963e-05, $\phi_{\ell'}$=2.5777
Q^2=19.5484, x_B=0.835149, t=-3.04905, W^2=4.73853, y=0.970436
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.0261 p_x= -0.0000 p_y= 0.0000 p_z= 2.0261
P' E= 0.9380 p_x= 0.0000 p_y= -0.0000 p_z= 0.0097
q_γ E= 2.0358 p_x= 0.0000 p_y= 0.0000 p_z= -2.0358
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.294\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.706\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_40
d_{\rm GHZ}=2.64788e-08
region: polarization_cluster_02_local_minimum_40
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0040
lepton mixing angle: alpha_e=0.76726665 rad
incoming lepton: 0.719811|+> +0.69417|->
proton mixing angle: alpha_p=2.3743383 rad
incoming proton: -0.719819|+> +0.694161|->
s=24.9961, $\sqrt{s}$=4.99961, $|\vec{P}|$=2.41181, $|\vec{P}'|$=0.0116328
$\theta_{p'}$=0.00959617, θ_γ=3.14159, $\phi_{p'}$=6.28284, ϕ_γ=6.2538
E_γ=2.03658, $\theta_{\ell'}$=5.51265e-05, $\phi_{\ell'}$=3.14125
Q^2=19.5352, x_B=0.834713, t=-3.03928, W^2=4.74815, y=0.970447
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4118 p_x= 0.0000 p_y= 0.0000 p_z= -2.4118
P E= 2.5878 p_x= 0.0000 p_y= 0.0000 p_z= 2.4118
ℓ' E= 2.0250 p_x= -0.0001 p_y= 0.0000 p_z= 2.0250
P' E= 0.9381 p_x= 0.0001 p_y= -0.0000 p_z= 0.0116
q_γ E= 2.0366 p_x= 0.0000 p_y= -0.0000 p_z= -2.0366
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.009\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.991\pi$
normalized amplitude magnitude $|\widetilde{A}|$

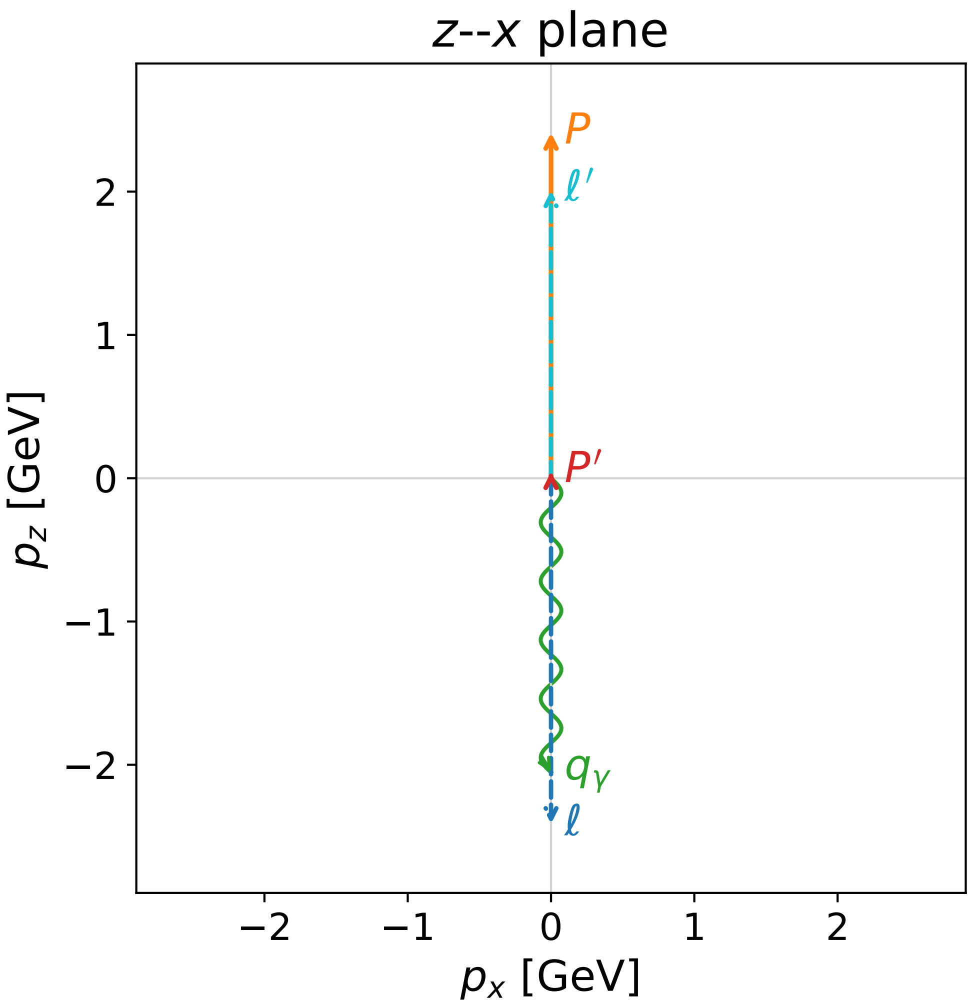

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_41
d_{\rm GHZ}=2.65018e-08
region: polarization_cluster_02_local_minimum_41
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0041
lepton mixing angle: alpha_e=0.70648598 rad
incoming lepton: 0.760648|+> +0.649165|->
proton mixing angle: alpha_p=2.4350875 rad
incoming proton: -0.760635|+> +0.649179|->
s=24.9206, $\sqrt{s}$=4.99205, $|\vec{P}|$=2.4079, $|\vec{P}'|$=0.101522
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=6.26287, ϕ_γ=1.79274
E_γ=1.97352, $\theta_{\ell'}$=3.12924e-07, $\phi_{\ell'}$=3.12128
Q^2=19.986, x_B=0.857428, t=-3.6054, W^2=4.20309, y=0.969575
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4079 p_x= 0.0000 p_y= 0.0000 p_z= -2.4079
P E= 2.5841 p_x= 0.0000 p_y= 0.0000 p_z= 2.4079
ℓ' E= 2.0750 p_x= -0.0000 p_y= 0.0000 p_z= 2.0750
P' E= 0.9435 p_x= 0.0000 p_y= -0.0000 p_z= -0.1015
q_γ E= 1.9735 p_x= -0.0000 p_y= 0.0000 p_z= -1.9735
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.577\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.577\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_42
d_{\rm GHZ}=2.65173e-08
region: polarization_cluster_02_local_minimum_42
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0042
lepton mixing angle: alpha_e=0.7472291 rad
incoming lepton: 0.733575|+> +0.679609|->
proton mixing angle: alpha_p=2.394356 rad
incoming proton: -0.73357|+> +0.679614|->
s=24.9854, $\sqrt{s}$=4.99854, $|\vec{P}|$=2.41126, $|\vec{P}'|$=0.0171201
$\theta_{p'}$=3.14159e-05, θ_γ=3.14159, $\phi_{p'}$=6.27545, ϕ_γ=2.74997
E_γ=2.03875, $\theta_{\ell'}$=2.66143e-07, $\phi_{\ell'}$=3.13386
Q^2=19.4987, x_B=0.833497, t=-3.01229, W^2=4.77498, y=0.970476
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4113 p_x= 0.0000 p_y= 0.0000 p_z= -2.4113
P E= 2.5873 p_x= 0.0000 p_y= 0.0000 p_z= 2.4113
ℓ' E= 2.0216 p_x= -0.0000 p_y= 0.0000 p_z= 2.0216
P' E= 0.9382 p_x= 0.0000 p_y= -0.0000 p_z= 0.0171
q_γ E= 2.0388 p_x= -0.0000 p_y= 0.0000 p_z= -2.0388
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.125\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.875\pi$
normalized amplitude magnitude $|\widetilde{A}|$



coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_49
d_{\rm GHZ}=2.67844e-08
region: polarization_cluster_02_local_minimum_49
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0049
lepton mixing angle: alpha_e=0.81607917 rad
incoming lepton: 0.685083|+> +0.728465|->
proton mixing angle: alpha_p=2.3255025 rad
incoming proton: -0.685075|+> +0.728473|->
s=24.9993, $\sqrt{s}$=4.99993, $|\vec{P}|$=2.41198, $|\vec{P}'|$=0.0563059
$\theta_{p'}$=2.80434e-05, θ_γ=3.14159, $\phi_{p'}$=1.88915, ϕ_γ=5.07566
E_γ=2.05828, $\theta_{\ell'}$=7.88636e-07, $\phi_{\ell'}$=5.03075
Q^2=19.3148, x_B=0.824895, t=-2.83243, W^2=4.9799, y=0.970788
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.0020 p_x= 0.0000 p_y= -0.0000 p_z= 2.0020
P' E= 0.9397 p_x= -0.0000 p_y= 0.0000 p_z= 0.0563
q_γ E= 2.0583 p_x= 0.0000 p_y= -0.0000 p_z= -2.0583
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.616\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.384\pi$
normalized amplitude magnitude $|\widetilde{A}|$

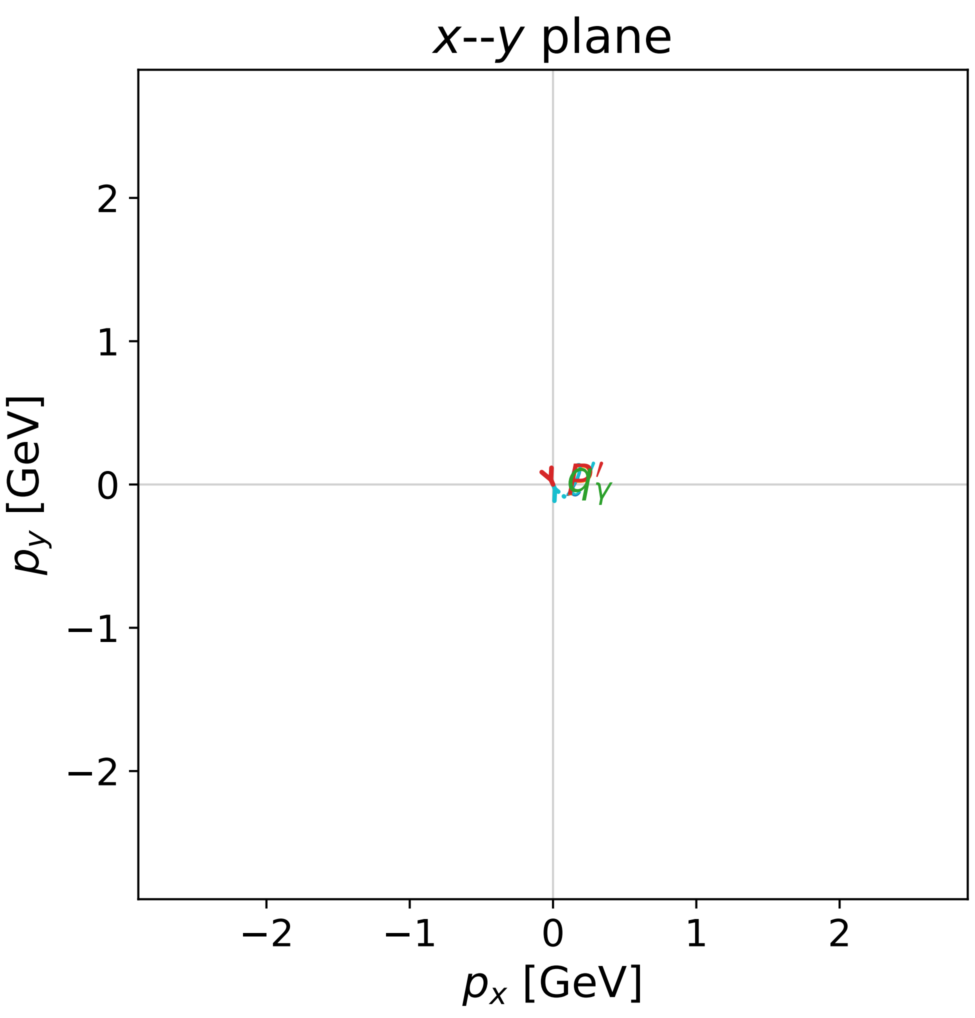
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

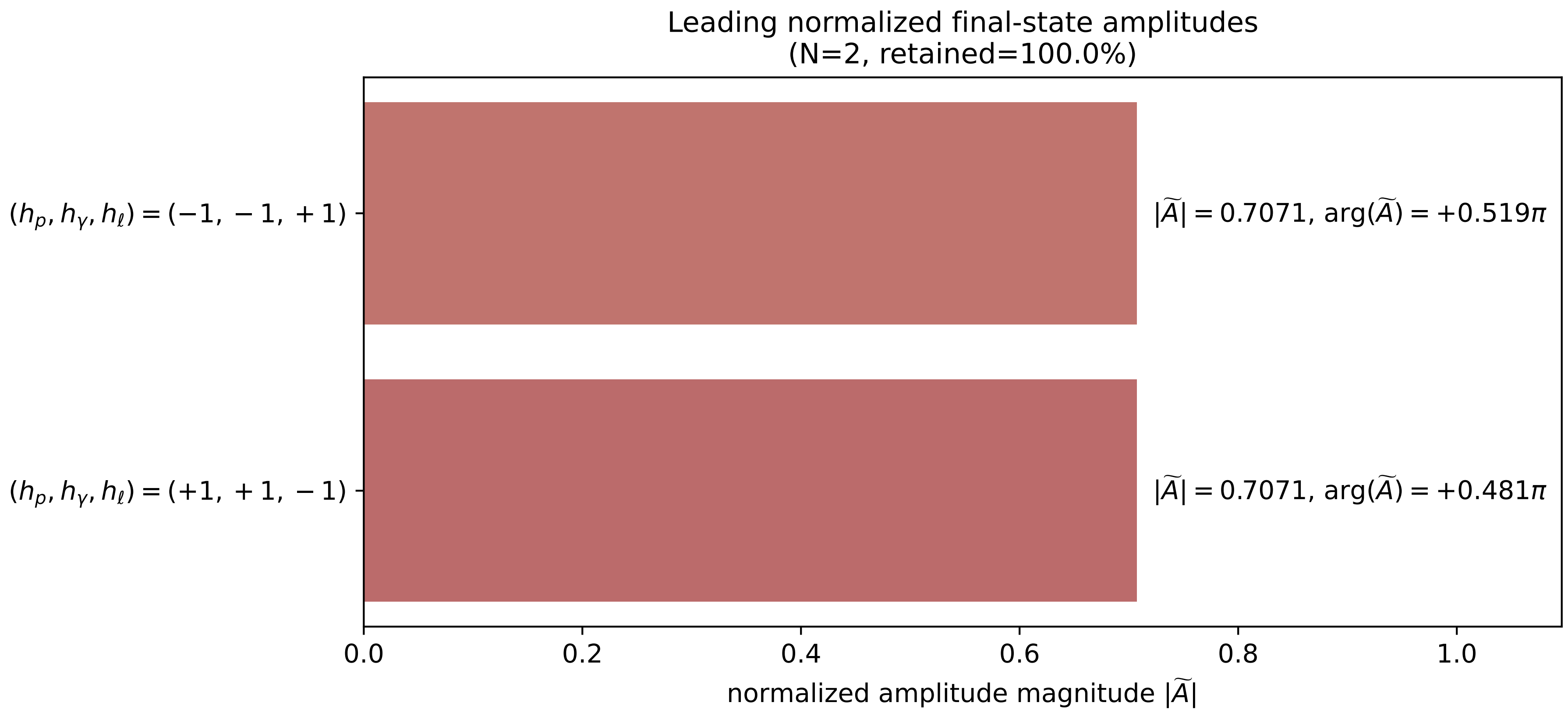


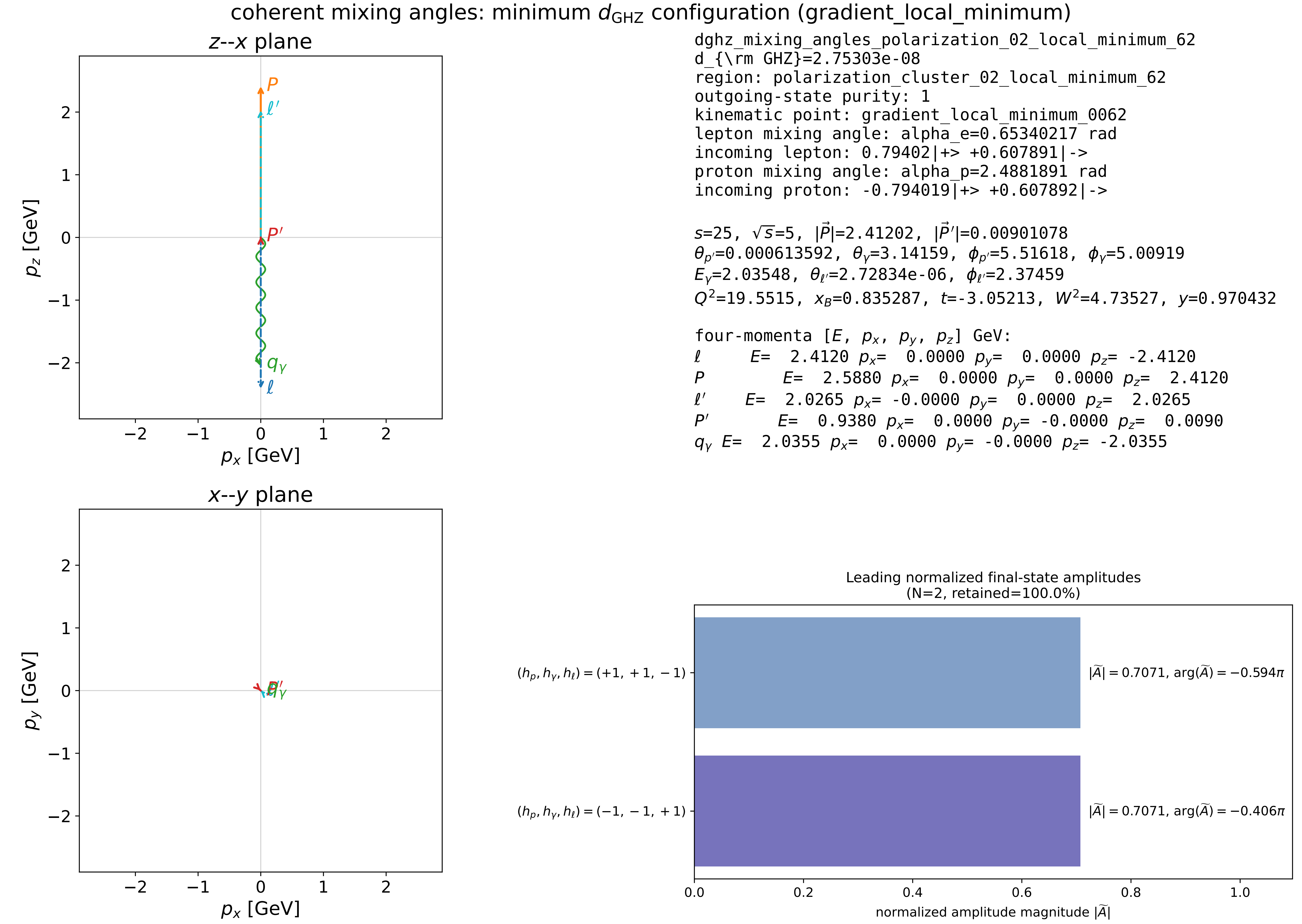
dghz_mixing_angles_polarization_02_local_minimum_54
d_{\rm GHZ}=2.70595e-08
region: polarization_cluster_02_local_minimum_54
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0054
lepton mixing angle: alpha_e=0.85741662 rad
incoming lepton: 0.654393|+> +0.756155|->
proton mixing angle: alpha_p=2.2841843 rad
incoming proton: -0.654399|+> +0.756149|->

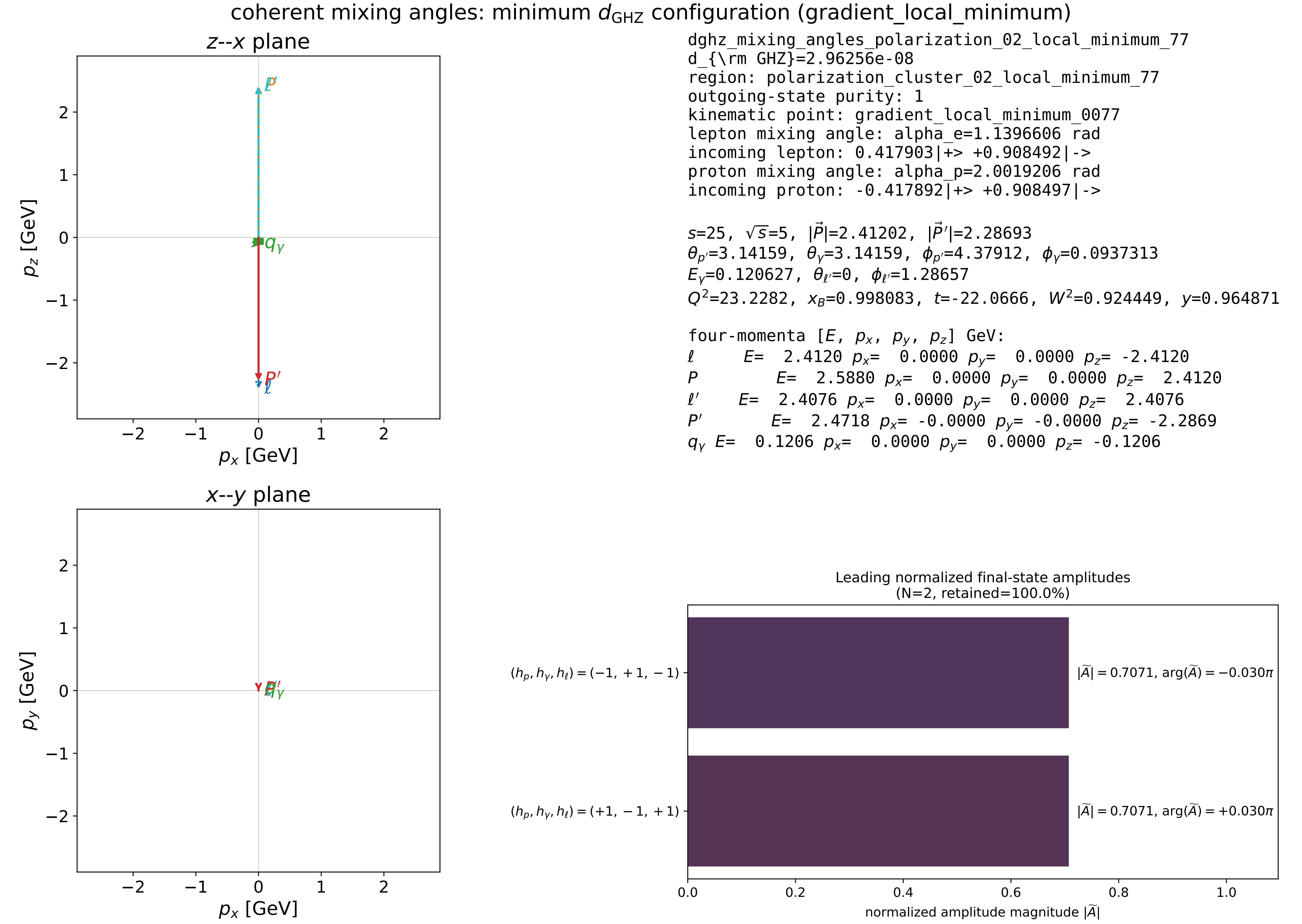
s=24.9965, $\sqrt{s}$=4.99965, $|\vec{P}|$=2.41184, $|\vec{P}'|$=0.0441066
$\theta_{p'}$=0.00122325, θ_γ=3.14159, $\phi_{p'}$=5.12862, ϕ_γ=1.63085
E_γ=2.05236, $\theta_{\ell'}$=2.68658e-05, $\phi_{\ell'}$=1.98702
Q^2=19.3743, x_B=0.827614, t=-2.88766, W^2=4.91537, y=0.970691

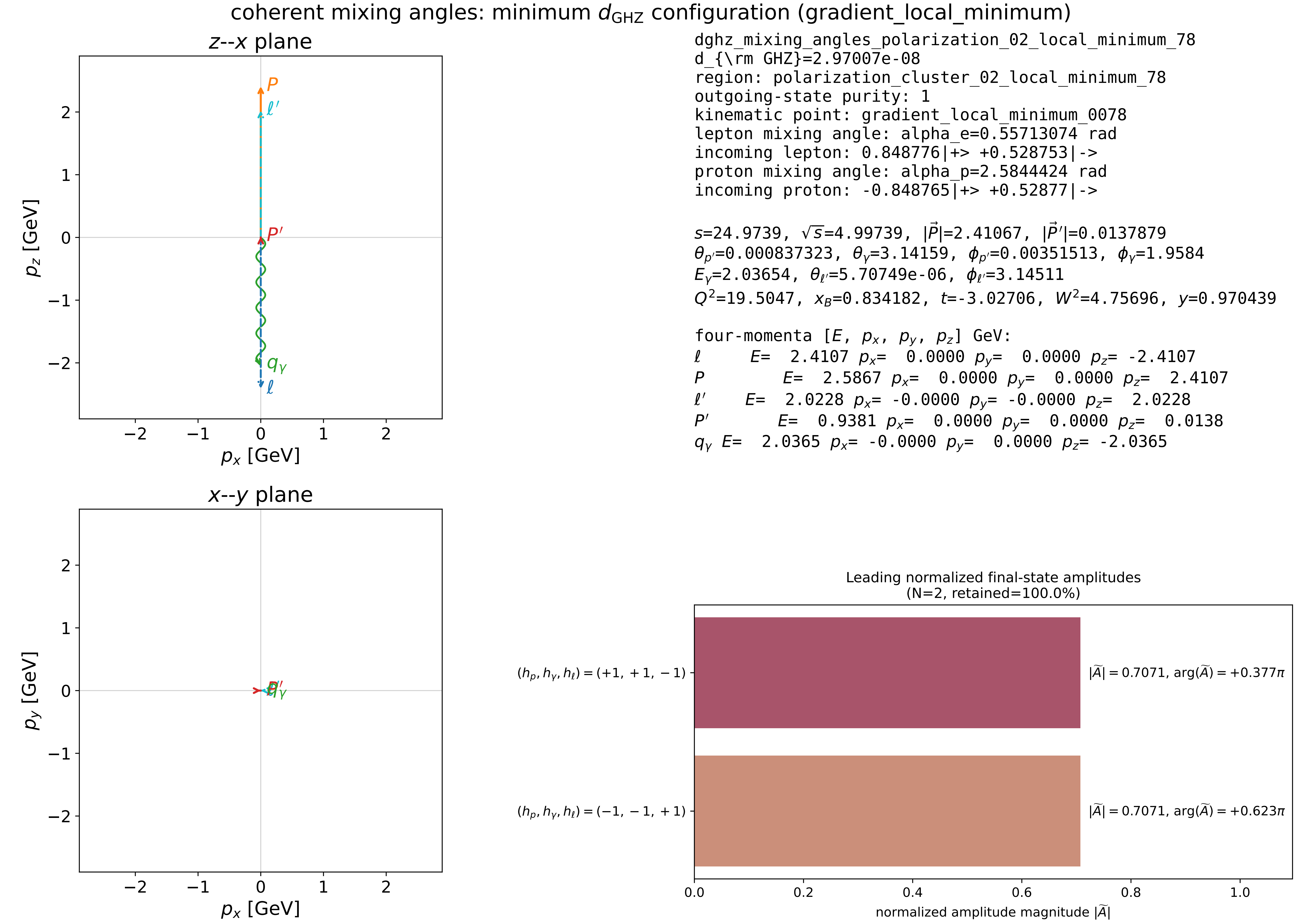
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4118 p_x= 0.0000 p_y= 0.0000 p_z= -2.4118
P E= 2.5878 p_x= 0.0000 p_y= 0.0000 p_z= 2.4118
ℓ' E= 2.0083 p_x= -0.0000 p_y= 0.0000 p_z= 2.0083
P' E= 0.9390 p_x= 0.0000 p_y= -0.0000 p_z= 0.0441
q_γ E= 2.0524 p_x= -0.0000 p_y= 0.0000 p_z= -2.0524



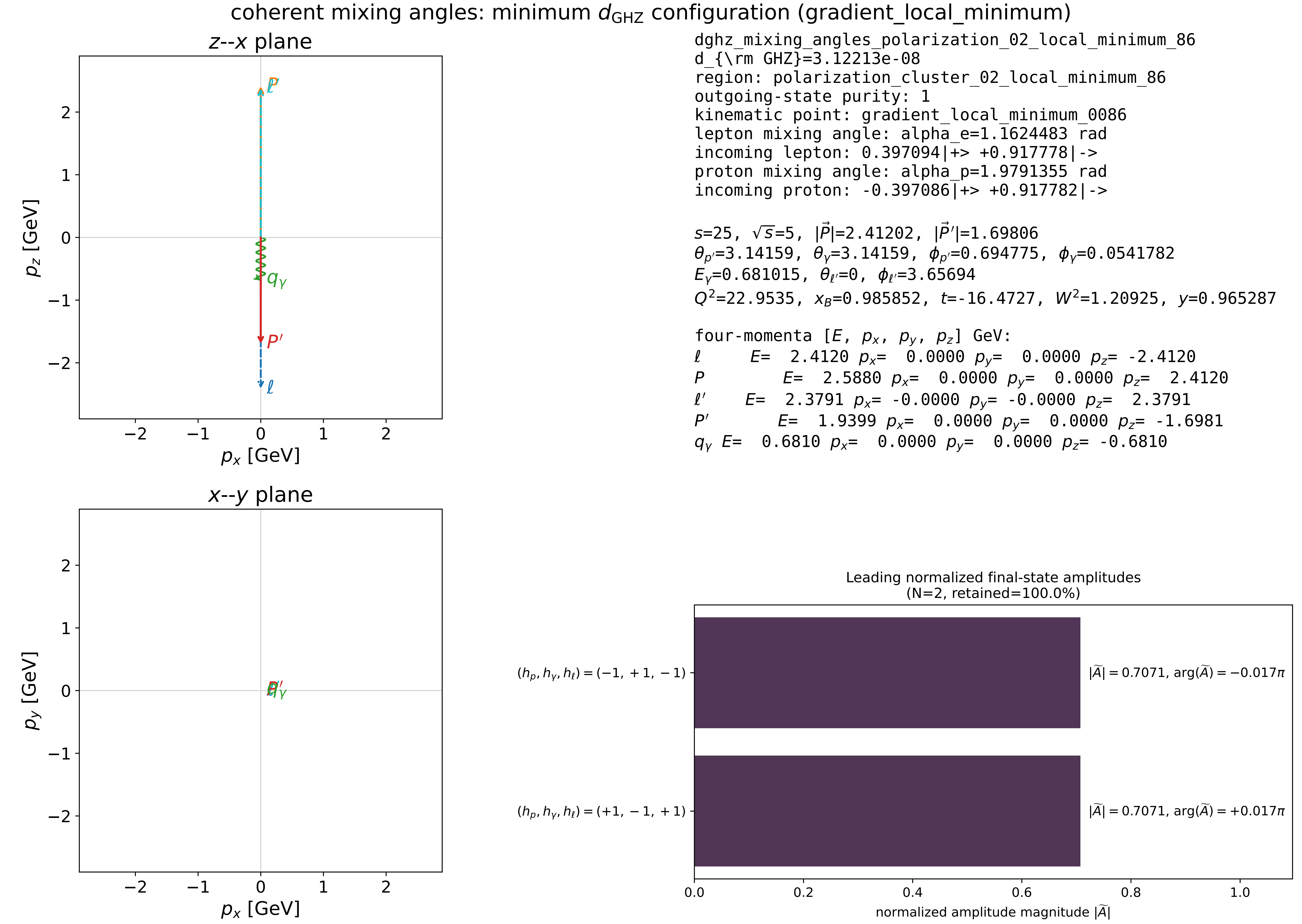


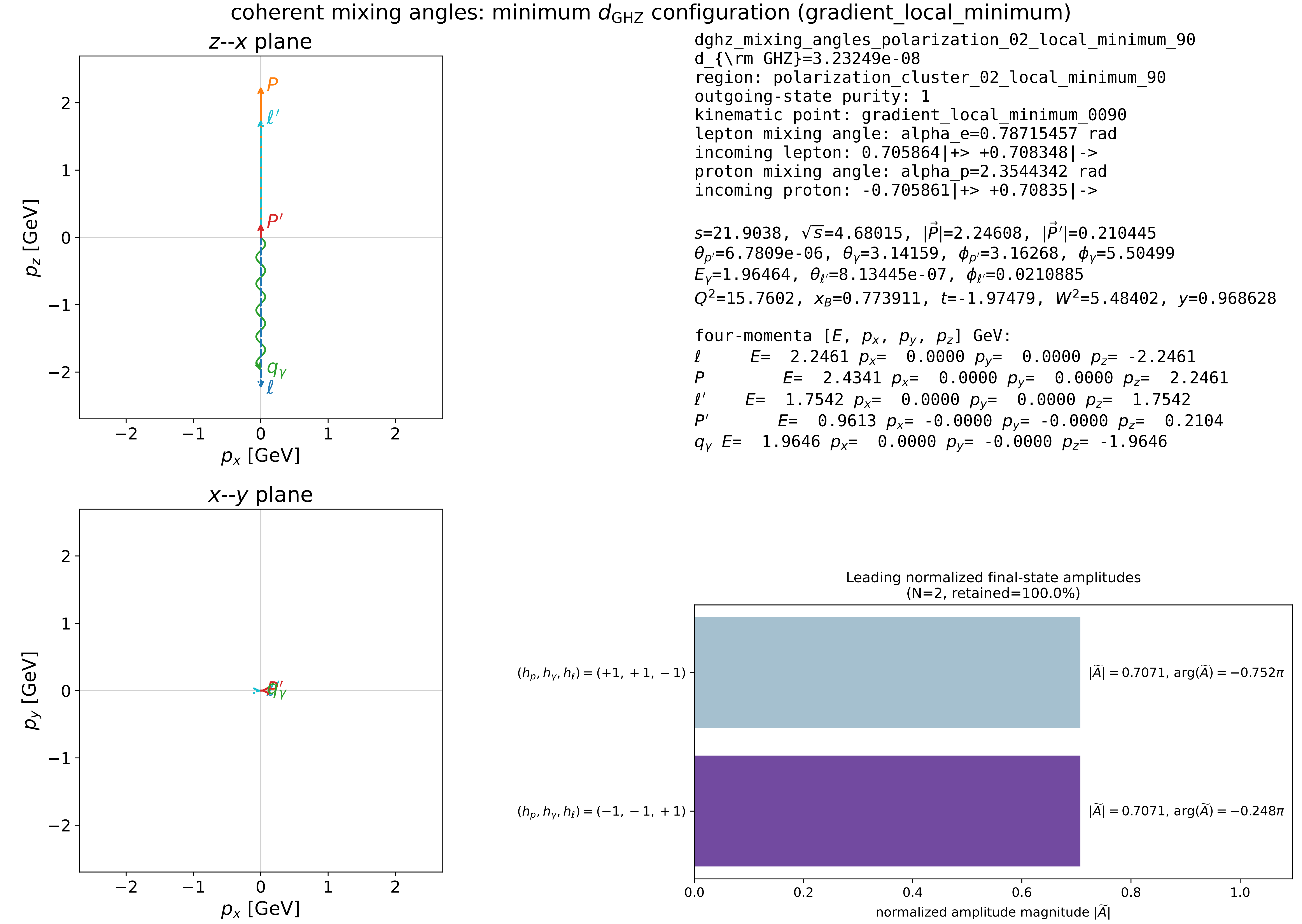

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_62
d_{\rm GHZ}=2.75303e-08
region: polarization_cluster_02_local_minimum_62
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0062
lepton mixing angle: alpha_e=0.65340217 rad
incoming lepton: 0.79402|+> +0.607891|->
proton mixing angle: alpha_p=2.4881891 rad
incoming proton: -0.794019|+> +0.607892|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.00901078
$\theta_{p'}$=0.000613592, θ_γ=3.14159, $\phi_{p'}$=5.51618, ϕ_γ=5.00919
E_γ=2.03548, $\theta_{\ell'}$=2.72834e-06, $\phi_{\ell'}$=2.37459
Q^2=19.5515, x_B=0.835287, t=-3.05213, W^2=4.73527, y=0.970432
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.0265 p_x= -0.0000 p_y= 0.0000 p_z= 2.0265
P' E= 0.9380 p_x= 0.0000 p_y= -0.0000 p_z= 0.0090
q_γ E= 2.0355 p_x= 0.0000 p_y= -0.0000 p_z= -2.0355
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.594\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.406\pi$
normalized amplitude magnitude $|\widetilde{A}|$

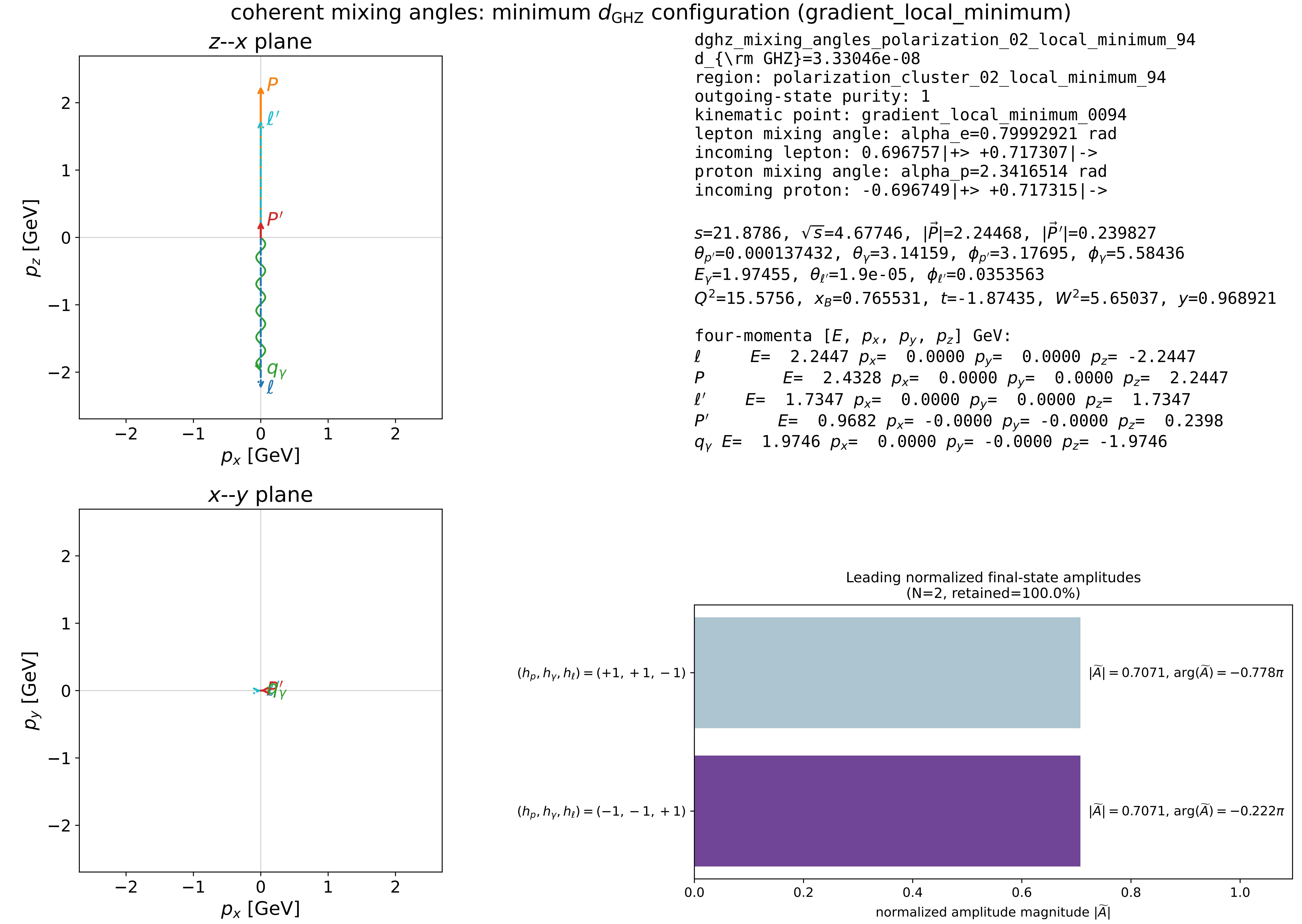

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
ℓ' P q_γ P' ℓ
dghz_mixing_angles_polarization_02_local_minimum_77
d_{\rm GHZ}=2.96256e-08
region: polarization_cluster_02_local_minimum_77
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0077
lepton mixing angle: alpha_e=1.1396606 rad
incoming lepton: 0.417903|+> +0.908492|->
proton mixing angle: alpha_p=2.0019206 rad
incoming proton: -0.417892|+> +0.908497|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=2.28693
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=4.37912, ϕ_γ=0.0937313
E_γ=0.120627, $\theta_{\ell'}$=0, $\phi_{\ell'}$=1.28657
Q^2=23.2282, x_B=0.998083, t=-22.0666, W^2=0.924449, y=0.964871
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.4076 p_x= 0.0000 p_y= 0.0000 p_z= 2.4076
P' E= 2.4718 p_x= -0.0000 p_y= -0.0000 p_z= -2.2869
q_γ E= 0.1206 p_x= 0.0000 p_y= 0.0000 p_z= -0.1206
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.030\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.030\pi$
normalized amplitude magnitude $|\widetilde{A}|$



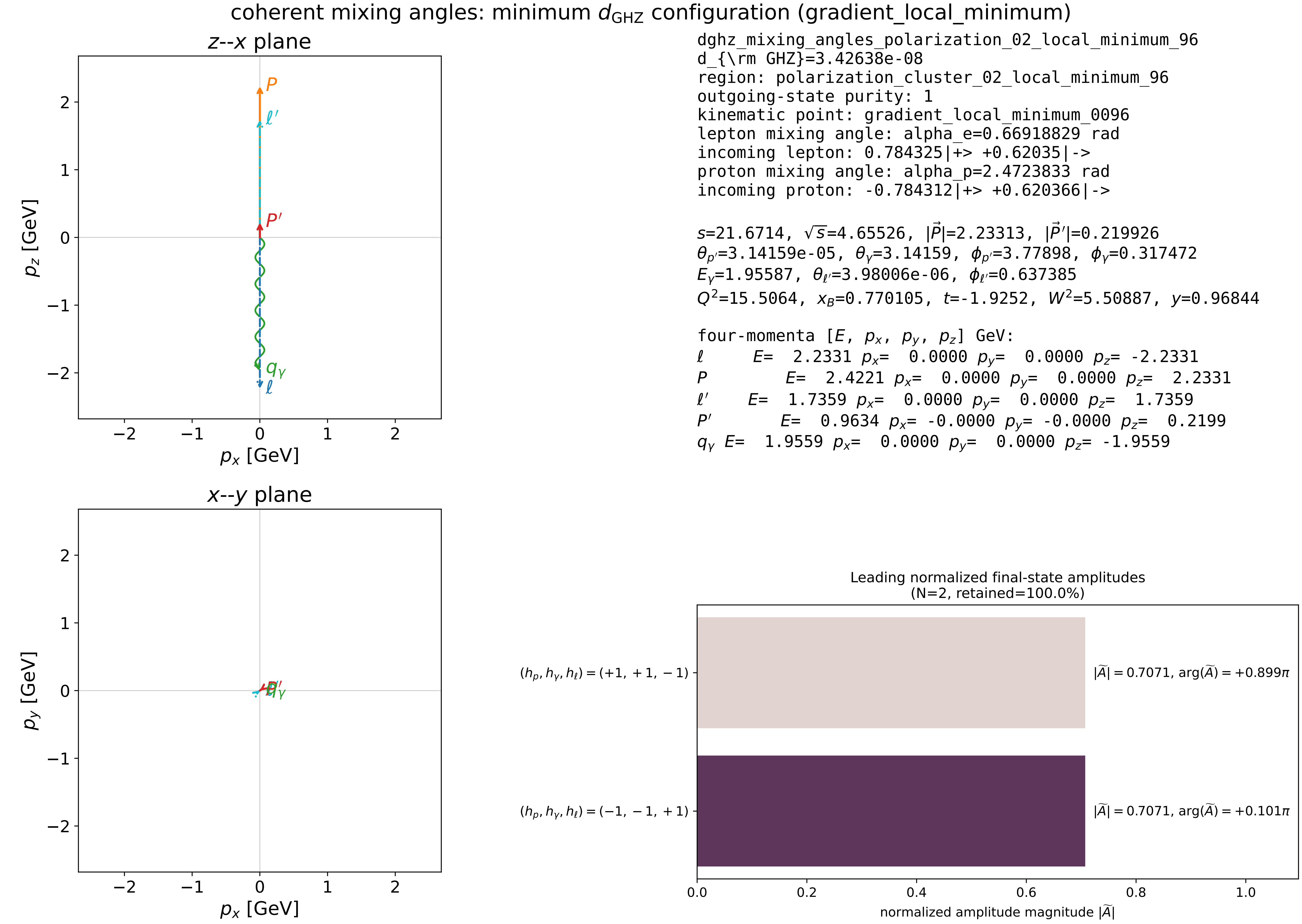
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
dghz_mixing_angles_polarization_02_local_minimum_78
d_{\rm GHZ}=2.97007e-08
region: polarization_cluster_02_local_minimum_78
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0078
lepton mixing angle: alpha_e=0.55713074 rad
incoming lepton: 0.848776|+> +0.528753|->
proton mixing angle: alpha_p=2.5844424 rad
incoming proton: -0.848765|+> +0.52877|->
s=24.9739, $\sqrt{s}$=4.99739, $|\vec{P}|$=2.41067, $|\vec{P}'|$=0.0137879
$\theta_{p'}$=0.000837323, θ_γ=3.14159, $\phi_{p'}$=0.00351513, ϕ_γ=1.9584
E_γ=2.03654, $\theta_{\ell'}$=5.70749e-06, $\phi_{\ell'}$=3.14511
Q^2=19.5047, x_B=0.834182, t=-3.02706, W^2=4.75696, y=0.970439
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4107 p_x= 0.0000 p_y= 0.0000 p_z= -2.4107
P E= 2.5867 p_x= 0.0000 p_y= 0.0000 p_z= 2.4107
ℓ' E= 2.0228 p_x= -0.0000 p_y= -0.0000 p_z= 2.0228
P' E= 0.9381 p_x= 0.0000 p_y= 0.0000 p_z= 0.0138
q_γ E= 2.0365 p_x= -0.0000 p_y= 0.0000 p_z= -2.0365
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.377\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.623\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
q_γ
P'
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_86
d_{\rm GHZ}=3.12213e-08
region: polarization_cluster_02_local_minimum_86
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0086
lepton mixing angle: alpha_e=1.1624483 rad
incoming lepton: 0.397094|+> +0.917778|->
proton mixing angle: alpha_p=1.9791355 rad
incoming proton: -0.397086|+> +0.917782|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=1.69806
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=0.694775, ϕ_γ=0.0541782
E_γ=0.681015, $\theta_{\ell'}$=0, $\phi_{\ell'}$=3.65694
Q^2=22.9535, x_B=0.985852, t=-16.4727, W^2=1.20925, y=0.965287
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.3791 p_x= -0.0000 p_y= -0.0000 p_z= 2.3791
P' E= 1.9399 p_x= 0.0000 p_y= 0.0000 p_z= -1.6981
q_γ E= 0.6810 p_x= 0.0000 p_y= 0.0000 p_z= -0.6810
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.017\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.017\pi$
normalized amplitude magnitude $|\widetilde{A}|$

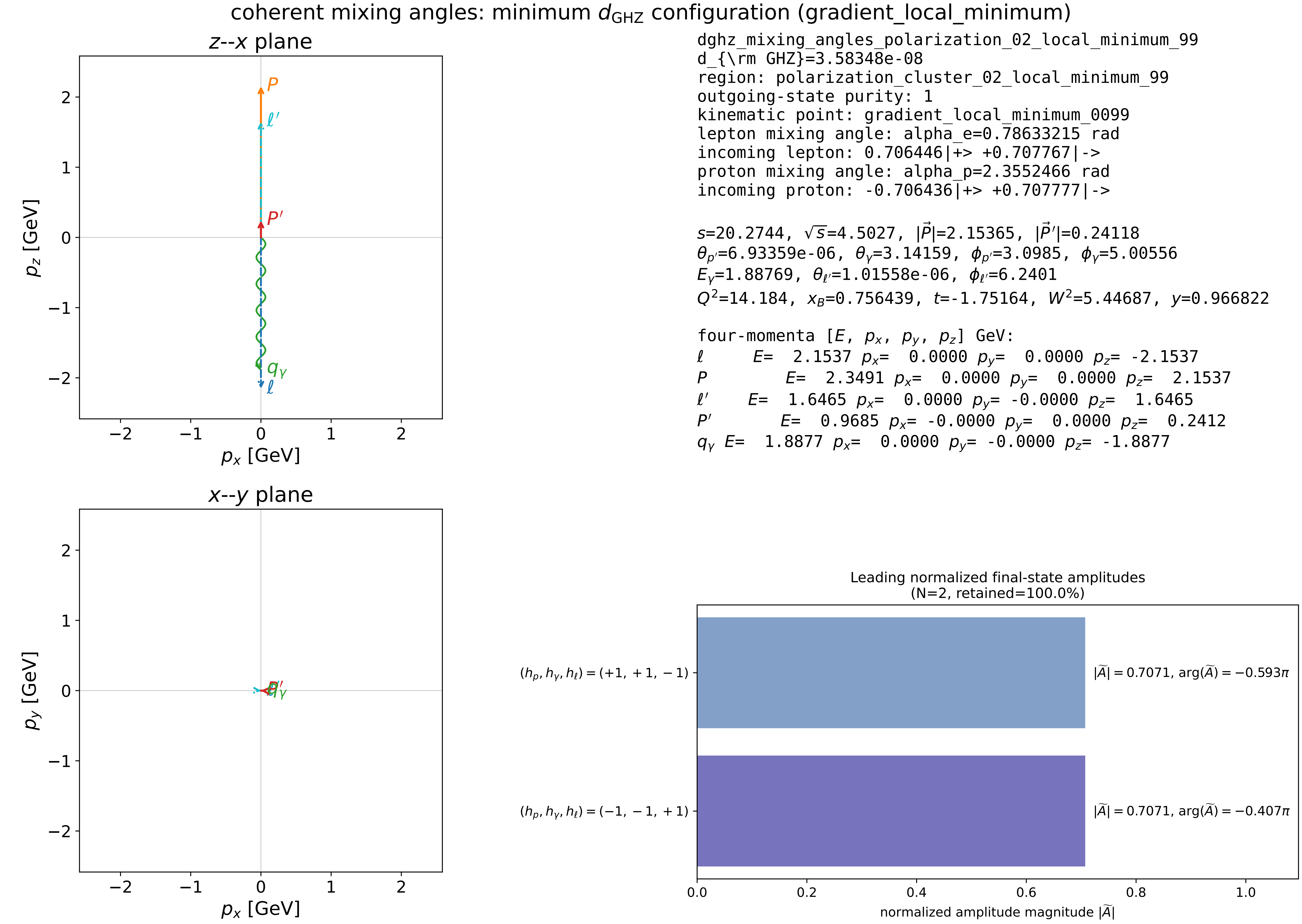

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_90
d_{\rm GHZ}=3.23249e-08
region: polarization_cluster_02_local_minimum_90
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0090
lepton mixing angle: alpha_e=0.78715457 rad
incoming lepton: 0.705864|+> +0.708348|->
proton mixing angle: alpha_p=2.3544342 rad
incoming proton: -0.705861|+> +0.70835|->
s=21.9038, $\sqrt{s}$=4.68015, $|\vec{P}|$=2.24608, $|\vec{P}'|$=0.210445
$\theta_{p'}$=6.7809e-06, θ_γ=3.14159, $\phi_{p'}$=3.16268, ϕ_γ=5.50499
E_γ=1.96464, $\theta_{\ell'}$=8.13445e-07, $\phi_{\ell'}$=0.0210885
Q^2=15.7602, x_B=0.773911, t=-1.97479, W^2=5.48402, y=0.968628
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2461 p_x= 0.0000 p_y= 0.0000 p_z= -2.2461
P E= 2.4341 p_x= 0.0000 p_y= 0.0000 p_z= 2.2461
ℓ' E= 1.7542 p_x= 0.0000 p_y= 0.0000 p_z= 1.7542
P' E= 0.9613 p_x= -0.0000 p_y= -0.0000 p_z= 0.2104
q_γ E= 1.9646 p_x= 0.0000 p_y= -0.0000 p_z= -1.9646
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.752\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.248\pi$
normalized amplitude magnitude $|\widetilde{A}|$



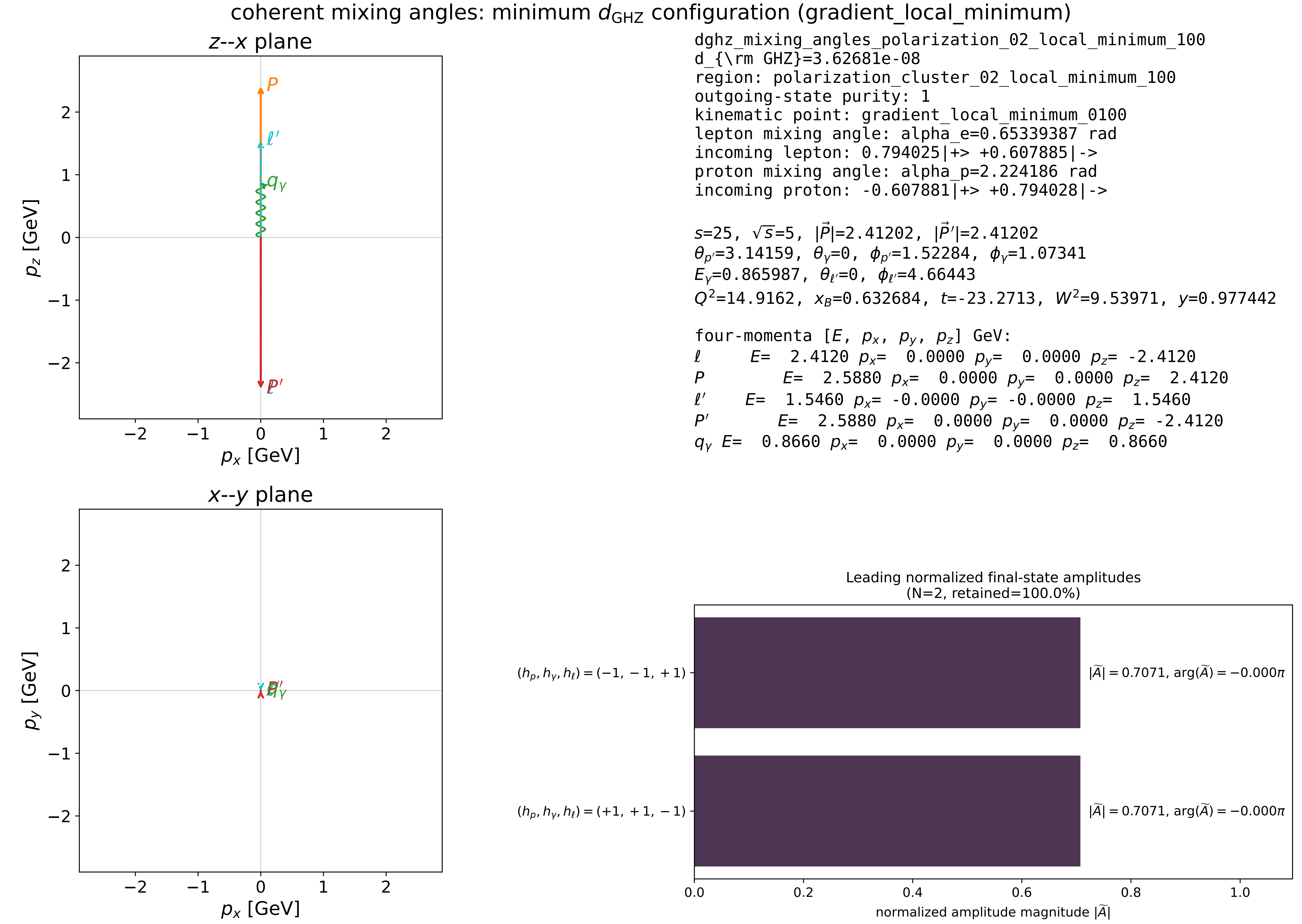
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_94
d_{\rm GHZ}=3.33046e-08
region: polarization_cluster_02_local_minimum_94
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0094
lepton mixing angle: alpha_e=0.79992921 rad
incoming lepton: 0.696757|+> +0.717307|->
proton mixing angle: alpha_p=2.3416514 rad
incoming proton: -0.696749|+> +0.717315|->
s=21.8786, $\sqrt{s}$=4.67746, $|\vec{P}|$=2.24468, $|\vec{P}'|$=0.239827
$\theta_{p'}$=0.000137432, θ_γ=3.14159, $\phi_{p'}$=3.17695, ϕ_γ=5.58436
E_γ=1.97455, $\theta_{\ell'}$=1.9e-05, $\phi_{\ell'}$=0.0353563
Q^2=15.5756, x_B=0.765531, t=-1.87435, W^2=5.65037, y=0.968921
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2447 p_x= 0.0000 p_y= 0.0000 p_z= -2.2447
P E= 2.4328 p_x= 0.0000 p_y= 0.0000 p_z= 2.2447
ℓ' E= 1.7347 p_x= 0.0000 p_y= 0.0000 p_z= 1.7347
P' E= 0.9682 p_x= -0.0000 p_y= -0.0000 p_z= 0.2398
q_γ E= 1.9746 p_x= 0.0000 p_y= -0.0000 p_z= -1.9746
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.778\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.222\pi$
normalized amplitude magnitude $|\widetilde{A}|$



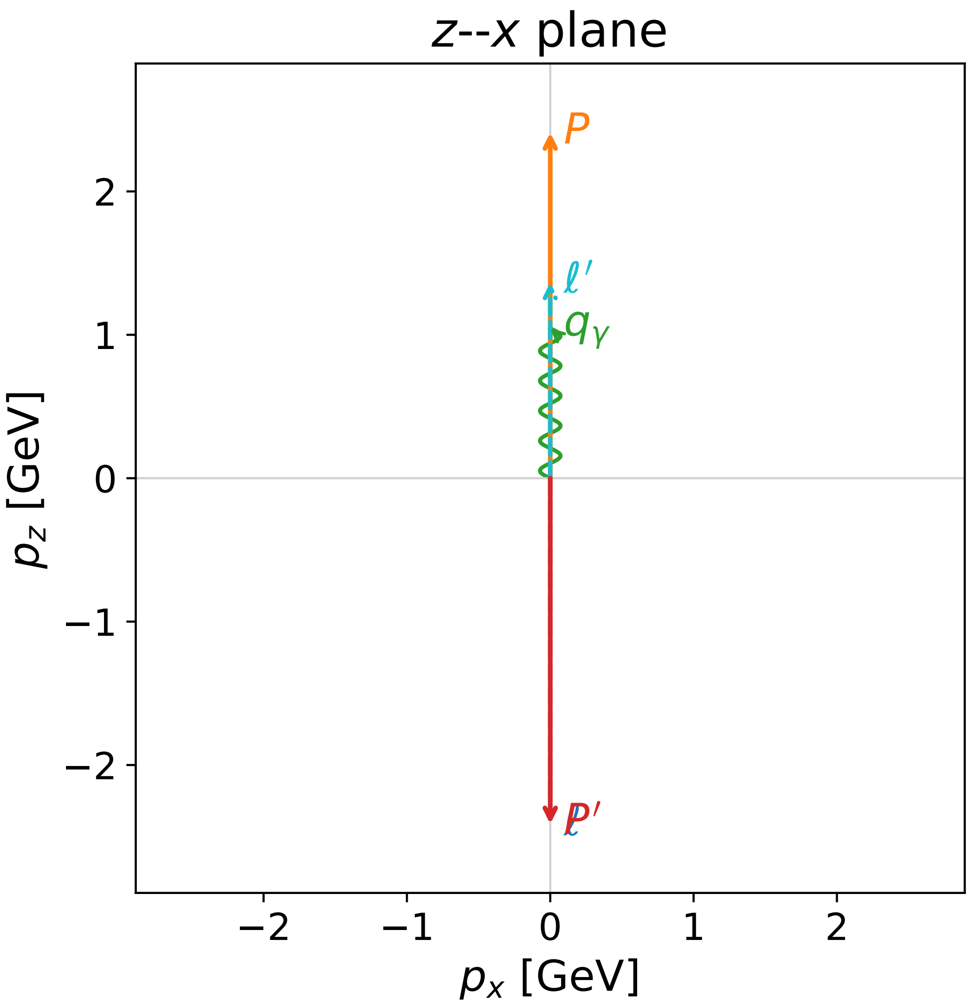
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_96
d_{\rm GHZ}=3.42638e-08
region: polarization_cluster_02_local_minimum_96
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0096
lepton mixing angle: alpha_e=0.66918829 rad
incoming lepton: 0.784325|+> +0.62035|->
proton mixing angle: alpha_p=2.4723833 rad
incoming proton: -0.784312|+> +0.620366|->
s=21.6714, $\sqrt{s}$=4.65526, $|\vec{P}|$=2.23313, $|\vec{P}'|$=0.219926
$\theta_{p'}$=3.14159e-05, θ_γ=3.14159, $\phi_{p'}$=3.77898, ϕ_γ=0.317472
E_γ=1.95587, $\theta_{\ell'}$=3.98006e-06, $\phi_{\ell'}$=0.637385
Q^2=15.5064, x_B=0.770105, t=-1.9252, W^2=5.50887, y=0.96844
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2331 p_x= 0.0000 p_y= 0.0000 p_z= -2.2331
P E= 2.4221 p_x= 0.0000 p_y= 0.0000 p_z= 2.2331
ℓ' E= 1.7359 p_x= 0.0000 p_y= 0.0000 p_z= 1.7359
P' E= 0.9634 p_x= -0.0000 p_y= -0.0000 p_z= 0.2199
q_γ E= 1.9559 p_x= 0.0000 p_y= 0.0000 p_z= -1.9559
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.899\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.101\pi$
normalized amplitude magnitude $|\widetilde{A}|$



coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_99
d_{\rm GHZ}=3.58348e-08
region: polarization_cluster_02_local_minimum_99
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0099
lepton mixing angle: alpha_e=0.78633215 rad
incoming lepton: 0.706446|+> +0.707767|->
proton mixing angle: alpha_p=2.3552466 rad
incoming proton: -0.706436|+> +0.707777|->
s=20.2744, $\sqrt{s}$=4.5027, $|\vec{P}|$=2.15365, $|\vec{P}'|$=0.24118
$\theta_{p'}$=6.93359e-06, θ_γ=3.14159, $\phi_{p'}$=3.0985, ϕ_γ=5.00556
E_γ=1.88769, $\theta_{\ell'}$=1.01558e-06, $\phi_{\ell'}$=6.2401
Q^2=14.184, x_B=0.756439, t=-1.75164, W^2=5.44687, y=0.966822
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.1537 p_x= 0.0000 p_y= 0.0000 p_z= -2.1537
P E= 2.3491 p_x= 0.0000 p_y= 0.0000 p_z= 2.1537
ℓ' E= 1.6465 p_x= 0.0000 p_y= -0.0000 p_z= 1.6465
P' E= 0.9685 p_x= -0.0000 p_y= 0.0000 p_z= 0.2412
q_γ E= 1.8877 p_x= 0.0000 p_y= -0.0000 p_z= -1.8877
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.593\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.407\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
q_γ
P'
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_100
d_{\rm GHZ}=3.62681e-08
region: polarization_cluster_02_local_minimum_100
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0100
lepton mixing angle: alpha_e=0.65339387 rad
incoming lepton: 0.794025|+> +0.607885|->
proton mixing angle: alpha_p=2.224186 rad
incoming proton: -0.607881|+> +0.794028|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=2.41202
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=1.52284, ϕ_γ=1.07341
E_γ=0.865987, $\theta_{\ell'}$=0, $\phi_{\ell'}$=4.66443
Q^2=14.9162, x_B=0.632684, t=-23.2713, W^2=9.53971, y=0.977442
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 1.5460 p_x= -0.0000 p_y= -0.0000 p_z= 1.5460
P' E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
q_γ E= 0.8660 p_x= 0.0000 p_y= 0.0000 p_z= 0.8660
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.000\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

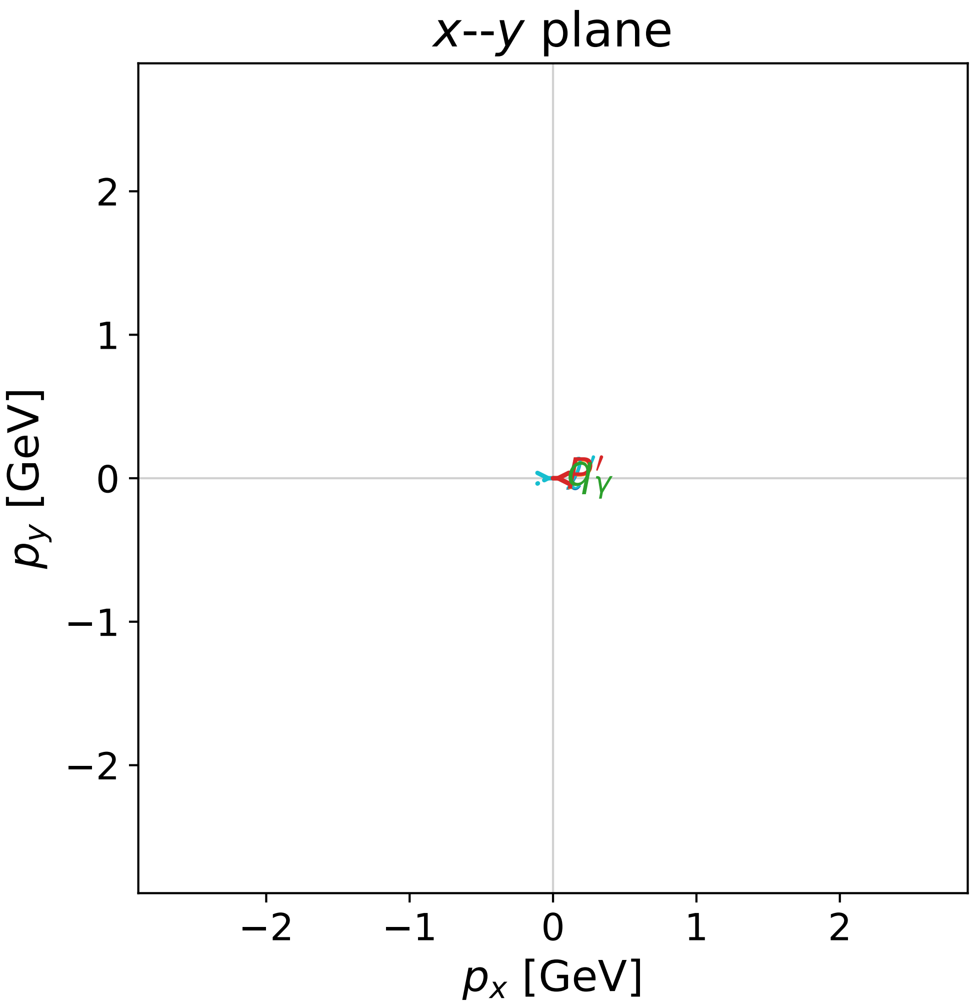
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

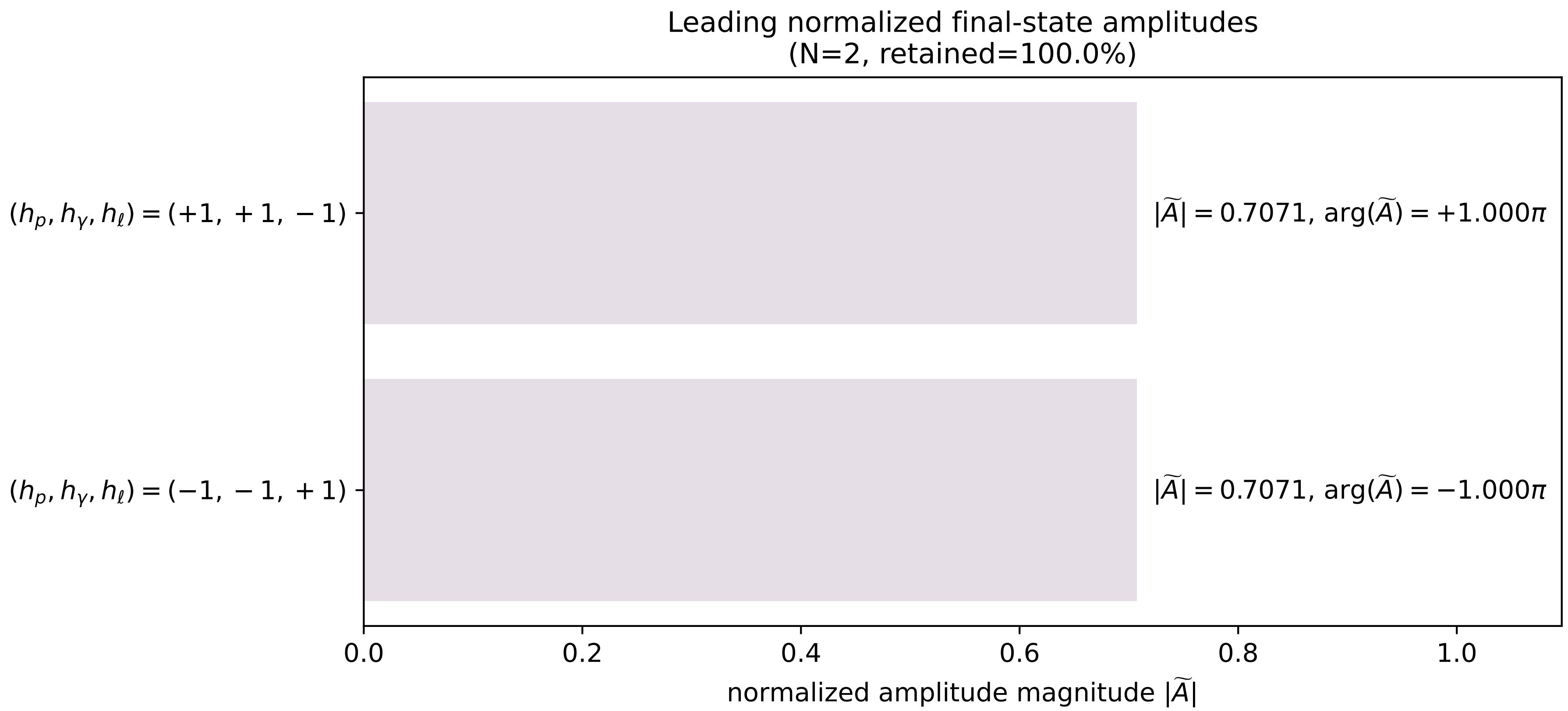


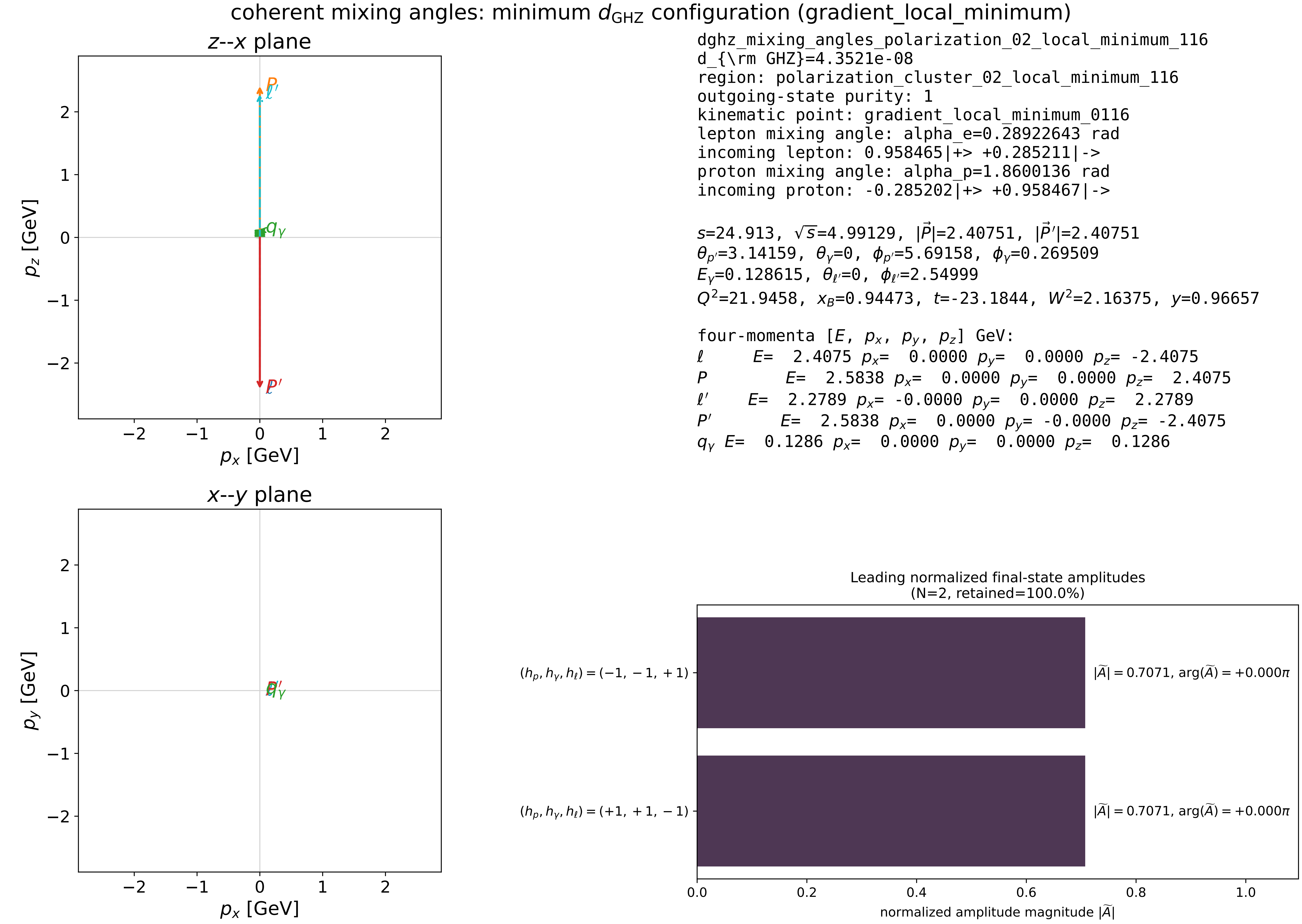
dghz_mixing_angles_polarization_02_local_minimum_110
d_{\rm GHZ}=4.05095e-08
region: polarization_cluster_02_local_minimum_110
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0110
lepton mixing angle: alpha_e=0.68998436 rad
incoming lepton: 0.771256|+> +0.636525|->
proton mixing angle: alpha_p=2.2608057 rad
incoming proton: -0.636544|+> +0.77124|->

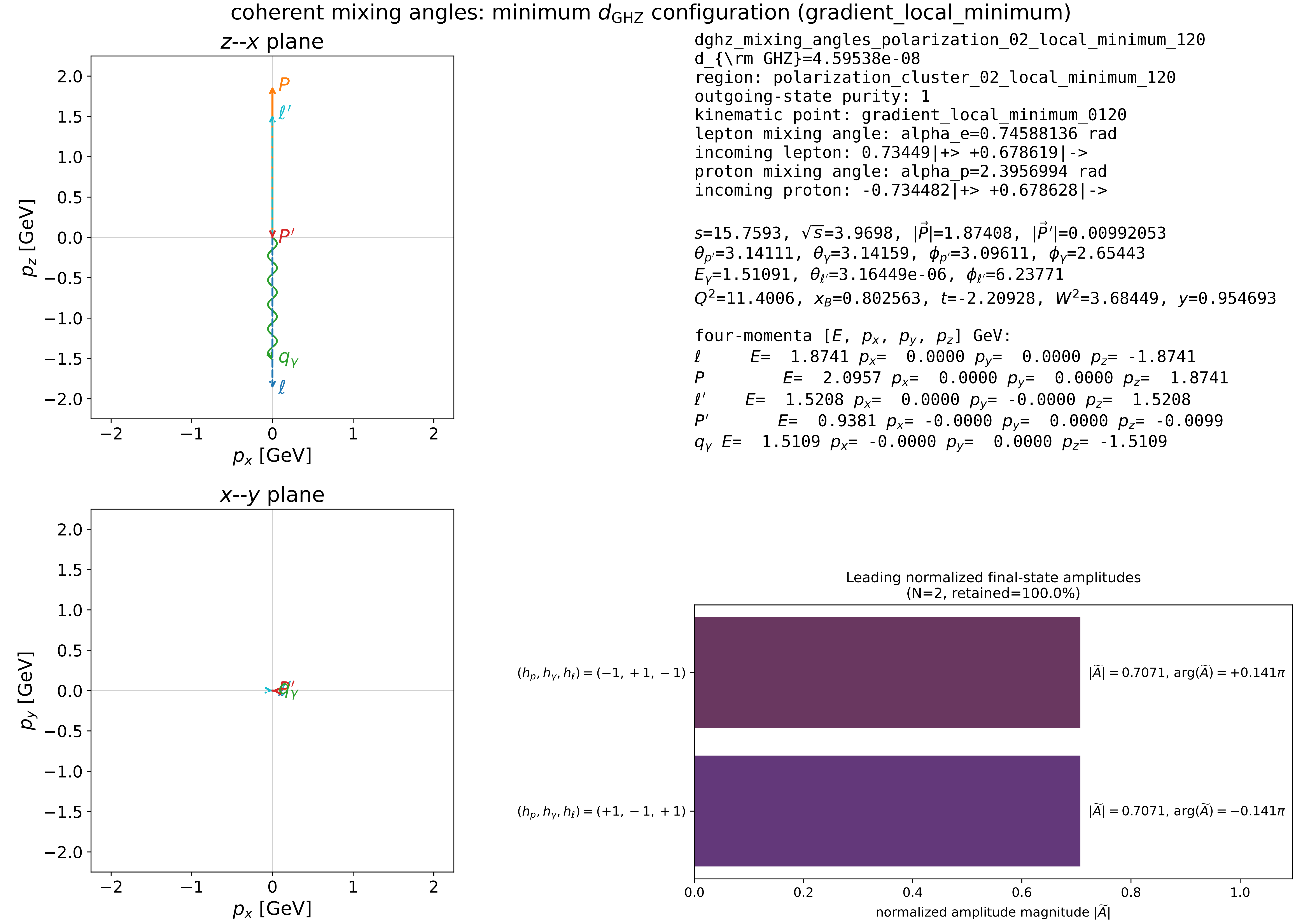
s=24.9589, $\sqrt{s}$=4.99589, $|\vec{P}|$=2.40989, $|\vec{P}'|$=2.40989
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=2.7054, ϕ_γ=1.95022
E_γ=1.04501, $\theta_{\ell'}$=0, $\phi_{\ell'}$=5.84699
Q^2=13.1568, x_B=0.557533, t=-23.2303, W^2=11.3213, y=0.980035

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4099 p_x= 0.0000 p_y= 0.0000 p_z= -2.4099
P E= 2.5860 p_x= 0.0000 p_y= 0.0000 p_z= 2.4099
ℓ' E= 1.3649 p_x= 0.0000 p_y= -0.0000 p_z= 1.3649
P' E= 2.5860 p_x= -0.0000 p_y= 0.0000 p_z= -2.4099
q_γ E= 1.0450 p_x= -0.0000 p_y= 0.0000 p_z= 1.0450

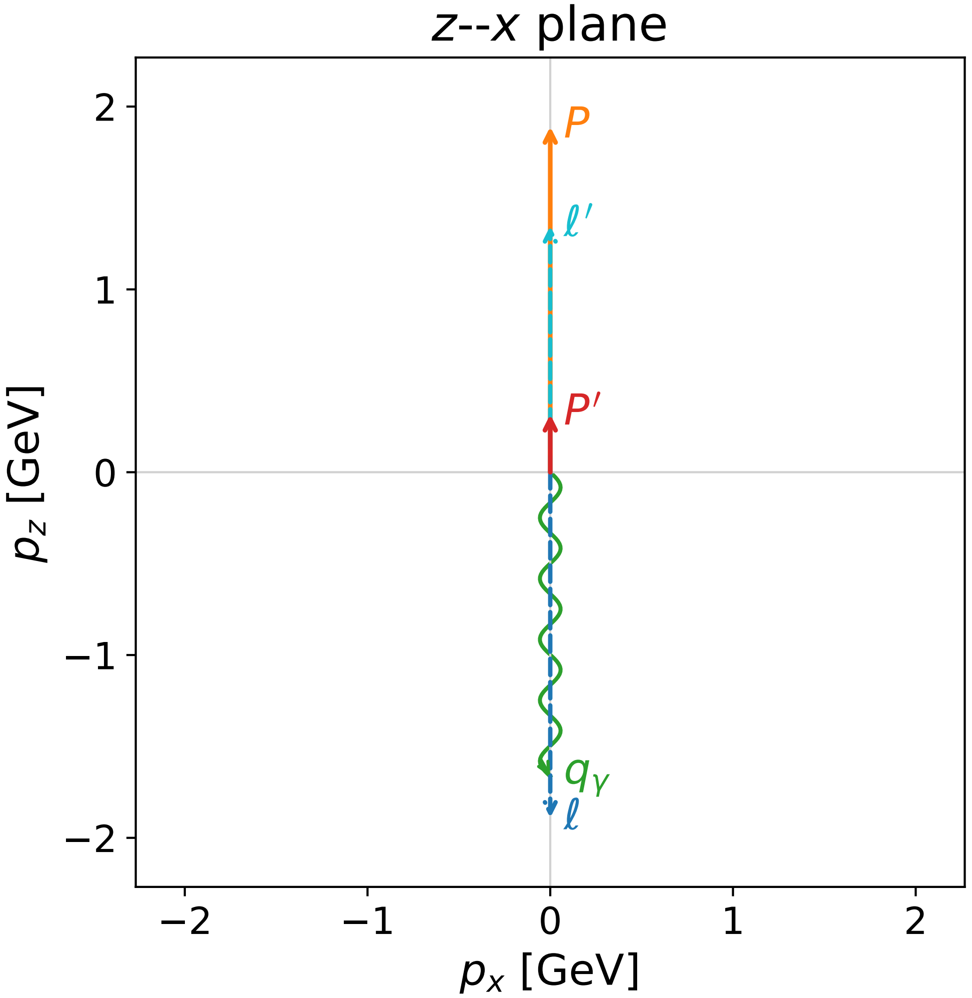





coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
q_γ
P'
ℓ
dghz_mixing_angles_polarization_02_local_minimum_116
d_{\rm GHZ}=4.3521e-08
region: polarization_cluster_02_local_minimum_116
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0116
lepton mixing angle: alpha_e=0.28922643 rad
incoming lepton: 0.958465|+> +0.285211|->
proton mixing angle: alpha_p=1.8600136 rad
incoming proton: -0.285202|+> +0.958467|->
s=24.913, $\sqrt{s}$=4.99129, $|\vec{P}|$=2.40751, $|\vec{P}'|$=2.40751
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=5.69158, ϕ_γ=0.269509
E_γ=0.128615, $\theta_{\ell'}$=0, $\phi_{\ell'}$=2.54999
Q^2=21.9458, x_B=0.94473, t=-23.1844, W^2=2.16375, y=0.96657
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4075 p_x= 0.0000 p_y= 0.0000 p_z= -2.4075
P E= 2.5838 p_x= 0.0000 p_y= 0.0000 p_z= 2.4075
ℓ' E= 2.2789 p_x= -0.0000 p_y= 0.0000 p_z= 2.2789
P' E= 2.5838 p_x= 0.0000 p_y= -0.0000 p_z= -2.4075
q_γ E= 0.1286 p_x= 0.0000 p_y= 0.0000 p_z= 0.1286
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.000\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_120
d_{\rm GHZ}=4.59538e-08
region: polarization_cluster_02_local_minimum_120
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0120
lepton mixing angle: alpha_e=0.74588136 rad
incoming lepton: 0.73449|+> +0.678619|->
proton mixing angle: alpha_p=2.3956994 rad
incoming proton: -0.734482|+> +0.678628|->
s=15.7593, $\sqrt{s}$=3.9698, $|\vec{P}|$=1.87408, $|\vec{P}'|$=0.00992053
$\theta_{p'}$=3.14111, θ_γ=3.14159, $\phi_{p'}$=3.09611, ϕ_γ=2.65443
E_γ=1.51091, $\theta_{\ell'}$=3.16449e-06, $\phi_{\ell'}$=6.23771
Q^2=11.4006, x_B=0.802563, t=-2.20928, W^2=3.68449, y=0.954693
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.8741 p_x= 0.0000 p_y= 0.0000 p_z= -1.8741
P E= 2.0957 p_x= 0.0000 p_y= 0.0000 p_z= 1.8741
ℓ' E= 1.5208 p_x= 0.0000 p_y= -0.0000 p_z= 1.5208
P' E= 0.9381 p_x= -0.0000 p_y= 0.0000 p_z= -0.0099
q_γ E= 1.5109 p_x= -0.0000 p_y= 0.0000 p_z= -1.5109
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.141\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.141\pi$
normalized amplitude magnitude $|\widetilde{A}|$

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)




```
dghz_mixing_angles_polarization_02_local_minimum_127
d_{\rm GHZ}=5.10304e-08
region: polarization_cluster_02_local_minimum_127
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0127
lepton mixing angle: alpha_e=0.78216053 rad
incoming lepton: 0.709392|+> +0.704814|->
proton mixing angle: alpha_p=2.3594272 rad
incoming proton: -0.709389|+> +0.704817|->
```

s=16.0046, $\sqrt{s}$=4.00057, $|\vec{P}|$=1.89032, $|\vec{P}'|$=0.316348

$\theta_{p'}$=9.29374e-05, θ_γ=3.14159, $\phi_{p'}$=6.26205, ϕ_γ=1.81607

E_γ=1.6635, $\theta_{\ell'}$=2.18242e-05, $\phi_{\ell'}$=3.12046

Q^2=10.1862, x_B=0.700944, t=-1.22222, W^2=5.22577, y=0.960822

four-momenta [E, p_x, p_y, p_z] GeV:

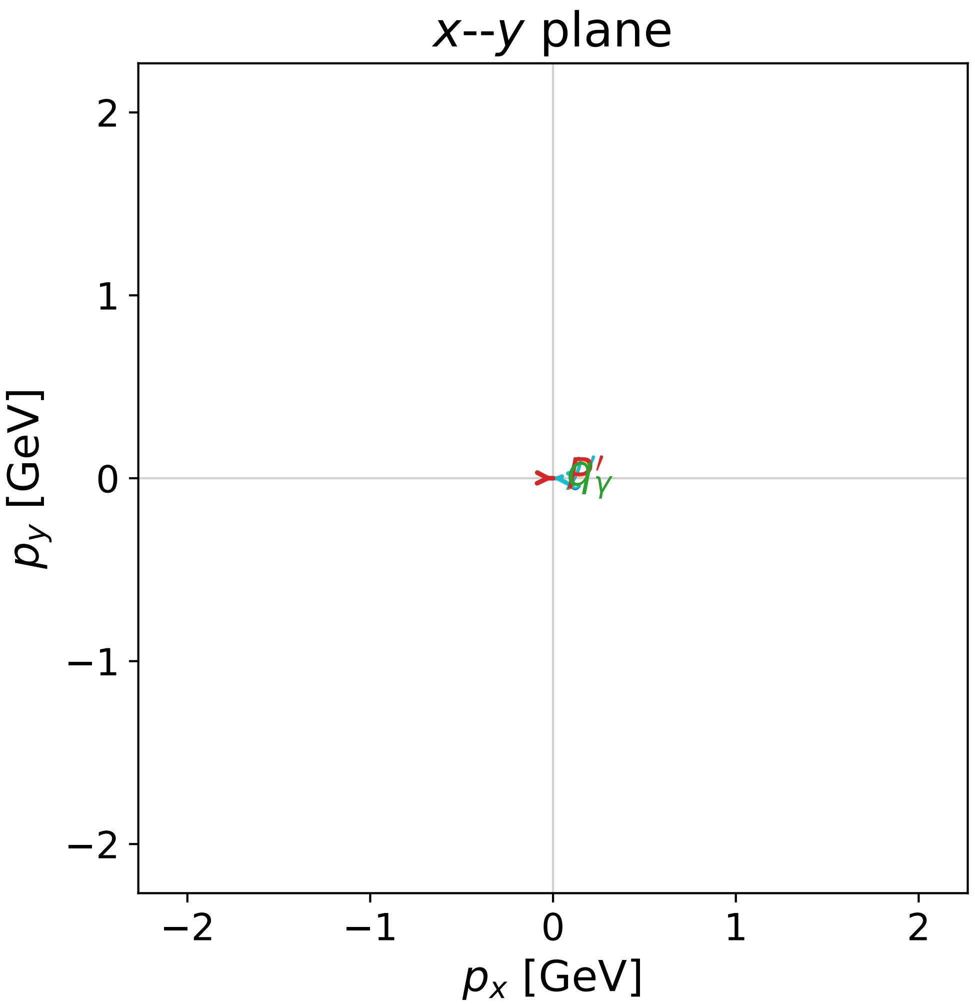
ℓ E= 1.8903 p_x= 0.0000 p_y= 0.0000 p_z= -1.8903

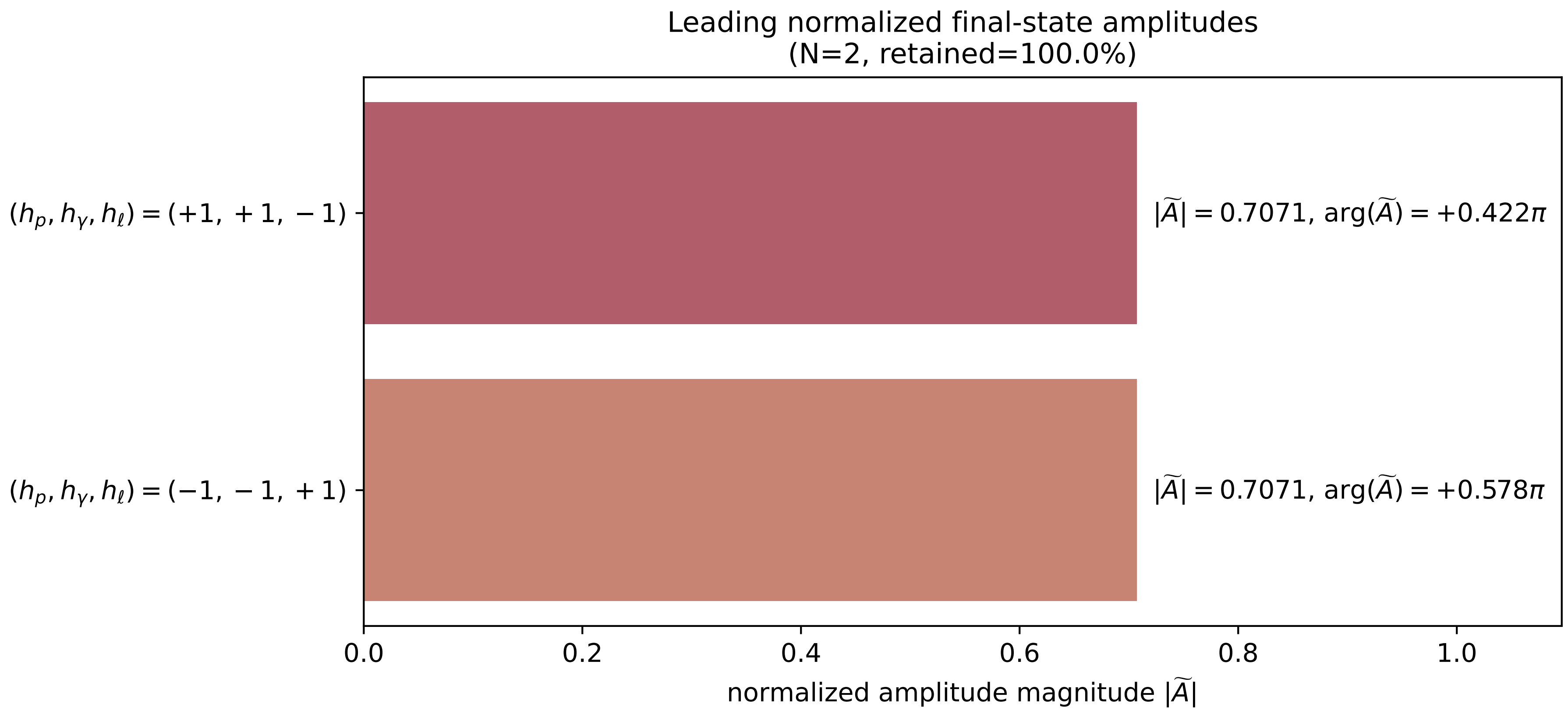
P E= 2.1102 p_x= 0.0000 p_y= 0.0000 p_z= 1.8903

ℓ' E= 1.3472 p_x= -0.0000 p_y= 0.0000 p_z= 1.3472

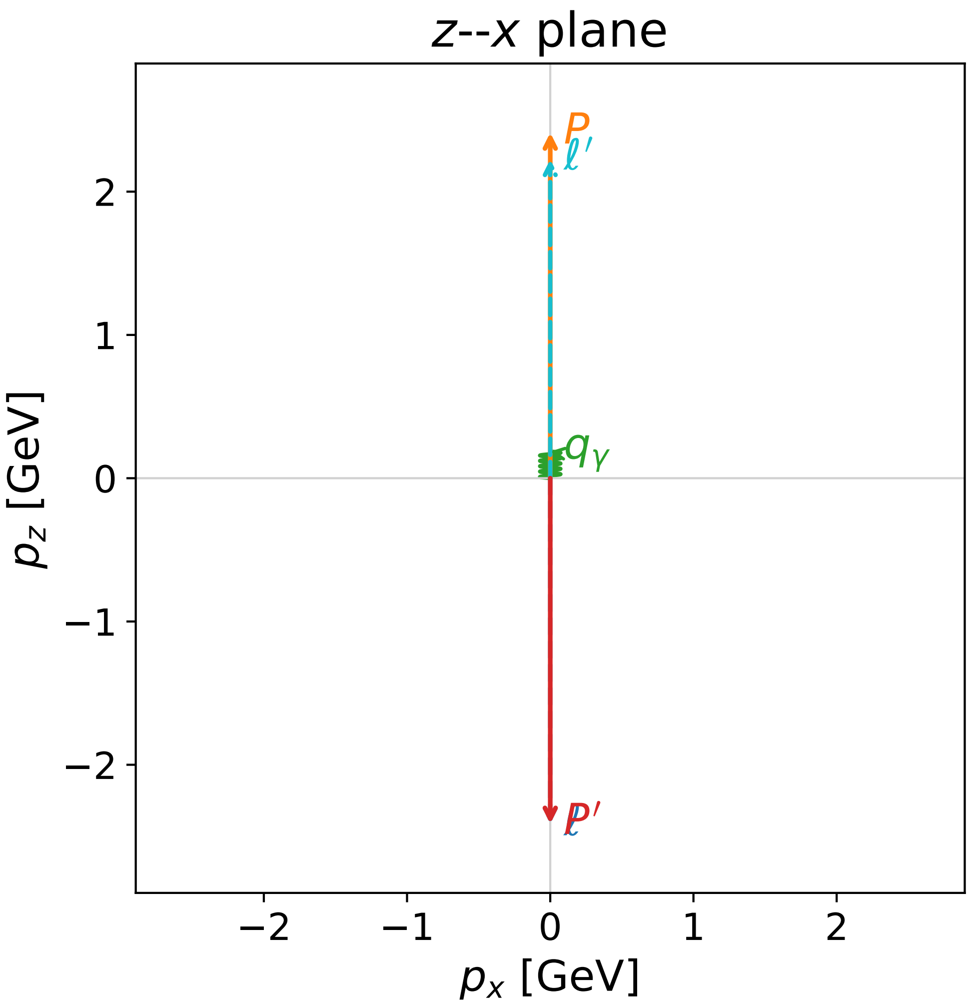
P' E= 0.9899 p_x= 0.0000 p_y= -0.0000 p_z= 0.3163

q_γ E= 1.6635 p_x= -0.0000 p_y= 0.0000 p_z= -1.6635





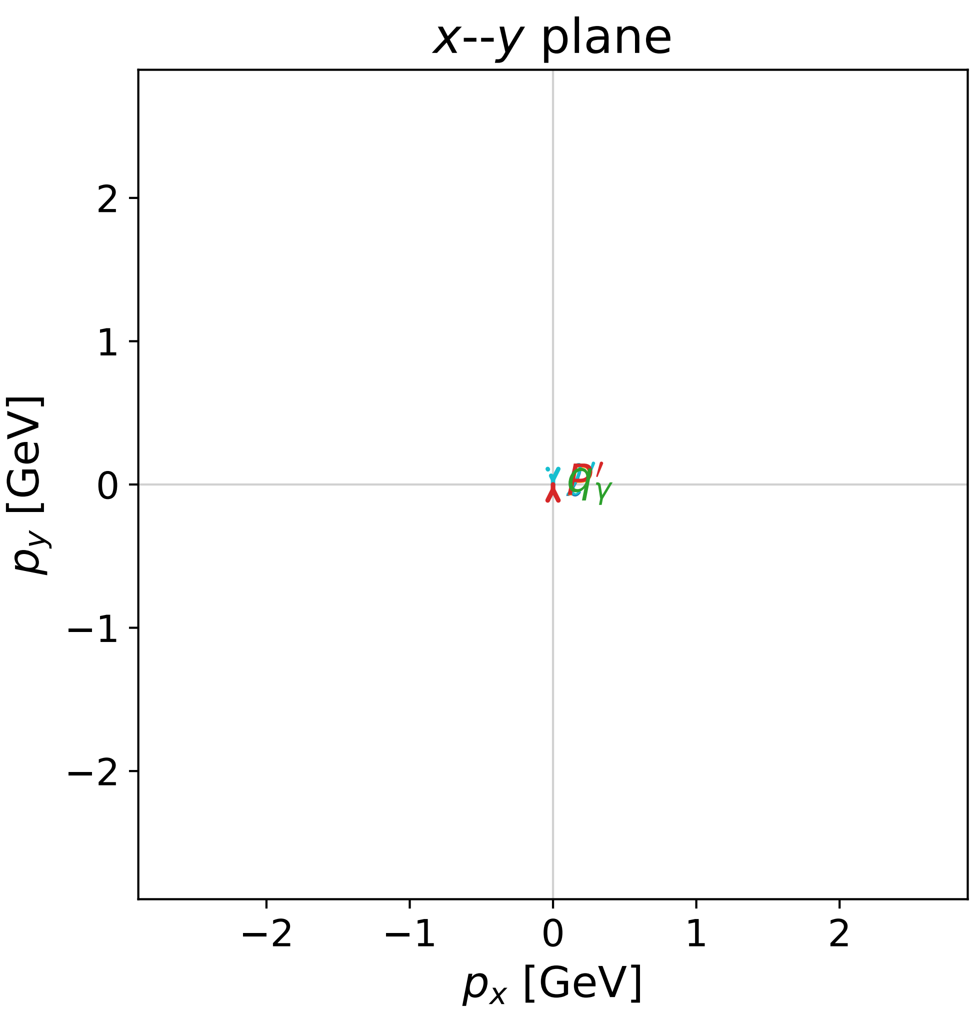
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

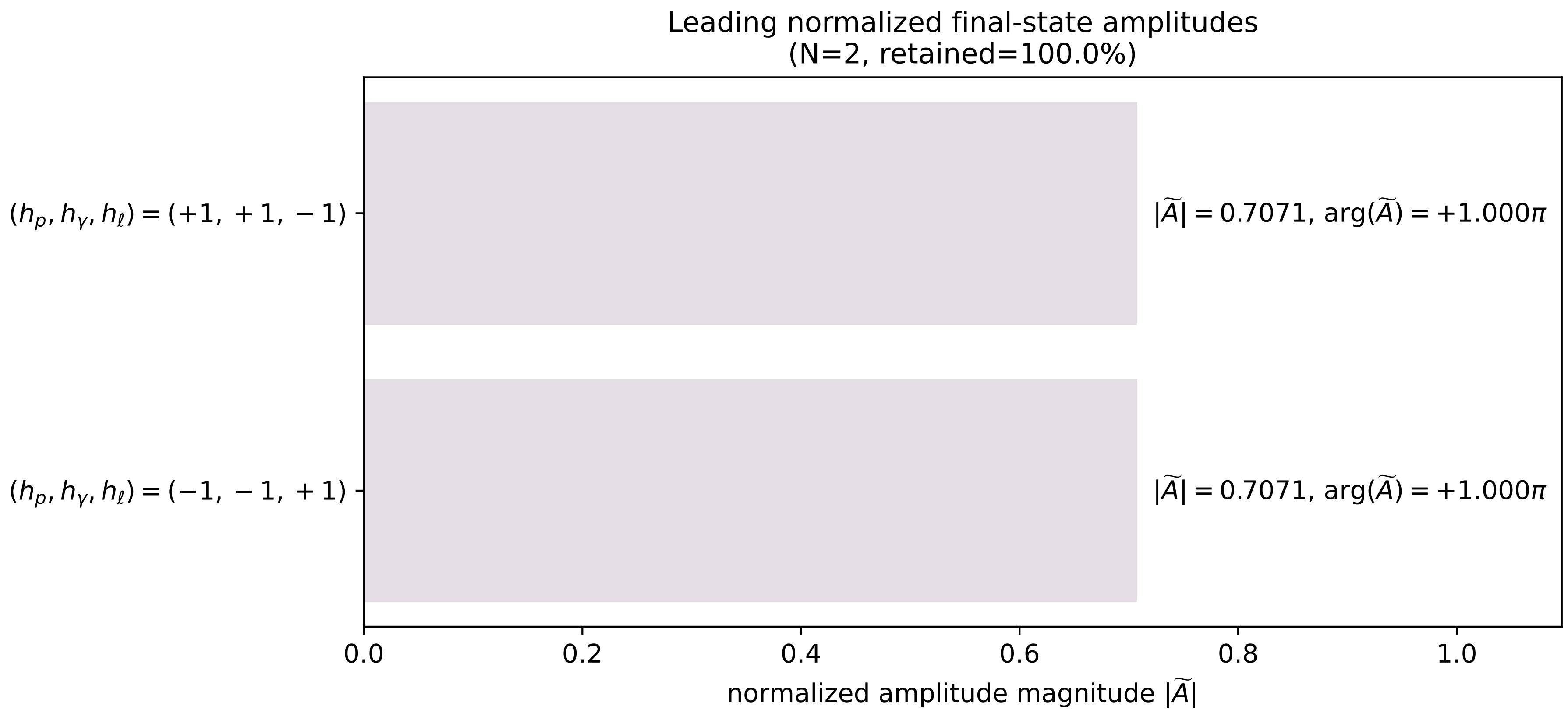


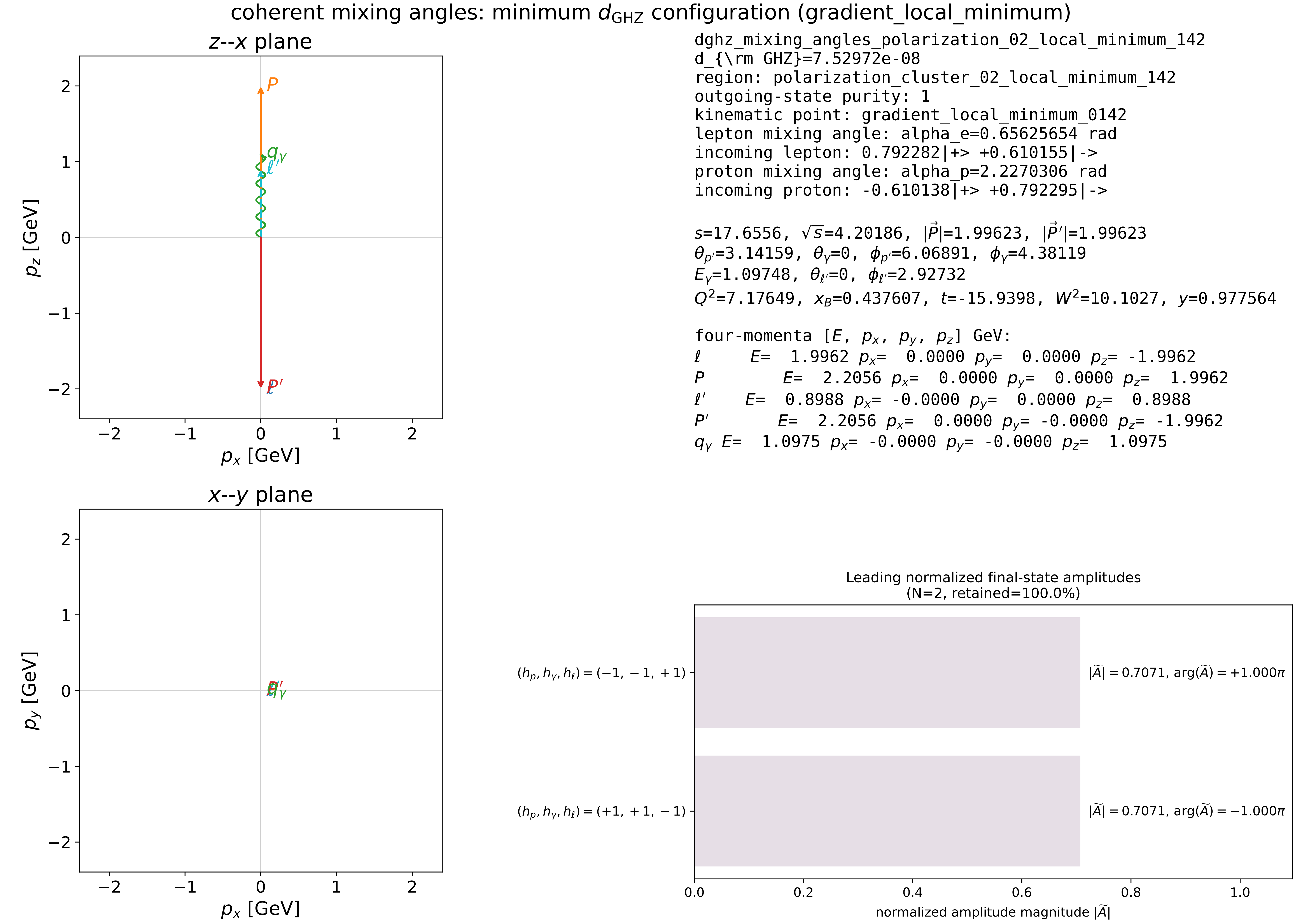
dghz_mixing_angles_polarization_02_local_minimum_136
d_{\rm GHZ}=6.21675e-08
region: polarization_cluster_02_local_minimum_136
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0136
lepton mixing angle: alpha_e=0.20100578 rad
incoming lepton: 0.979866|+> +0.199655|->
proton mixing angle: alpha_p=1.7718087 rad
incoming proton: -0.199661|+> +0.979865|->

s=24.9997, $\sqrt{s}$=4.99997, $|\vec{P}|$=2.412, $|\vec{P}'|$=2.412
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=1.40635, ϕ_γ=3.24397
E_γ=0.186808, $\theta_{\ell'}$=0, $\phi_{\ell'}$=4.54794
Q^2=21.4686, x_B=0.919952, t=-23.2709, W^2=2.74791, y=0.967532

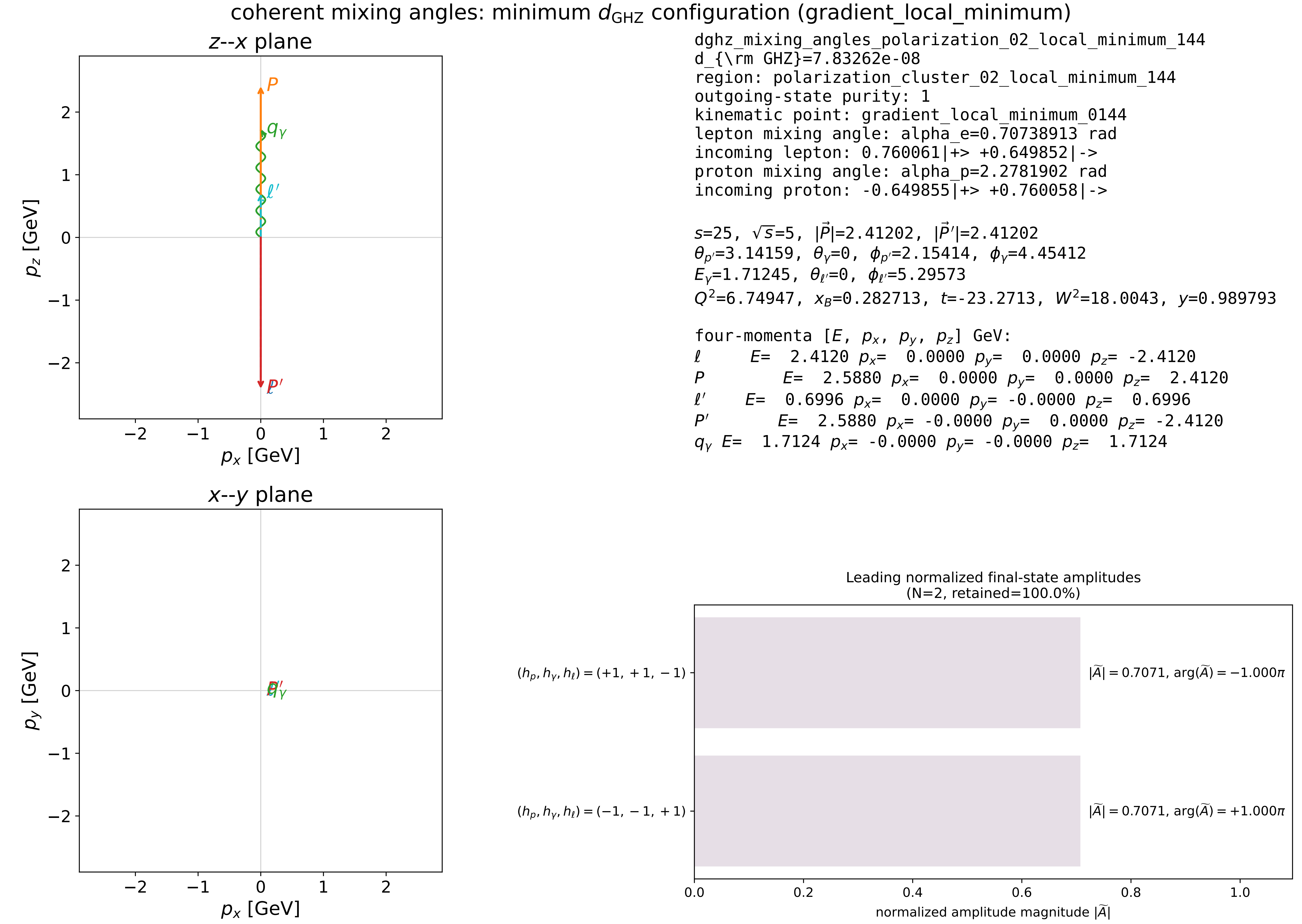
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.2252 p_x= -0.0000 p_y= -0.0000 p_z= 2.2252
P' E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
q_γ E= 0.1868 p_x= -0.0000 p_y= -0.0000 p_z= 0.1868



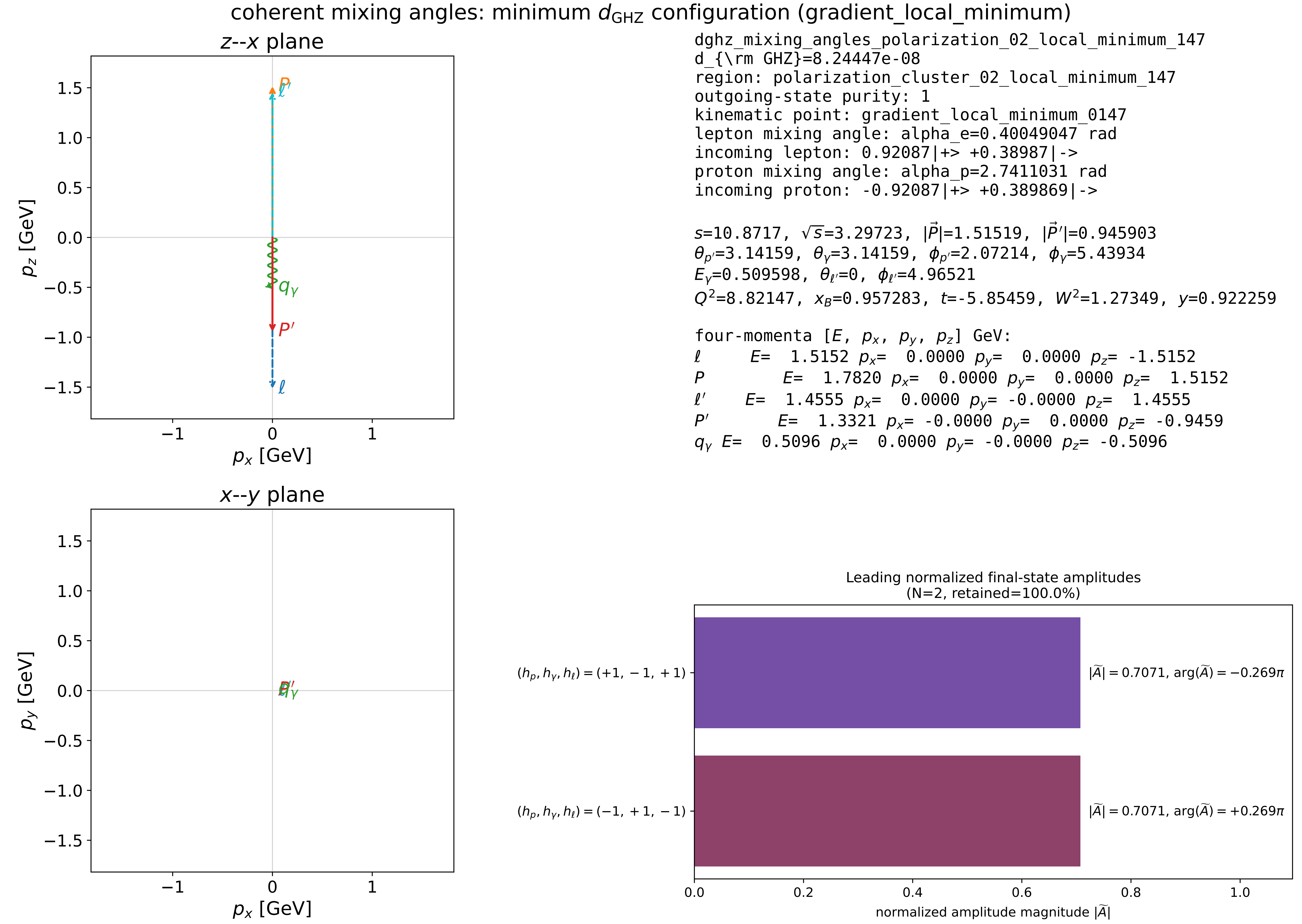




coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
q_γ
ℓ'
P'
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_142
d_{\rm GHZ}=7.52972e-08
region: polarization_cluster_02_local_minimum_142
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0142
lepton mixing angle: alpha_e=0.65625654 rad
incoming lepton: 0.792282|+> +0.610155|->
proton mixing angle: alpha_p=2.2270306 rad
incoming proton: -0.610138|+> +0.792295|->
s=17.6556, $\sqrt{s}$=4.20186, $|\vec{P}|$=1.99623, $|\vec{P}'|$=1.99623
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=6.06891, ϕ_γ=4.38119
E_γ=1.09748, $\theta_{\ell'}$=0, $\phi_{\ell'}$=2.92732
Q^2=7.17649, x_B=0.437607, t=-15.9398, W^2=10.1027, y=0.977564
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.9962 p_x= 0.0000 p_y= 0.0000 p_z= -1.9962
P E= 2.2056 p_x= 0.0000 p_y= 0.0000 p_z= 1.9962
ℓ' E= 0.8988 p_x= -0.0000 p_y= 0.0000 p_z= 0.8988
P' E= 2.2056 p_x= 0.0000 p_y= -0.0000 p_z= -1.9962
q_γ E= 1.0975 p_x= -0.0000 p_y= -0.0000 p_z= 1.0975
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +1.000\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

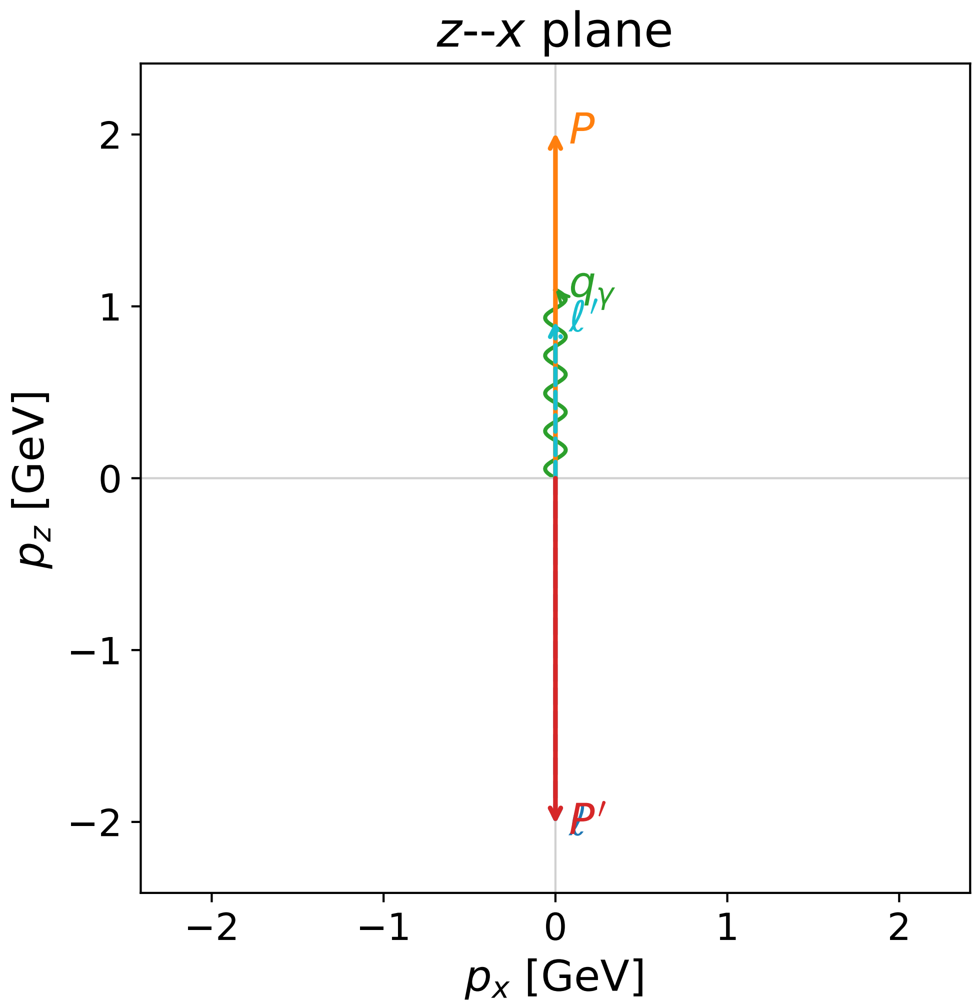


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
q_γ
ℓ'
P'
dghz_mixing_angles_polarization_02_local_minimum_144
d_{\rm GHZ}=7.83262e-08
region: polarization_cluster_02_local_minimum_144
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0144
lepton mixing angle: alpha_e=0.70738913 rad
incoming lepton: 0.760061|+> +0.649852|->
proton mixing angle: alpha_p=2.2781902 rad
incoming proton: -0.649855|+> +0.760058|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=2.41202
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=2.15414, ϕ_γ=4.45412
E_γ=1.71245, $\theta_{\ell'}$=0, $\phi_{\ell'}$=5.29573
Q^2=6.74947, x_B=0.282713, t=-23.2713, W^2=18.0043, y=0.989793
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 0.6996 p_x= 0.0000 p_y= -0.0000 p_z= 0.6996
P' E= 2.5880 p_x= -0.0000 p_y= 0.0000 p_z= -2.4120
q_γ E= 1.7124 p_x= -0.0000 p_y= -0.0000 p_z= 1.7124
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$



coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
x--y plane
p_x [GeV]
p_z [GeV]
p_y [GeV]
P
ℓ'
q_γ
P'
ℓ
dghz_mixing_angles_polarization_02_local_minimum_147
d_{\rm GHZ}=8.24447e-08
region: polarization_cluster_02_local_minimum_147
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0147
lepton mixing angle: alpha_e=0.40049047 rad
incoming lepton: 0.92087|+> +0.38987|->
proton mixing angle: alpha_p=2.7411031 rad
incoming proton: -0.92087|+> +0.389869|->
s=10.8717, $\sqrt{s}$=3.29723, $|\vec{P}|$=1.51519, $|\vec{P}'|$=0.945903
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=2.07214, ϕ_γ=5.43934
E_γ=0.509598, $\theta_{\ell'}$=0, $\phi_{\ell'}$=4.96521
Q^2=8.82147, x_B=0.957283, t=-5.85459, W^2=1.27349, y=0.922259
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.5152 p_x= 0.0000 p_y= 0.0000 p_z= -1.5152
P E= 1.7820 p_x= 0.0000 p_y= 0.0000 p_z= 1.5152
ℓ' E= 1.4555 p_x= 0.0000 p_y= -0.0000 p_z= 1.4555
P' E= 1.3321 p_x= -0.0000 p_y= 0.0000 p_z= -0.9459
q_γ E= 0.5096 p_x= 0.0000 p_y= -0.0000 p_z= -0.5096
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.269\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.269\pi$
normalized amplitude magnitude $|\widetilde{A}|$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

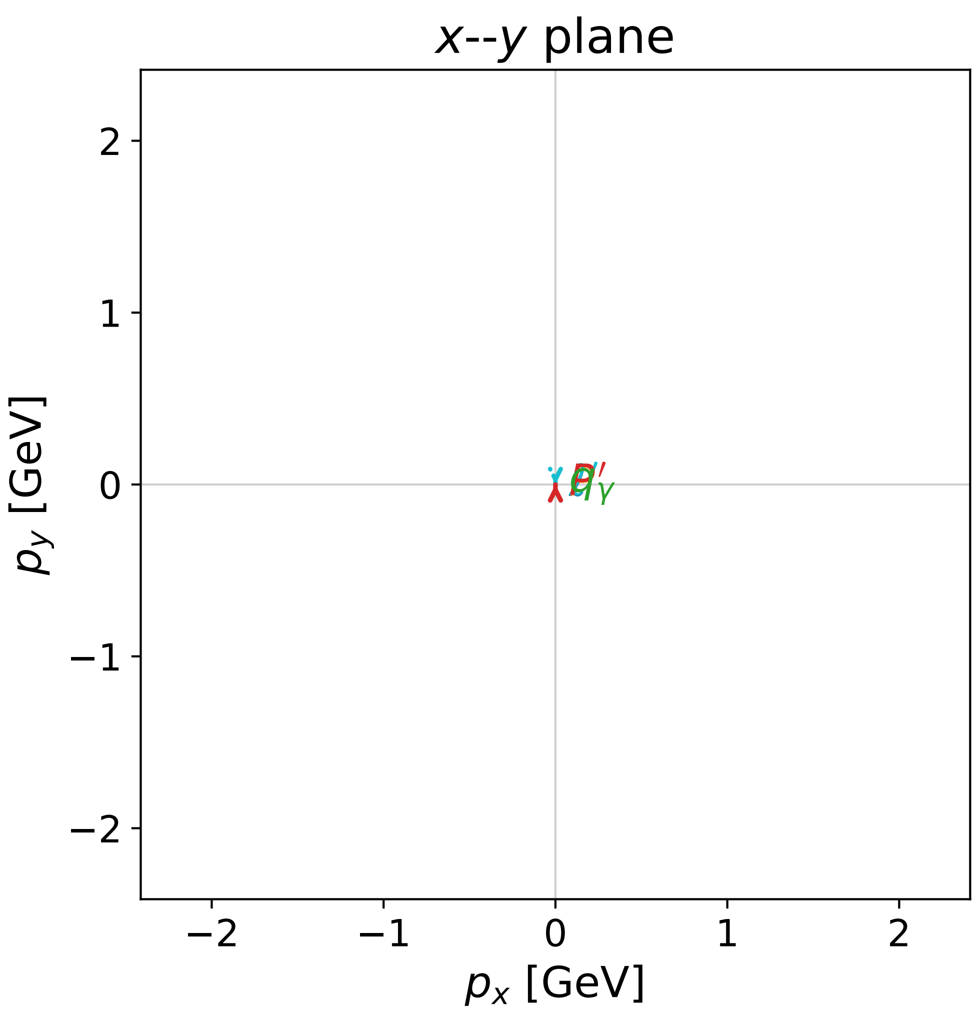


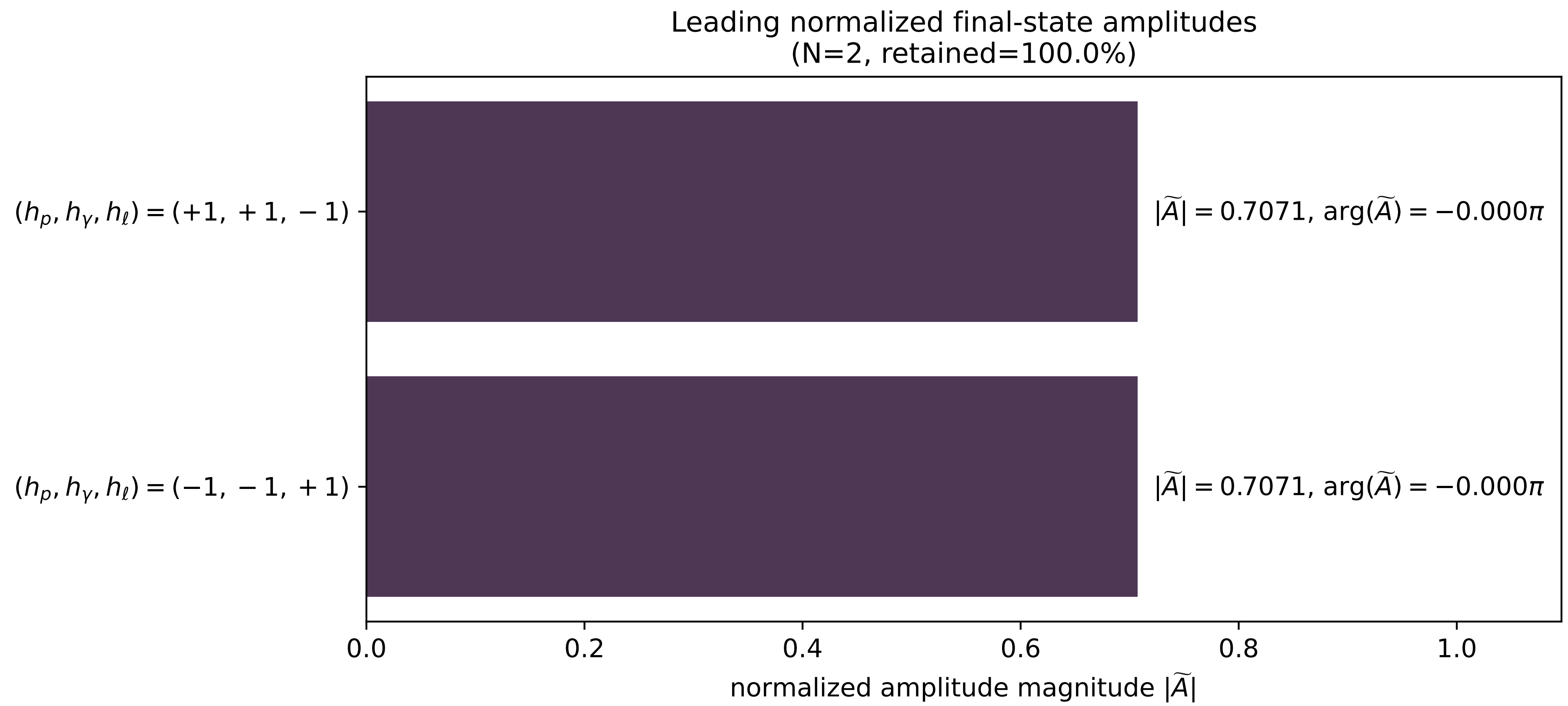

```
dghz_mixing_angles_polarization_02_local_minimum_148
d_{\rm GHZ}=8.27454e-08
region: polarization_cluster_02_local_minimum_148
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0148
lepton mixing angle: alpha_e=1.0537095 rad
incoming lepton: 0.49435|+> +0.869263|->
proton mixing angle: alpha_p=2.6245189 rad
incoming proton: -0.869269|+> +0.494339|->
```

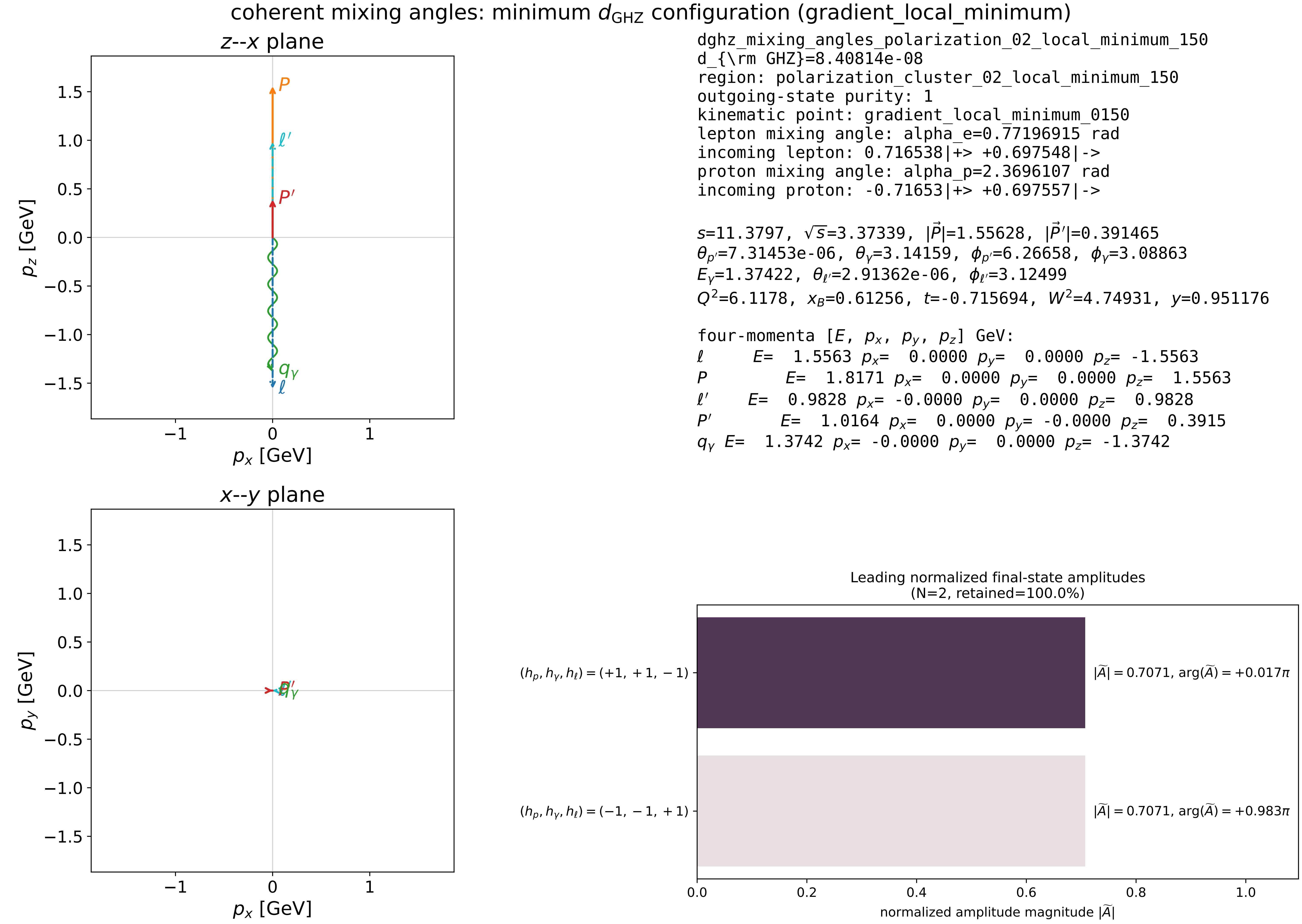
s=17.8907, $\sqrt{s}$=4.22973, $|\vec{P}|$=2.01086, $|\vec{P}'|$=2.01086
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=1.82192, ϕ_γ=5.84502
E_γ=1.09781, $\theta_{\ell'}$=0, $\phi_{\ell'}$=4.96351
Q^2=7.34406, x_B=0.44159, t=-16.1742, W^2=10.1667, y=0.97767

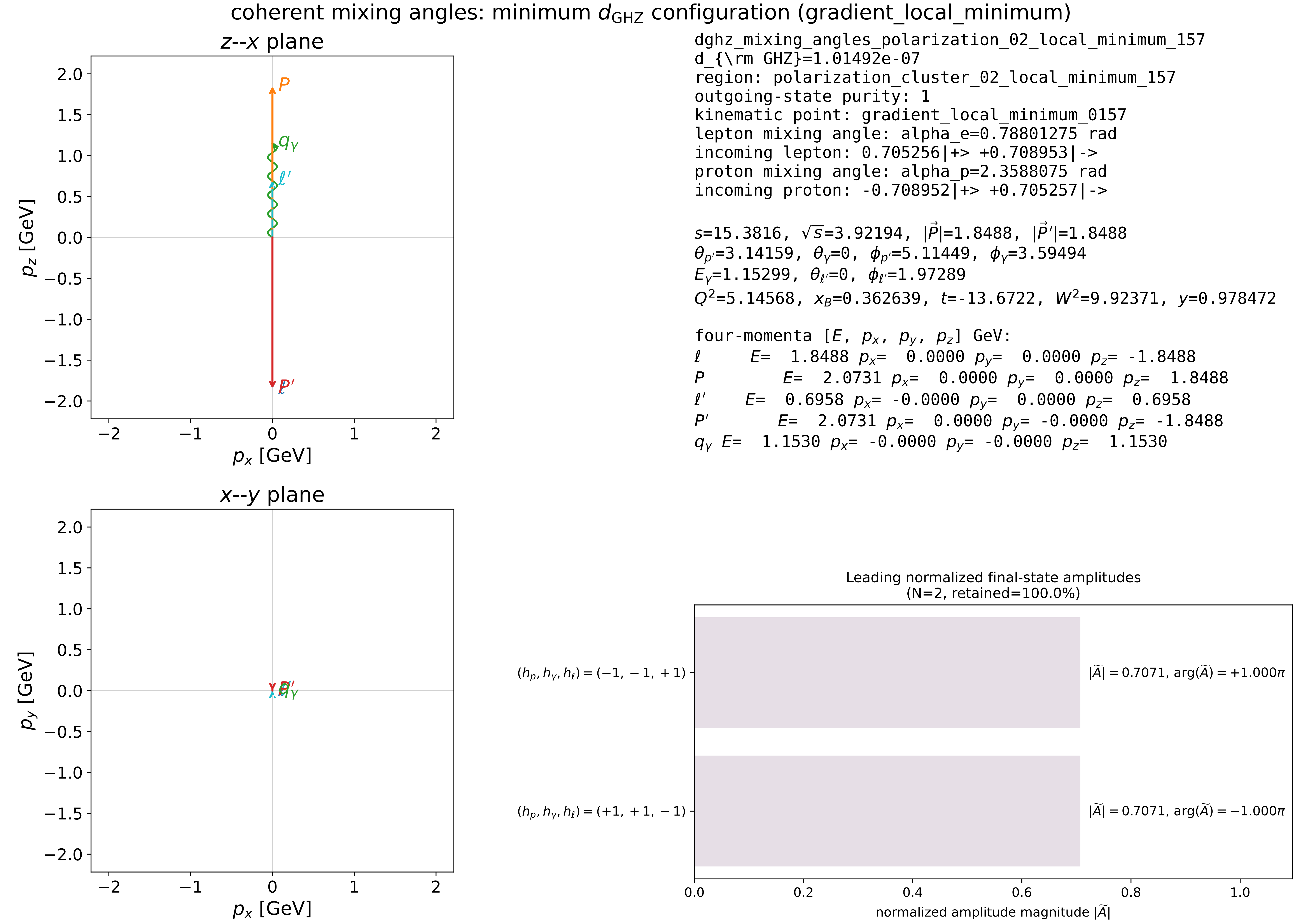
four-momenta [E, p_x, p_y, p_z] GeV:

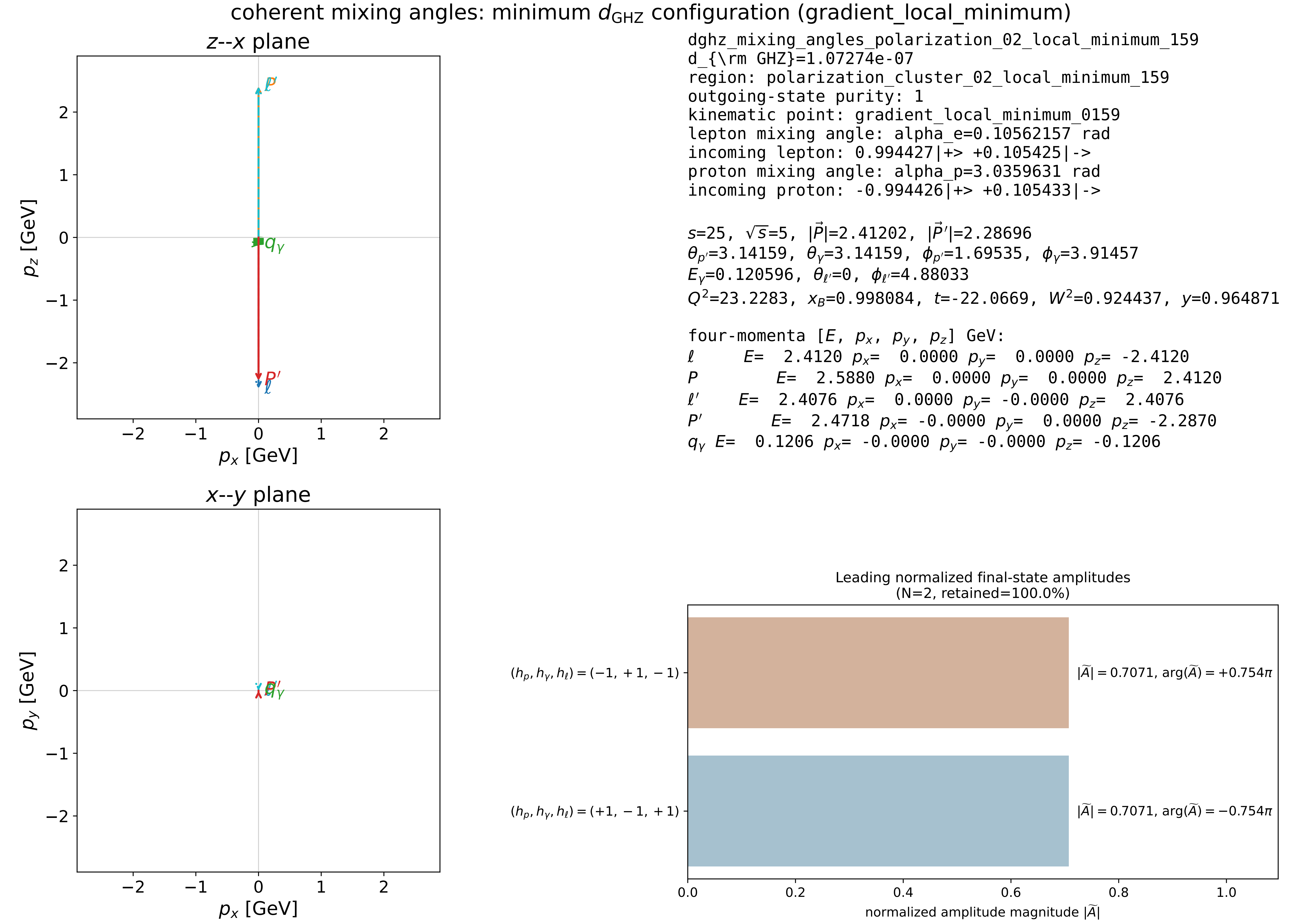
ℓ	E= 2.0109	p_x= 0.0000	p_y= 0.0000	p_z= -2.0109
P	E= 2.2189	p_x= 0.0000	p_y= 0.0000	p_z= 2.0109
ℓ'	E= 0.9130	p_x= 0.0000	p_y= -0.0000	p_z= 0.9130
P'	E= 2.2189	p_x= -0.0000	p_y= 0.0000	p_z= -2.0109
q_γ	E= 1.0978	p_x= 0.0000	p_y= -0.0000	p_z= 1.0978





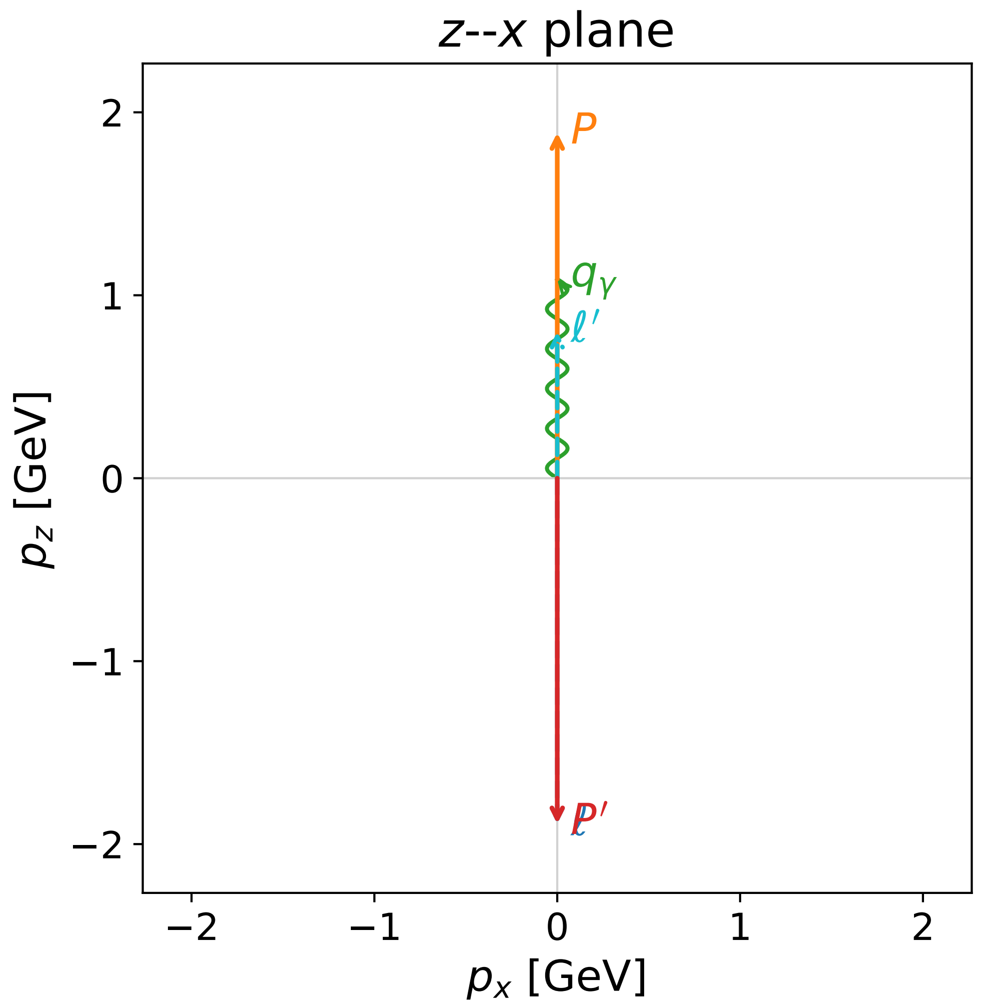

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_150
d_{\rm GHZ}=8.40814e-08
region: polarization_cluster_02_local_minimum_150
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0150
lepton mixing angle: alpha_e=0.77196915 rad
incoming lepton: 0.716538|+> +0.697548|->
proton mixing angle: alpha_p=2.3696107 rad
incoming proton: -0.71653|+> +0.697557|->
s=11.3797, $\sqrt{s}$=3.37339, $|\vec{P}|$=1.55628, $|\vec{P}'|$=0.391465
$\theta_{p'}$=7.31453e-06, θ_γ=3.14159, $\phi_{p'}$=6.26658, ϕ_γ=3.08863
E_γ=1.37422, $\theta_{\ell'}$=2.91362e-06, $\phi_{\ell'}$=3.12499
Q^2=6.1178, x_B=0.61256, t=-0.715694, W^2=4.74931, y=0.951176
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.5563 p_x= 0.0000 p_y= 0.0000 p_z= -1.5563
P E= 1.8171 p_x= 0.0000 p_y= 0.0000 p_z= 1.5563
ℓ' E= 0.9828 p_x= -0.0000 p_y= 0.0000 p_z= 0.9828
P' E= 1.0164 p_x= 0.0000 p_y= -0.0000 p_z= 0.3915
q_γ E= 1.3742 p_x= -0.0000 p_y= 0.0000 p_z= -1.3742
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.017\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.983\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
q_γ
ℓ'
P'
dghz_mixing_angles_polarization_02_local_minimum_157
d_{\rm GHZ}=1.01492e-07
region: polarization_cluster_02_local_minimum_157
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0157
lepton mixing angle: alpha_e=0.78801275 rad
incoming lepton: 0.705256|+> +0.708953|->
proton mixing angle: alpha_p=2.3588075 rad
incoming proton: -0.708952|+> +0.705257|->
s=15.3816, $\sqrt{s}$=3.92194, $|\vec{P}|$=1.8488, $|\vec{P}'|$=1.8488
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=5.11449, ϕ_γ=3.59494
E_γ=1.15299, $\theta_{\ell'}$=0, $\phi_{\ell'}$=1.97289
Q^2=5.14568, x_B=0.362639, t=-13.6722, W^2=9.92371, y=0.978472
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.8488 p_x= 0.0000 p_y= 0.0000 p_z= -1.8488
P E= 2.0731 p_x= 0.0000 p_y= 0.0000 p_z= 1.8488
ℓ' E= 0.6958 p_x= -0.0000 p_y= 0.0000 p_z= 0.6958
P' E= 2.0731 p_x= 0.0000 p_y= -0.0000 p_z= -1.8488
q_γ E= 1.1530 p_x= -0.0000 p_y= -0.0000 p_z= 1.1530
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +1.000\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$



coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
ℓ'
P
q_γ
P'
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_159
d_{\rm GHZ}=1.07274e-07
region: polarization_cluster_02_local_minimum_159
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0159
lepton mixing angle: alpha_e=0.10562157 rad
incoming lepton: 0.994427|+> +0.105425|->
proton mixing angle: alpha_p=3.0359631 rad
incoming proton: -0.994426|+> +0.105433|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=2.28696
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=1.69535, ϕ_γ=3.91457
E_γ=0.120596, $\theta_{\ell'}$=0, $\phi_{\ell'}$=4.88033
Q^2=23.2283, x_B=0.998084, t=-22.0669, W^2=0.924437, y=0.964871
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.4076 p_x= 0.0000 p_y= -0.0000 p_z= 2.4076
P' E= 2.4718 p_x= -0.0000 p_y= 0.0000 p_z= -2.2870
q_γ E= 0.1206 p_x= -0.0000 p_y= -0.0000 p_z= -0.1206
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.754\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.754\pi$
normalized amplitude magnitude $|\widetilde{A}|$

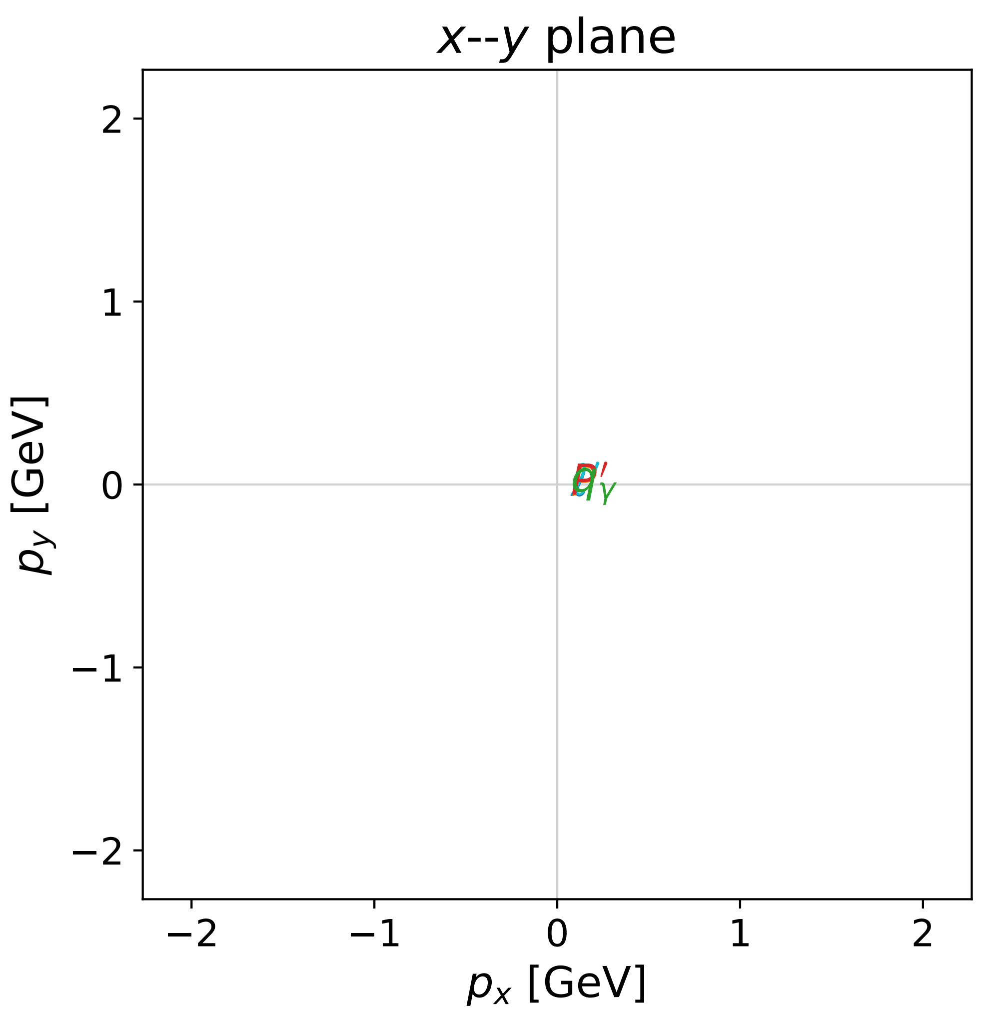
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

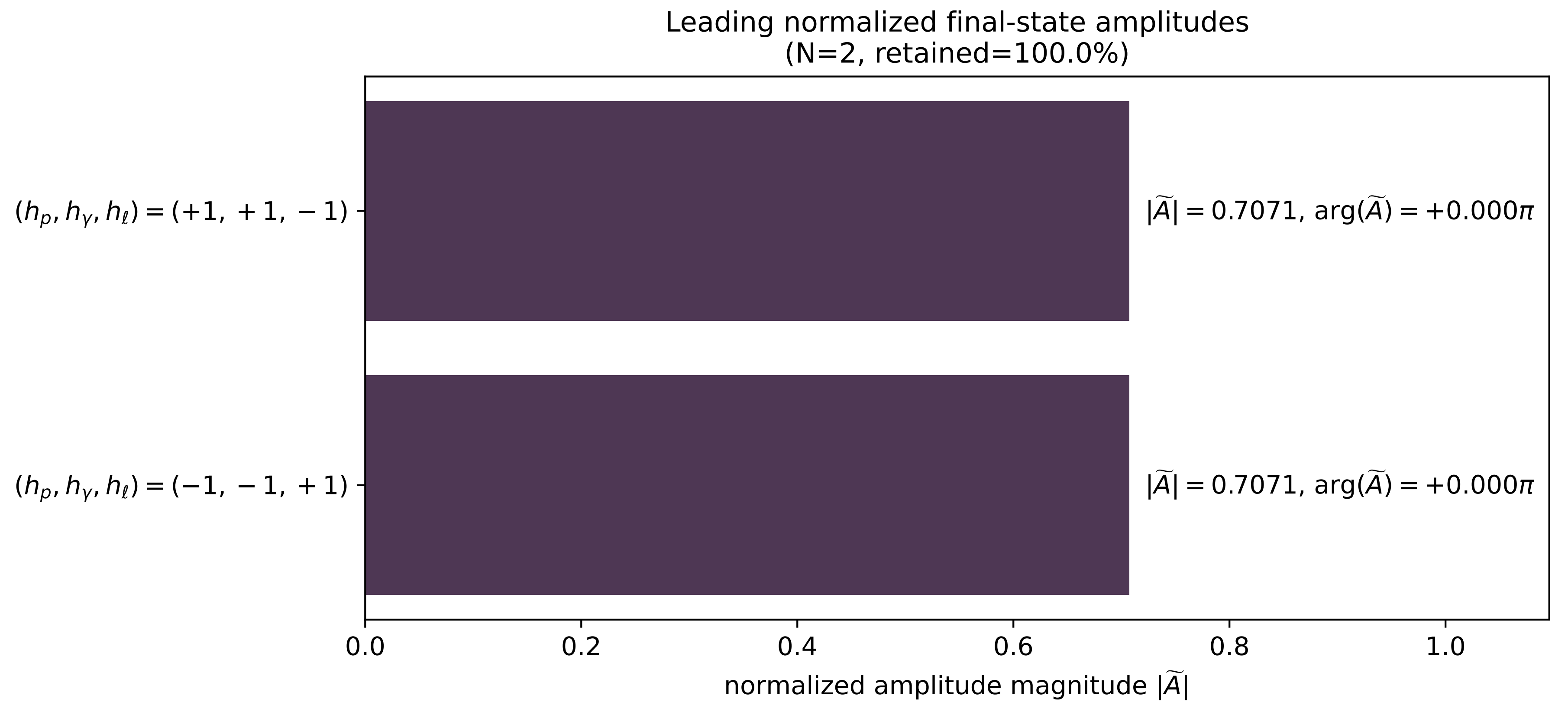


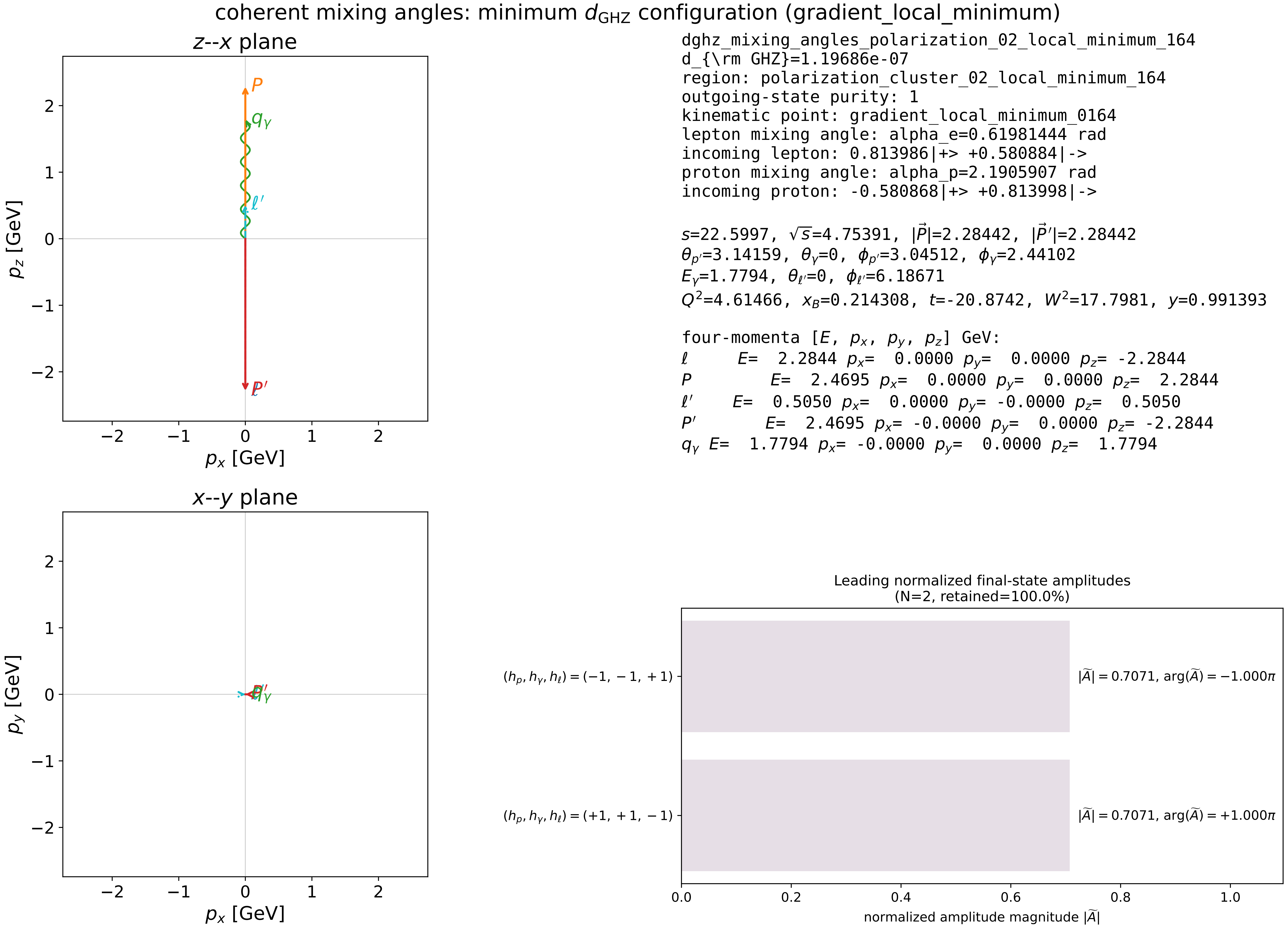
dghz_mixing_angles_polarization_02_local_minimum_160
d_{\rm GHZ}=1.12987e-07
region: polarization_cluster_02_local_minimum_160
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0160
lepton mixing angle: alpha_e=1.1362778 rad
incoming lepton: 0.420974|+> +0.907073|->
proton mixing angle: alpha_p=2.7070795 rad
incoming proton: -0.907075|+> +0.420969|->

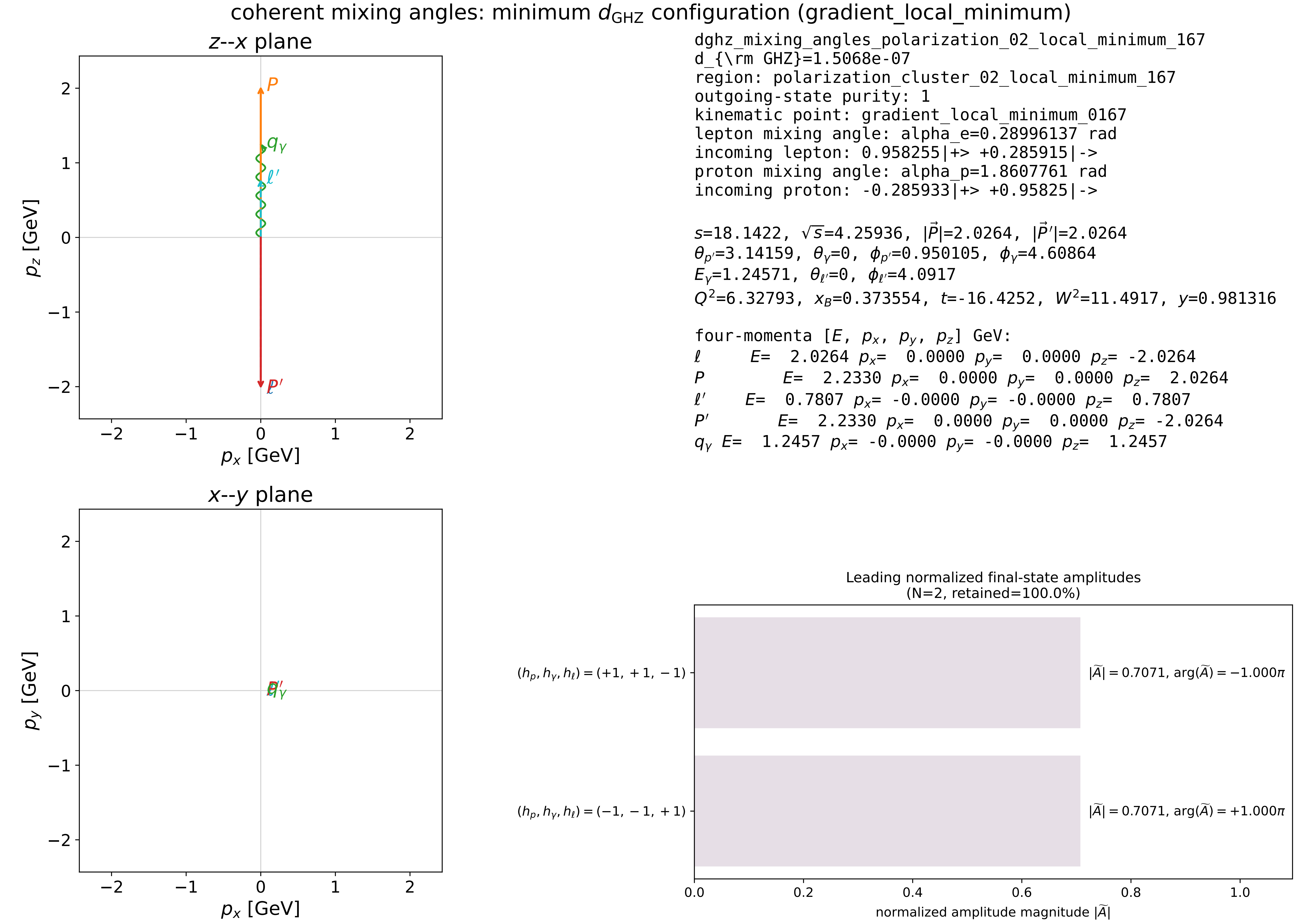
s=15.9848, $\sqrt{s}$=3.9981, $|\vec{P}|$=1.88901, $|\vec{P}'|$=1.88901
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=4.2262, ϕ_γ=6.10955
E_γ=1.08804, $\theta_{\ell'}$=0, $\phi_{\ell'}$=1.08461
Q^2=6.05225, x_B=0.410256, t=-14.2735, W^2=9.57998, y=0.976661

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.8890 p_x= 0.0000 p_y= 0.0000 p_z= -1.8890
P E= 2.1091 p_x= 0.0000 p_y= 0.0000 p_z= 1.8890
ℓ' E= 0.8010 p_x= 0.0000 p_y= 0.0000 p_z= 0.8010
P' E= 2.1091 p_x= -0.0000 p_y= -0.0000 p_z= -1.8890
q_γ E= 1.0880 p_x= 0.0000 p_y= -0.0000 p_z= 1.0880

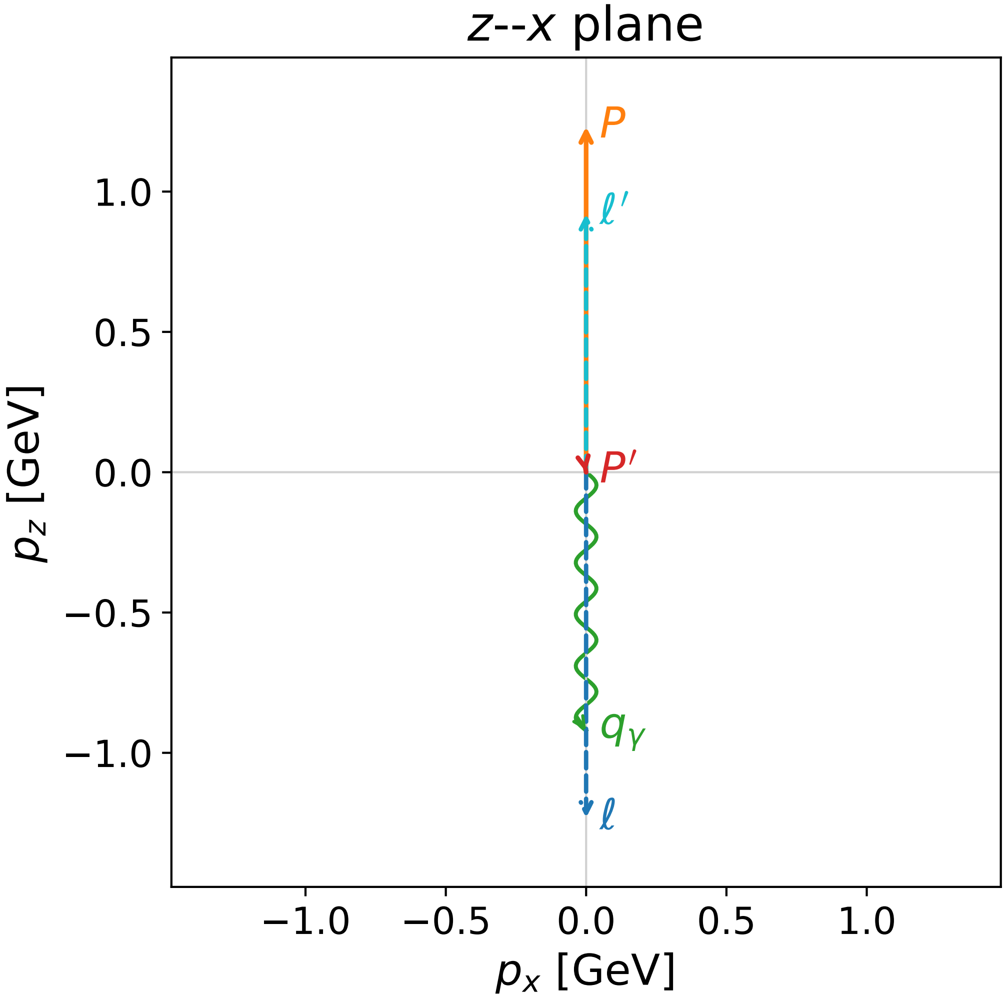





coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
q_γ
ℓ'
P'
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_164
d_{\rm GHZ}=1.19686e-07
region: polarization_cluster_02_local_minimum_164
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0164
lepton mixing angle: alpha_e=0.61981444 rad
incoming lepton: 0.813986|+> +0.580884|->
proton mixing angle: alpha_p=2.1905907 rad
incoming proton: -0.580868|+> +0.813998|->
s=22.5997, $\sqrt{s}$=4.75391, $|\vec{P}|$=2.28442, $|\vec{P}'|$=2.28442
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=3.04512, ϕ_γ=2.44102
E_γ=1.7794, $\theta_{\ell'}$=0, $\phi_{\ell'}$=6.18671
Q^2=4.61466, x_B=0.214308, t=-20.8742, W^2=17.7981, y=0.991393
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2844 p_x= 0.0000 p_y= 0.0000 p_z= -2.2844
P E= 2.4695 p_x= 0.0000 p_y= 0.0000 p_z= 2.2844
ℓ' E= 0.5050 p_x= 0.0000 p_y= -0.0000 p_z= 0.5050
P' E= 2.4695 p_x= -0.0000 p_y= 0.0000 p_z= -2.2844
q_γ E= 1.7794 p_x= -0.0000 p_y= 0.0000 p_z= 1.7794
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
q_γ
ℓ'
P'
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_167
d_{\rm GHZ}=1.5068e-07
region: polarization_cluster_02_local_minimum_167
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0167
lepton mixing angle: alpha_e=0.28996137 rad
incoming lepton: 0.958255|+> +0.285915|->
proton mixing angle: alpha_p=1.8607761 rad
incoming proton: -0.285933|+> +0.95825|->
s=18.1422, $\sqrt{s}$=4.25936, $|\vec{P}|$=2.0264, $|\vec{P}'|$=2.0264
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=0.950105, ϕ_γ=4.60864
E_γ=1.24571, $\theta_{\ell'}$=0, $\phi_{\ell'}$=4.0917
Q^2=6.32793, x_B=0.373554, t=-16.4252, W^2=11.4917, y=0.981316
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.0264 p_x= 0.0000 p_y= 0.0000 p_z= -2.0264
P E= 2.2330 p_x= 0.0000 p_y= 0.0000 p_z= 2.0264
ℓ' E= 0.7807 p_x= -0.0000 p_y= -0.0000 p_z= 0.7807
P' E= 2.2330 p_x= 0.0000 p_y= 0.0000 p_z= -2.0264
q_γ E= 1.2457 p_x= -0.0000 p_y= -0.0000 p_z= 1.2457
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

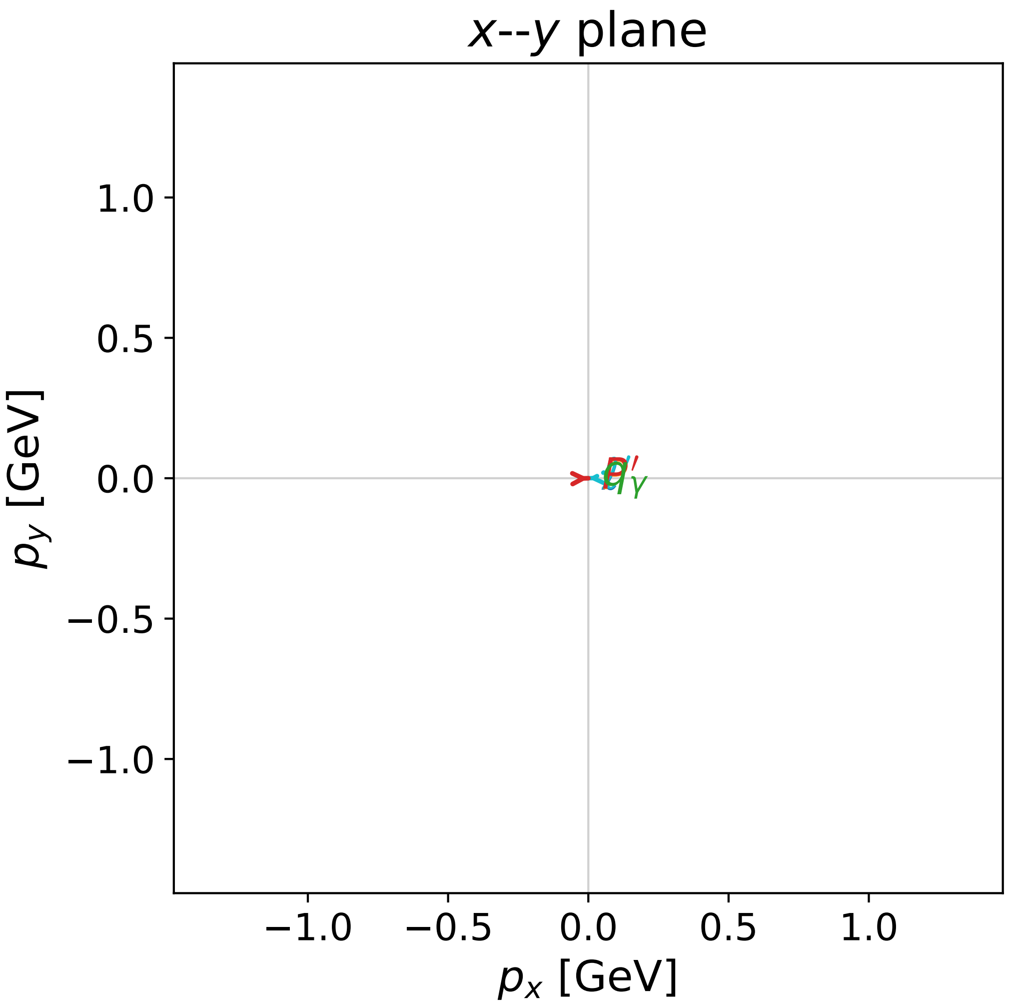
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

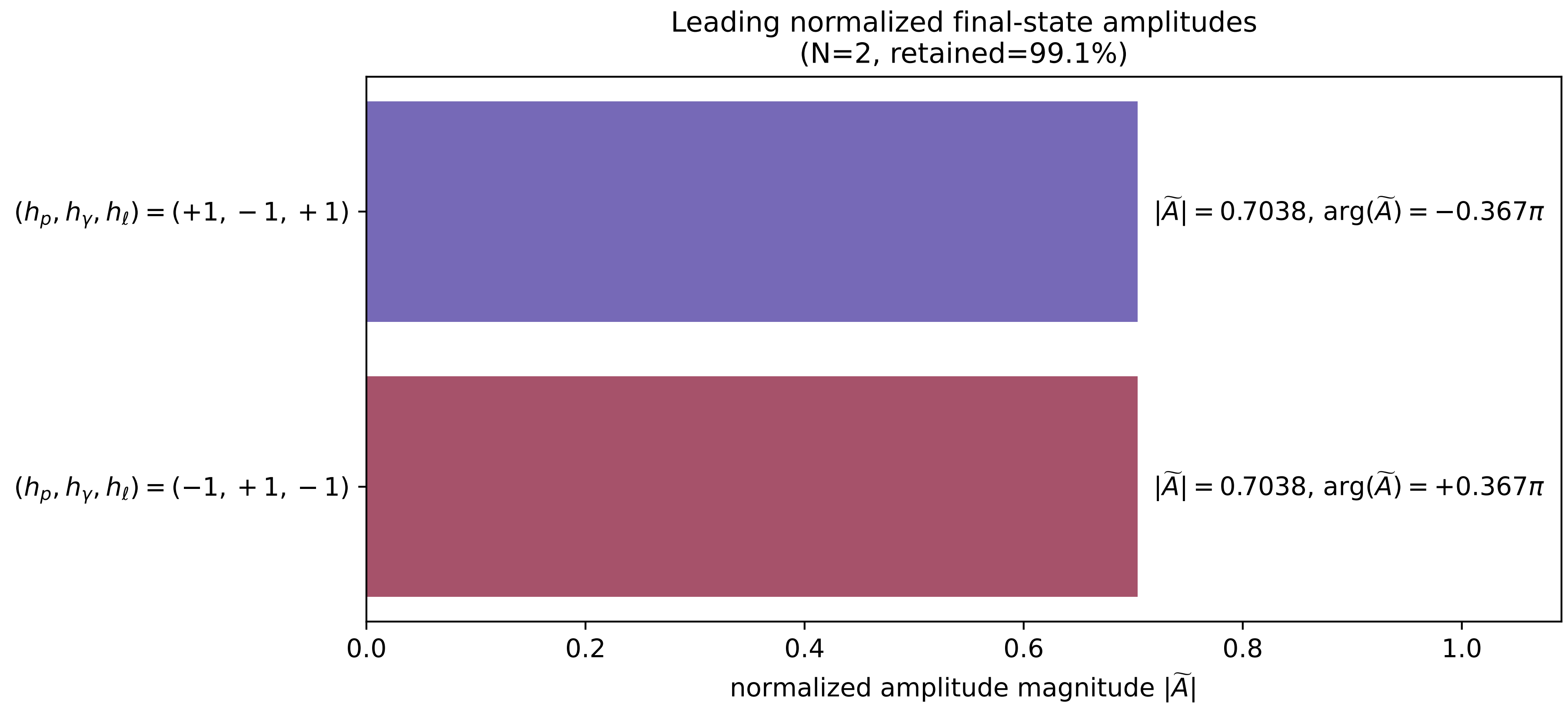


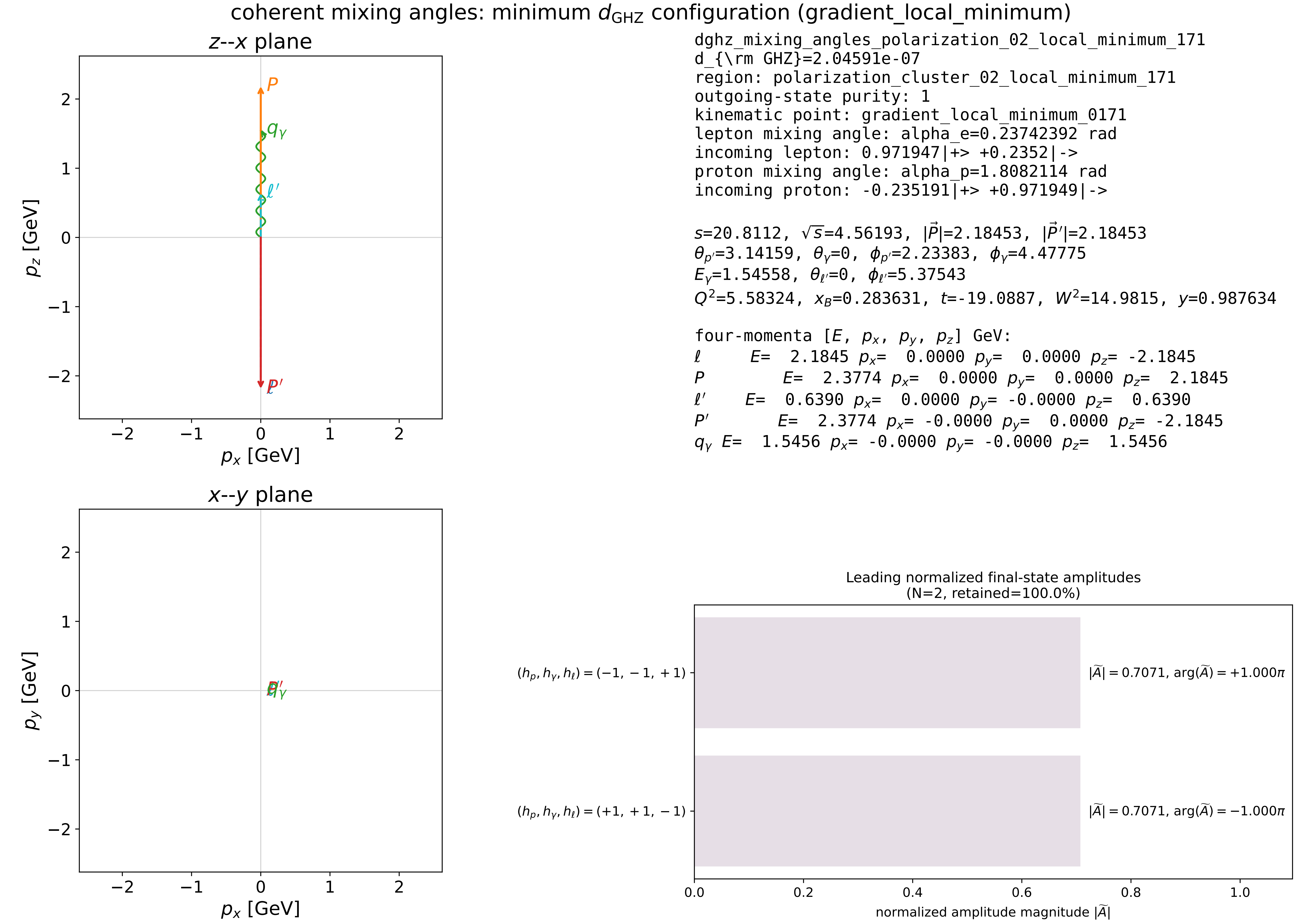
dghz_mixing_angles_polarization_02_local_minimum_168
d_{\rm GHZ}=1.70137e-07
region: polarization_cluster_02_local_minimum_168
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0168
lepton mixing angle: alpha_e=1.2036147 rad
incoming lepton: 0.358986|+> +0.933343|->
proton mixing angle: alpha_p=1.9380061 rad
incoming proton: -0.359013|+> +0.933333|->

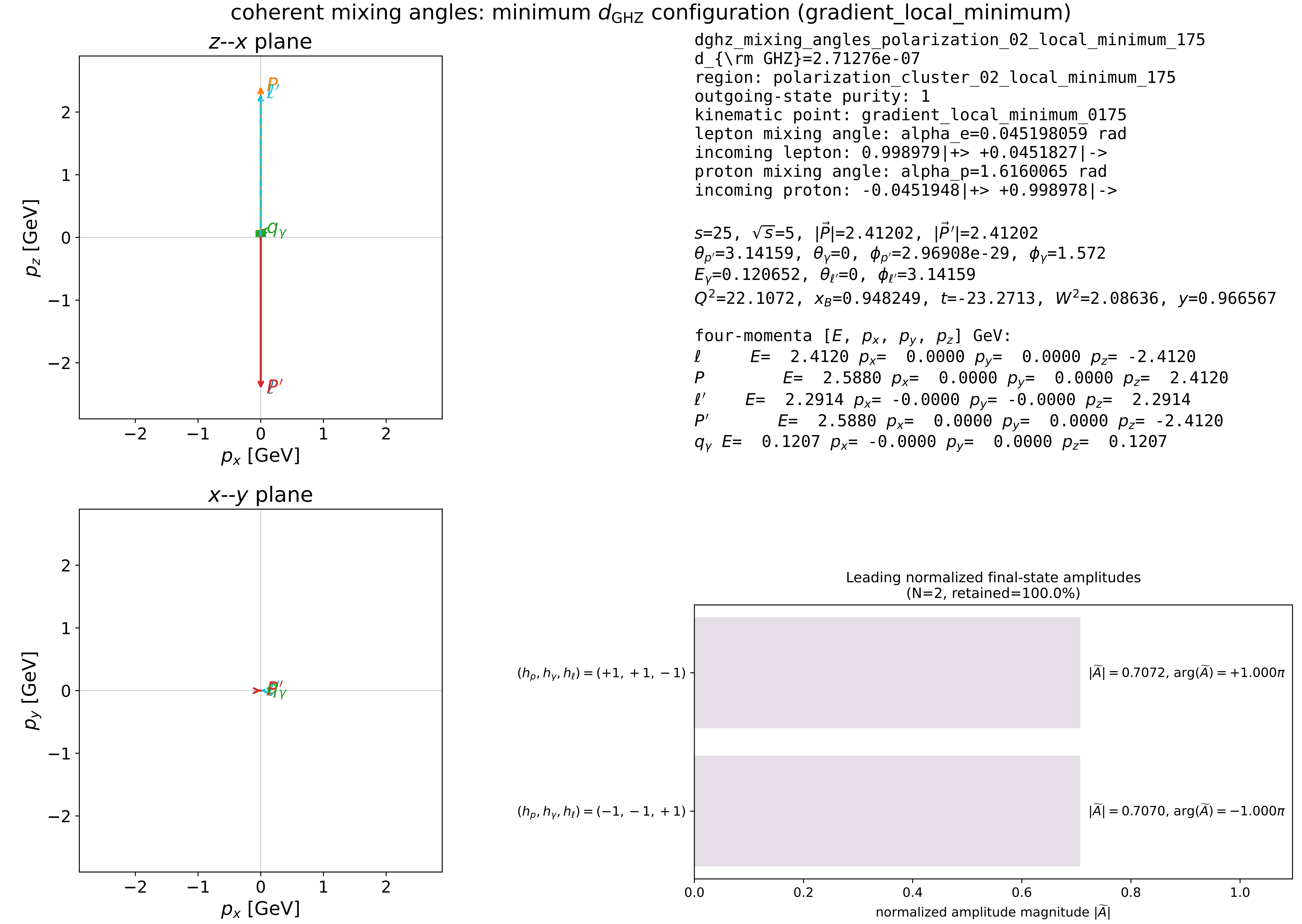
s=7.72597, $\sqrt{s}$=2.77956, $|\vec{P}|$=1.23151, $|\vec{P}'|$=0.000244016
$\theta_{p'}$=2.94845, θ_γ=3.14159, $\phi_{p'}$=0.027548, ϕ_γ=5.15721
E_γ=0.920662, $\theta_{\ell'}$=5.08603e-05, $\phi_{\ell'}$=3.16914
Q^2=4.5364, x_B=0.724304, t=-1.14505, W^2=2.60656, y=0.914842

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.2315 p_x= 0.0000 p_y= 0.0000 p_z= -1.2315
P E= 1.5481 p_x= 0.0000 p_y= 0.0000 p_z= 1.2315
ℓ' E= 0.9209 p_x= -0.0000 p_y= -0.0000 p_z= 0.9209
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= -0.0002
q_γ E= 0.9207 p_x= 0.0000 p_y= -0.0000 p_z= -0.9207



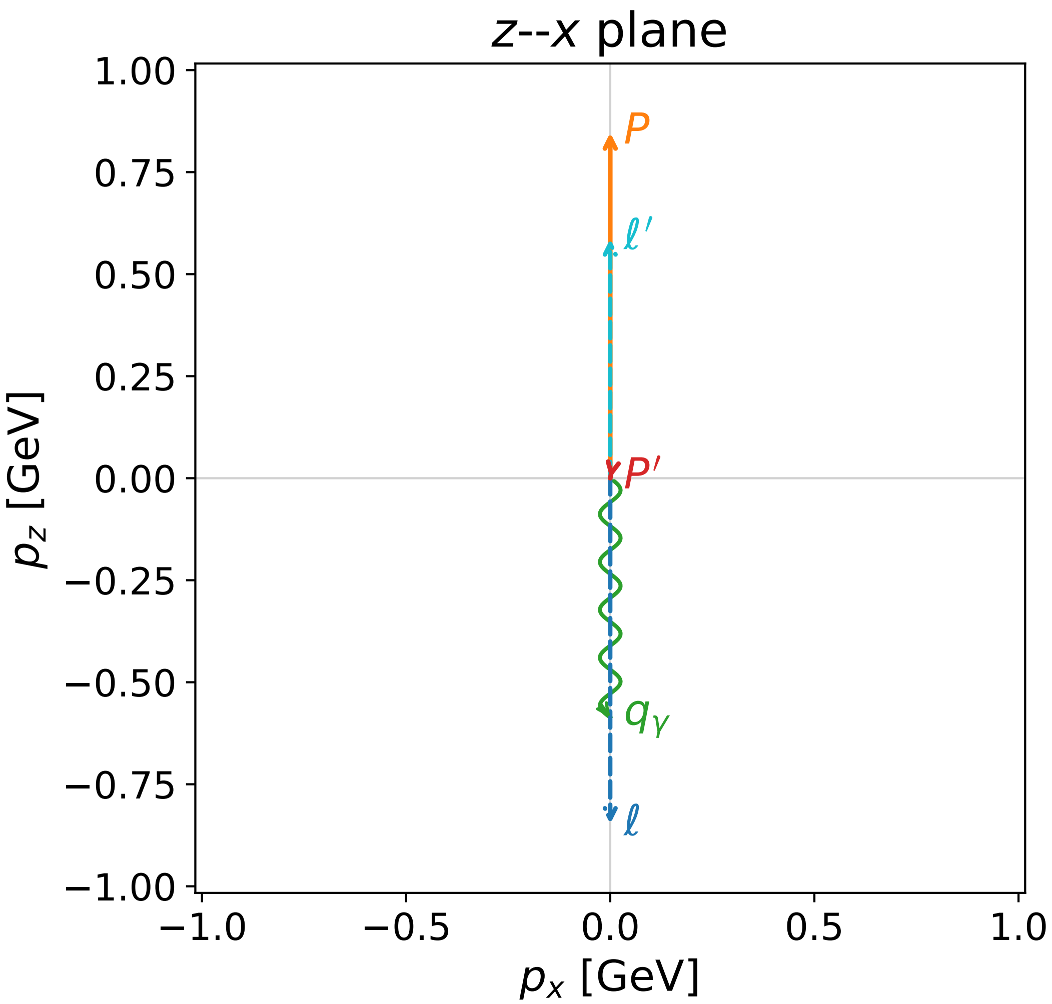



coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
q_γ
ℓ'
P'
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_171
d_{\rm GHZ}=2.04591e-07
region: polarization_cluster_02_local_minimum_171
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0171
lepton mixing angle: alpha_e=0.23742392 rad
incoming lepton: 0.971947|+> +0.2352|->
proton mixing angle: alpha_p=1.8082114 rad
incoming proton: -0.235191|+> +0.971949|->
s=20.8112, $\sqrt{s}$=4.56193, $|\vec{P}|$=2.18453, $|\vec{P}'|$=2.18453
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=2.23383, ϕ_γ=4.47775
E_γ=1.54558, $\theta_{\ell'}$=0, $\phi_{\ell'}$=5.37543
Q^2=5.58324, x_B=0.283631, t=-19.0887, W^2=14.9815, y=0.987634
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.1845 p_x= 0.0000 p_y= 0.0000 p_z= -2.1845
P E= 2.3774 p_x= 0.0000 p_y= 0.0000 p_z= 2.1845
ℓ' E= 0.6390 p_x= 0.0000 p_y= -0.0000 p_z= 0.6390
P' E= 2.3774 p_x= -0.0000 p_y= 0.0000 p_z= -2.1845
q_γ E= 1.5456 p_x= -0.0000 p_y= -0.0000 p_z= 1.5456
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +1.000\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$



coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
x--y plane
p_x [GeV]
p_z [GeV]
p_y [GeV]
P
ℓ'
q_γ
P'
ℓ
dghz_mixing_angles_polarization_02_local_minimum_175
d_{\rm GHZ}=2.71276e-07
region: polarization_cluster_02_local_minimum_175
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0175
lepton mixing angle: alpha_e=0.045198059 rad
incoming lepton: 0.998979|+> +0.0451827|->
proton mixing angle: alpha_p=1.6160065 rad
incoming proton: -0.0451948|+> +0.998978|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=2.41202
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=2.96908e-29, ϕ_γ=1.572
E_γ=0.120652, $\theta_{\ell'}$=0, $\phi_{\ell'}$=3.14159
Q^2=22.1072, x_B=0.948249, t=-23.2713, W^2=2.08636, y=0.966567
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.2914 p_x= -0.0000 p_y= -0.0000 p_z= 2.2914
P' E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
q_γ E= 0.1207 p_x= -0.0000 p_y= 0.0000 p_z= 0.1207
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7072$, $\arg(\widetilde{A}) = +1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7070$, $\arg(\widetilde{A}) = -1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

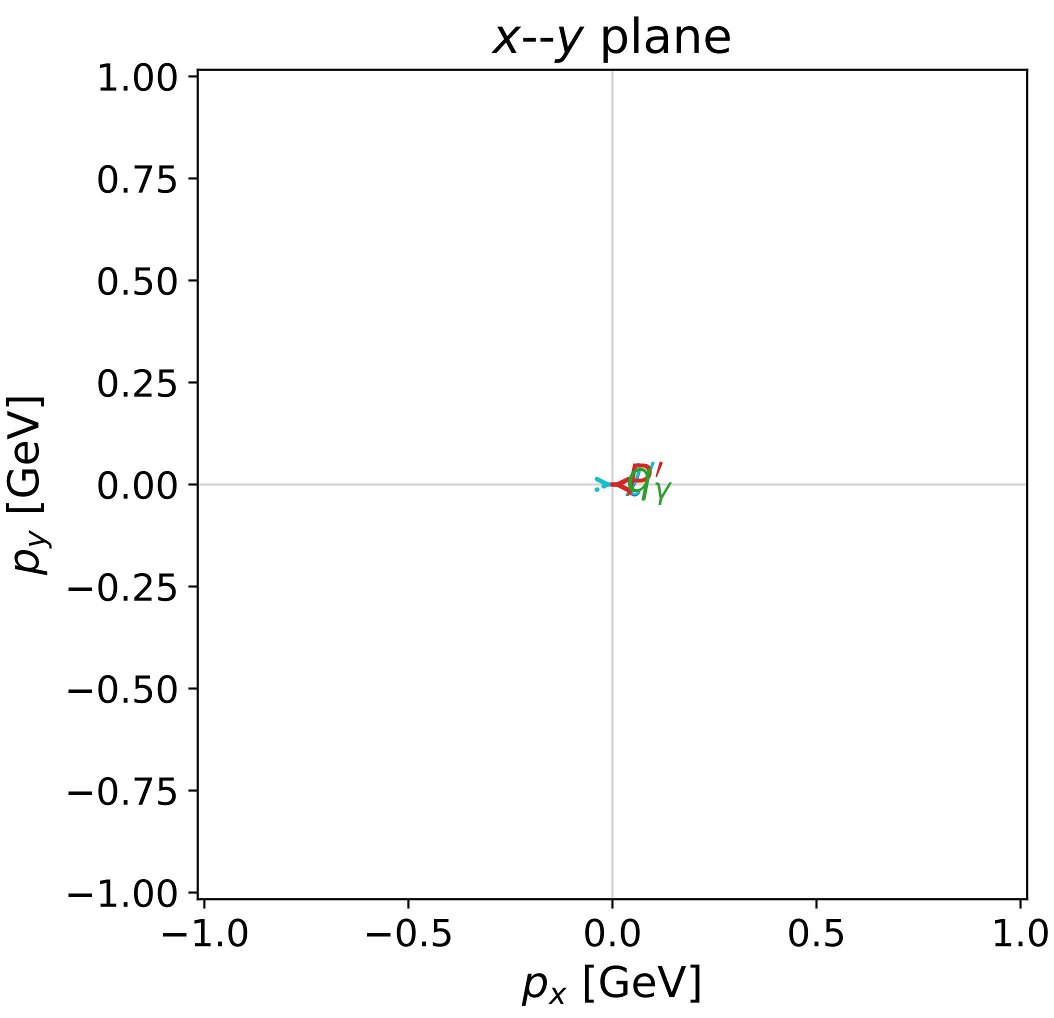
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

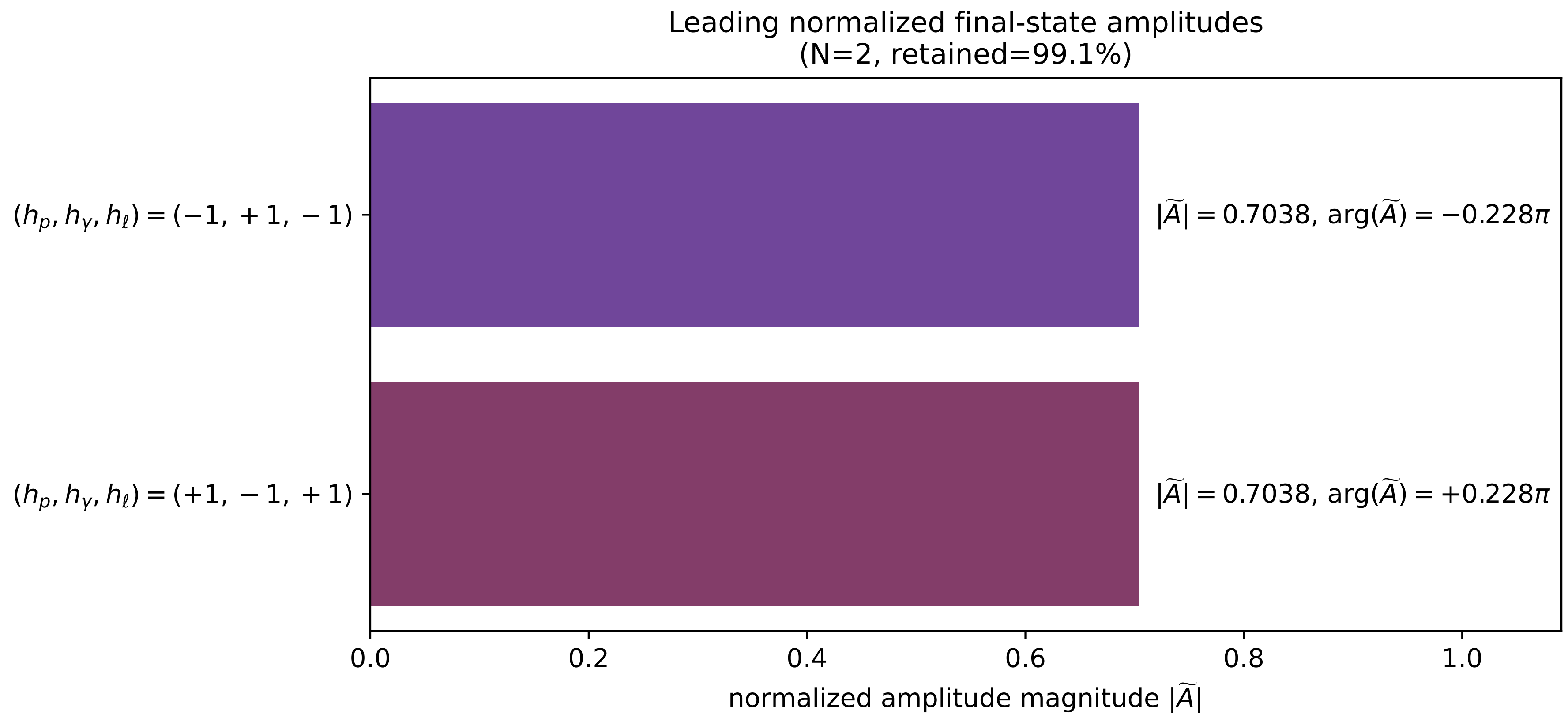



```
dghz_mixing_angles_polarization_02_local_minimum_182
d_{\rm GHZ}=6.61545e-07
region: polarization_cluster_02_local_minimum_182
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0182
lepton mixing angle: alpha_e=0.2035518 rad
incoming lepton: 0.979355|+> +0.202149|->
proton mixing angle: alpha_p=2.9380379 rad
incoming proton: -0.979354|+> +0.202152|->
```

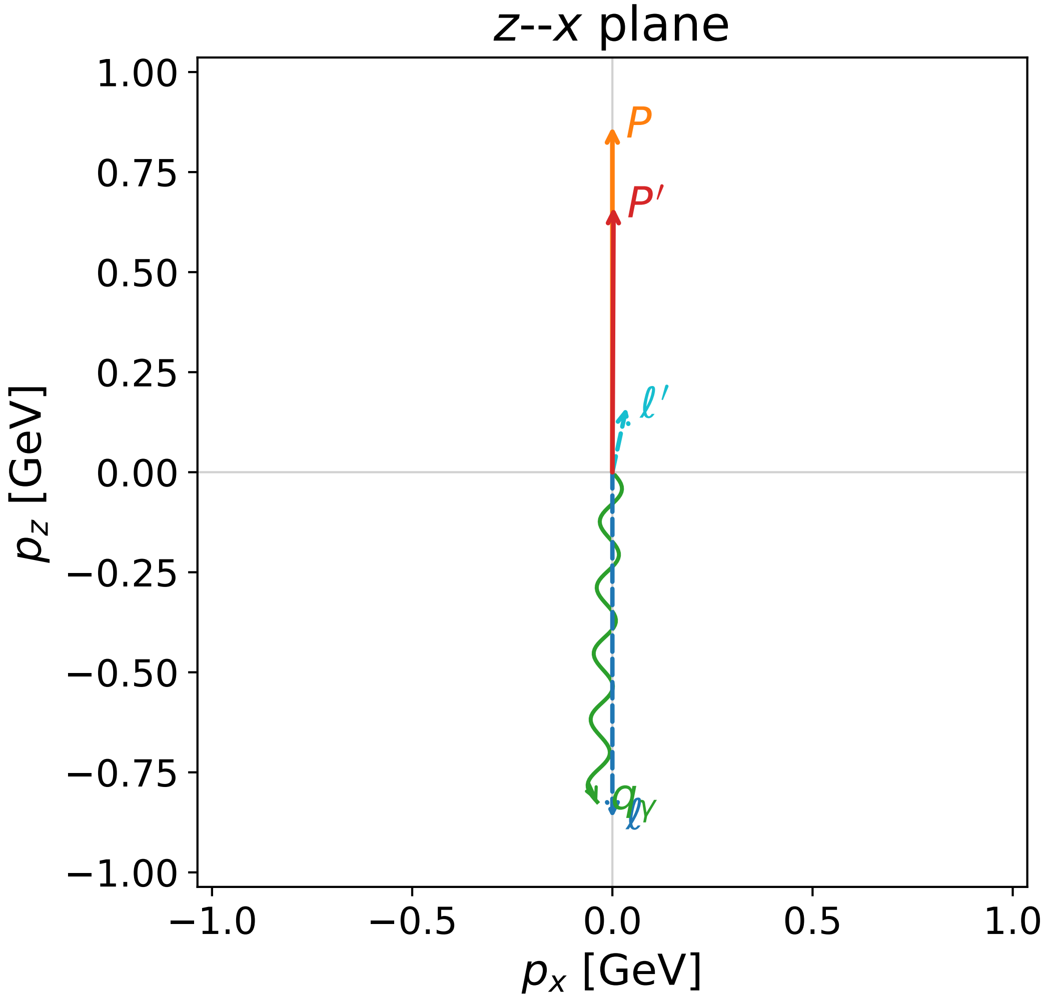
s=4.45467, $\sqrt{s}$=2.11061, $|\vec{P}|$=0.846871, $|\vec{P}'|$=1.86603e-05
$\theta_{p'}$=2.94793, θ_γ=3.14159, $\phi_{p'}$=3.13457, ϕ_γ=3.85199
E_γ=0.586295, $\theta_{\ell'}$=6.12523e-06, $\phi_{\ell'}$=6.27616
Q^2=1.98613, x_B=0.643594, t=-0.611116, W^2=1.97971, y=0.863258

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.8469 p_x= 0.0000 p_y= 0.0000 p_z= -0.8469
P E= 1.2637 p_x= 0.0000 p_y= 0.0000 p_z= 0.8469
ℓ' E= 0.5863 p_x= 0.0000 p_y= -0.0000 p_z= 0.5863
P' E= 0.9380 p_x= -0.0000 p_y= 0.0000 p_z= -0.0000
q_γ E= 0.5863 p_x= -0.0000 p_y= -0.0000 p_z= -0.5863





coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)




```
dghz_mixing_angles_polarization_02_local_minimum_185
d_{\rm GHZ}=3.76832e-06
region: polarization_cluster_02_local_minimum_185
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0185
lepton mixing angle: alpha_e=1.2208886 rad
incoming lepton: 0.342811|+> +0.939404|->
proton mixing angle: alpha_p=2.8239017 rad
incoming proton: -0.949959|+> +0.312374|->
```

s=4.57376, $\sqrt{s}$=2.13864, $|\vec{P}|$=0.863616, $|\vec{P}'|$=0.662371

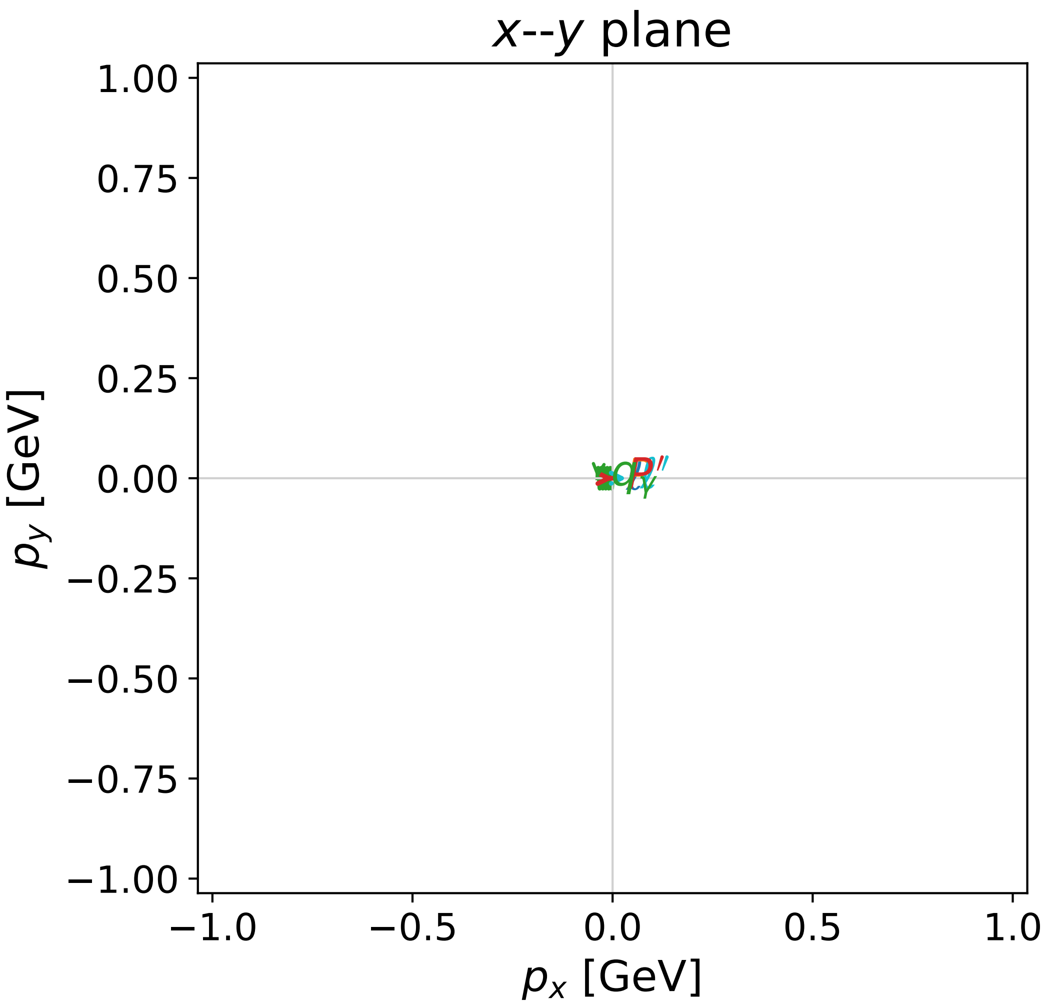
$\theta_{p'}$=0.00416983, θ_γ=3.09573, $\phi_{p'}$=6.28319, ϕ_γ=3.1416

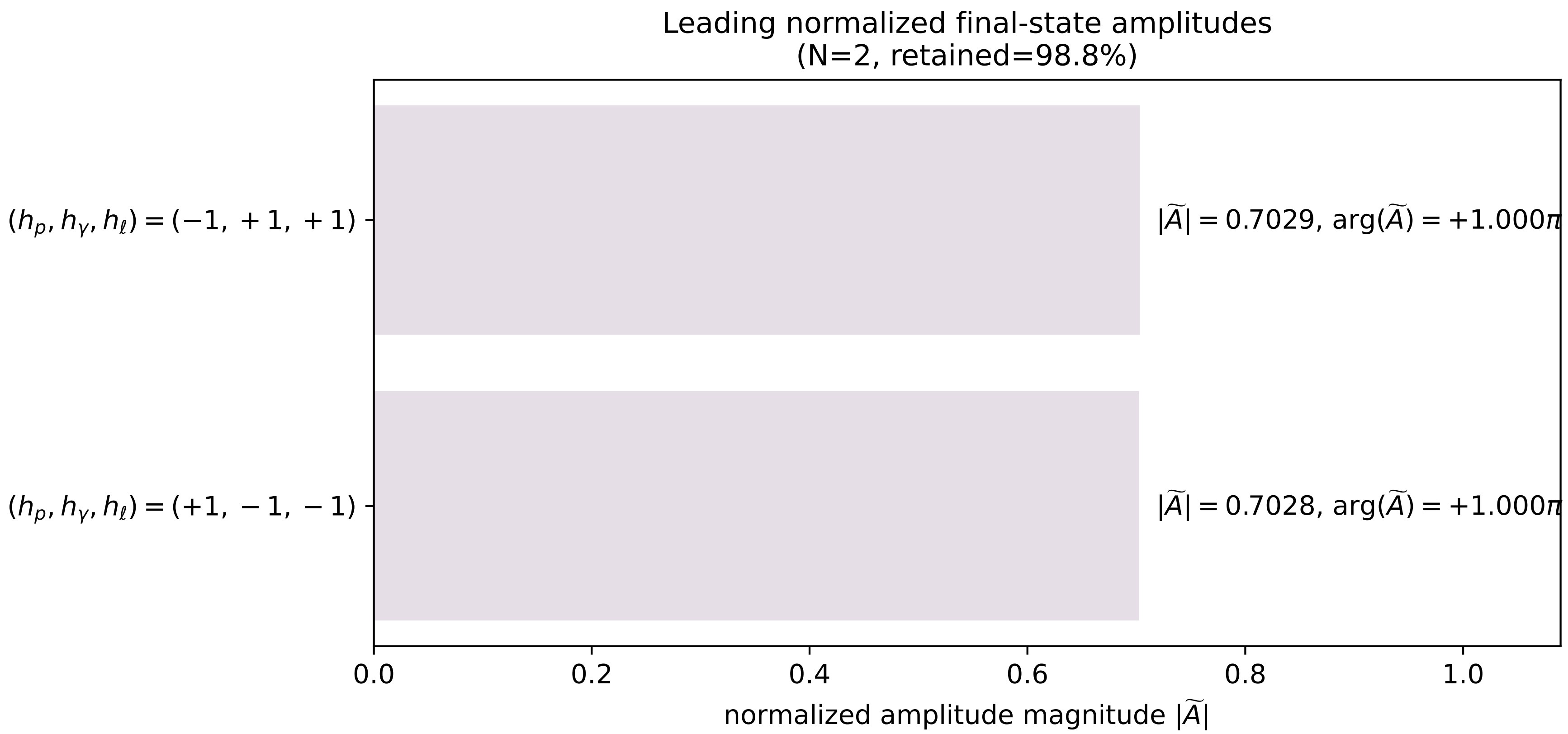
E_γ=0.824908, $\theta_{\ell'}$=0.213519, $\phi_{\ell'}$=6.85947e-06

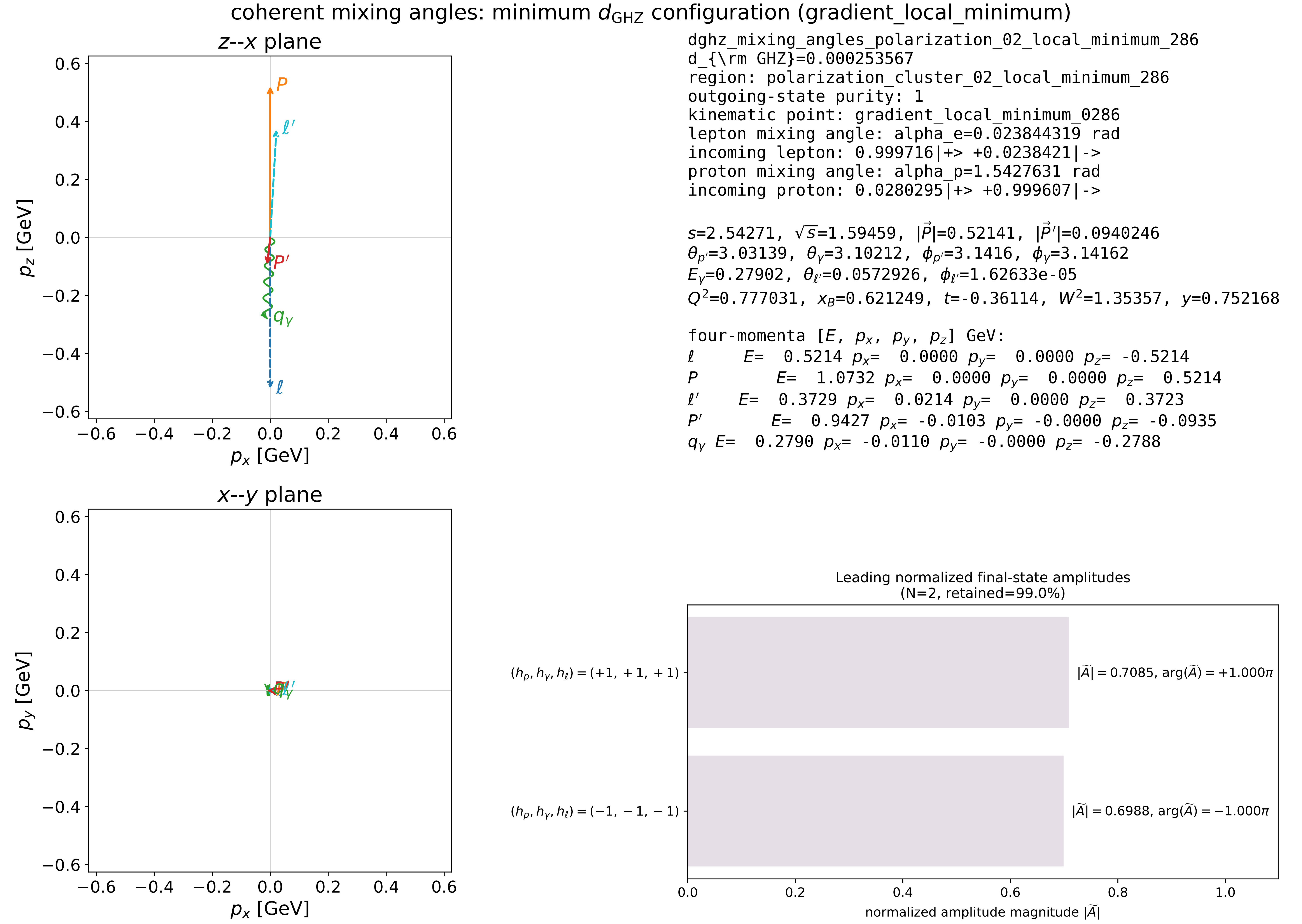
Q^2=0.564992, x_B=0.159094, t=-0.0244499, W^2=3.86616, y=0.961393

four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 0.8636	p_x= 0.0000	p_y= 0.0000	p_z= -0.8636
P	E= 1.2750	p_x= 0.0000	p_y= 0.0000	p_z= 0.8636
ℓ'	E= 0.1654	p_x= 0.0351	p_y= 0.0000	p_z= 0.1617
P'	E= 1.1483	p_x= 0.0028	p_y= -0.0000	p_z= 0.6624
q_γ	E= 0.8249	p_x= -0.0378	p_y= -0.0000	p_z= -0.8240

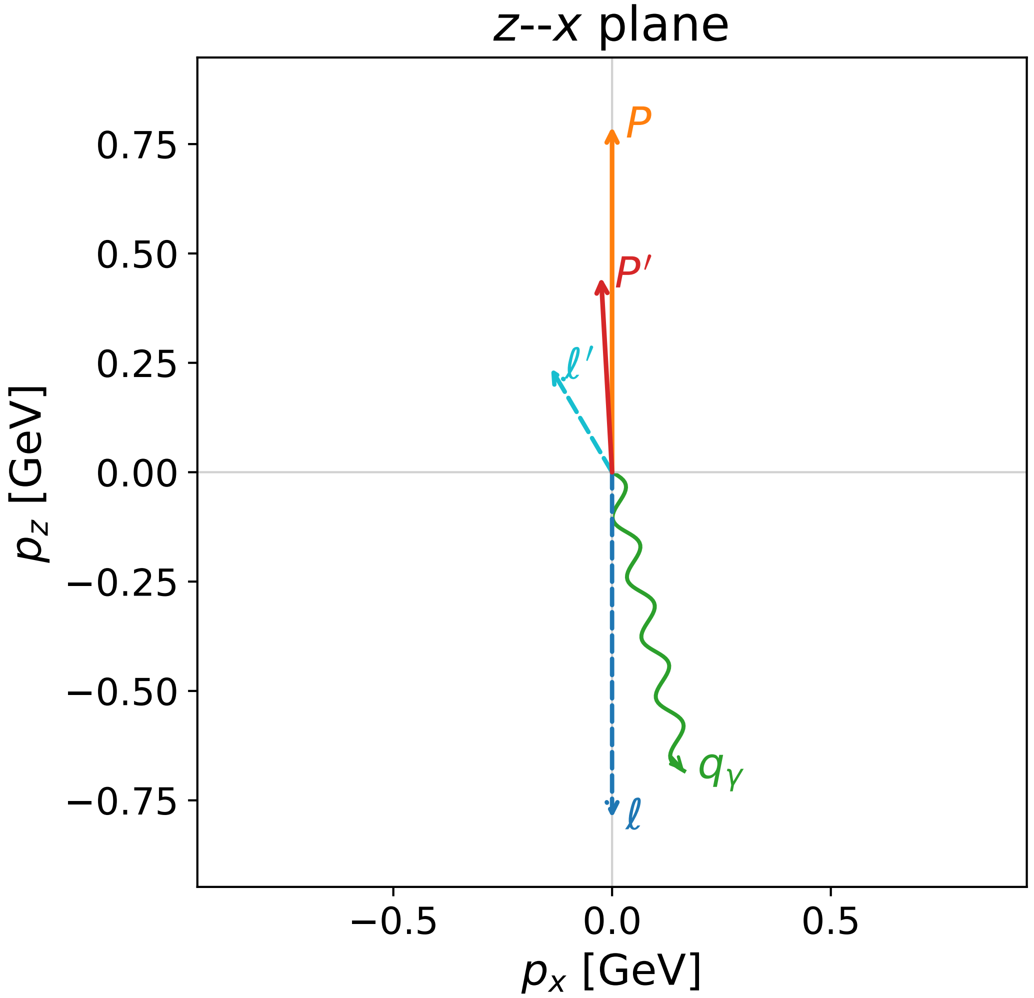






coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
z--x plane
x--y plane
p_z [GeV]
p_x [GeV]
p_y [GeV]
P
ℓ'
P'
q_γ
ℓ
dghz_mixing_angles_polarization_02_local_minimum_286
d_{\rm GHZ}=0.000253567
region: polarization_cluster_02_local_minimum_286
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0286
lepton mixing angle: alpha_e=0.023844319 rad
incoming lepton: 0.999716|+> +0.0238421|->
proton mixing angle: alpha_p=1.5427631 rad
incoming proton: 0.0280295|+> +0.999607|->
s=2.54271, $\sqrt{s}$=1.59459, $|\vec{P}|$=0.52141, $|\vec{P}'|$=0.0940246
$\theta_{p'}$=3.03139, θ_γ=3.10212, $\phi_{p'}$=3.1416, ϕ_γ=3.14162
E_γ=0.27902, $\theta_{\ell'}$=0.0572926, $\phi_{\ell'}$=1.62633e-05
Q^2=0.777031, x_B=0.621249, t=-0.36114, W^2=1.35357, y=0.752168
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.5214 p_x= 0.0000 p_y= 0.0000 p_z= -0.5214
P E= 1.0732 p_x= 0.0000 p_y= 0.0000 p_z= 0.5214
ℓ' E= 0.3729 p_x= 0.0214 p_y= 0.0000 p_z= 0.3723
P' E= 0.9427 p_x= -0.0103 p_y= -0.0000 p_z= -0.0935
q_γ E= 0.2790 p_x= -0.0110 p_y= -0.0000 p_z= -0.2788
Leading normalized final-state amplitudes
(N=2, retained=99.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$
$|\widetilde{A}| = 0.7085$, $\arg(\widetilde{A}) = +1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
$|\widetilde{A}| = 0.6988$, $\arg(\widetilde{A}) = -1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

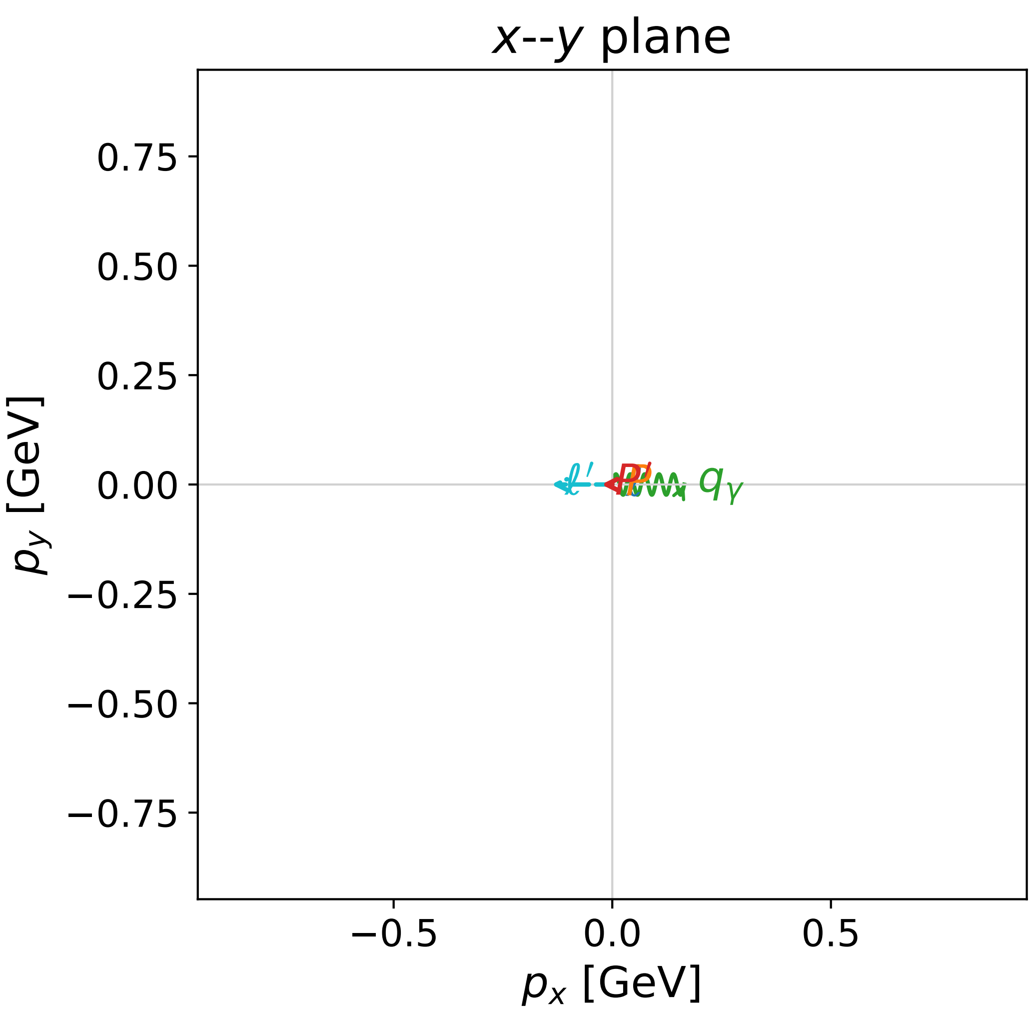
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

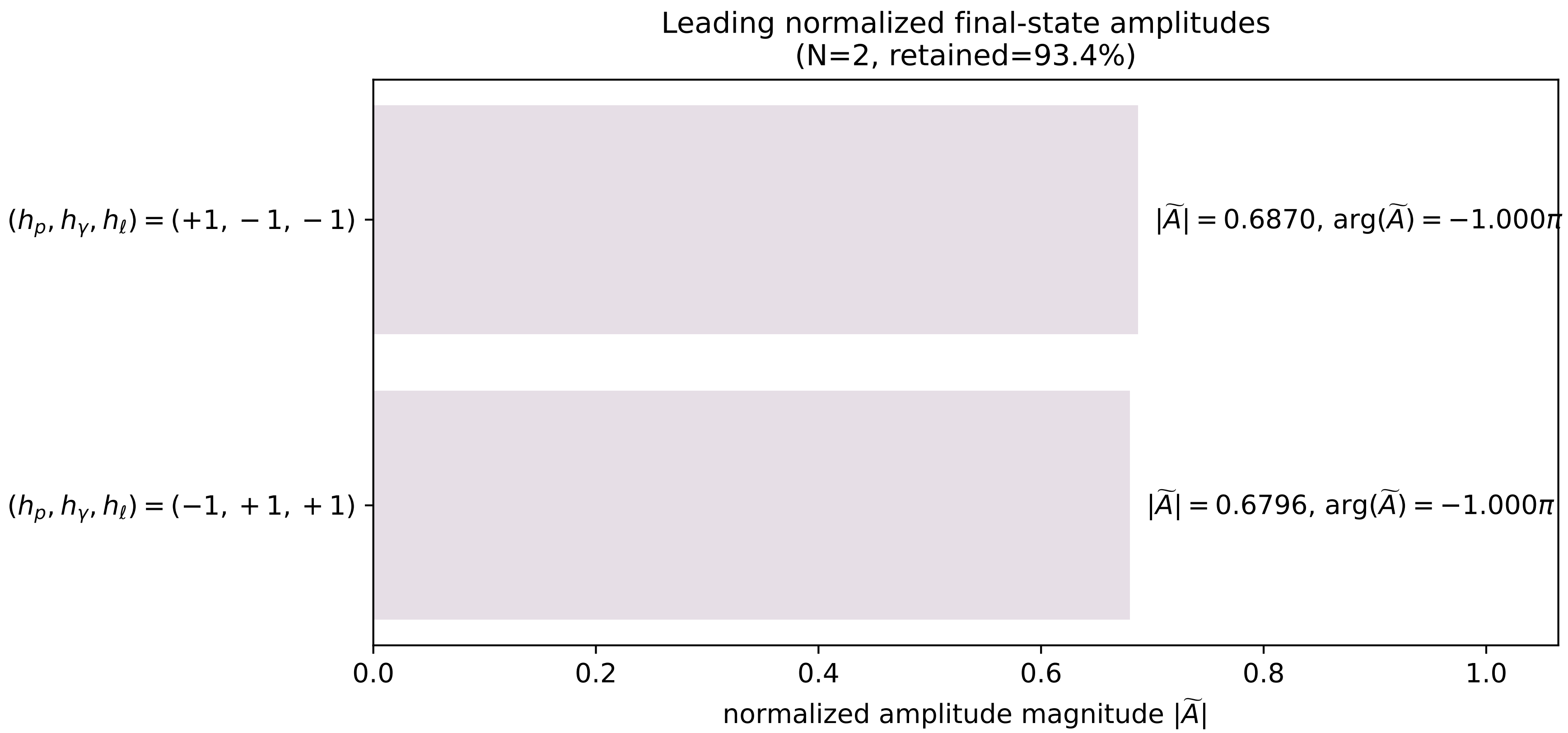


dghz_mixing_angles_polarization_02_local_minimum_296
d_{\rm GHZ}=0.000302995
region: polarization_cluster_02_local_minimum_296
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0296
lepton mixing angle: alpha_e=0.55513734 rad
incoming lepton: 0.849828|+> +0.52706|->
proton mixing angle: alpha_p=2.048015 rad
incoming proton: -0.45931|+> +0.888276|->

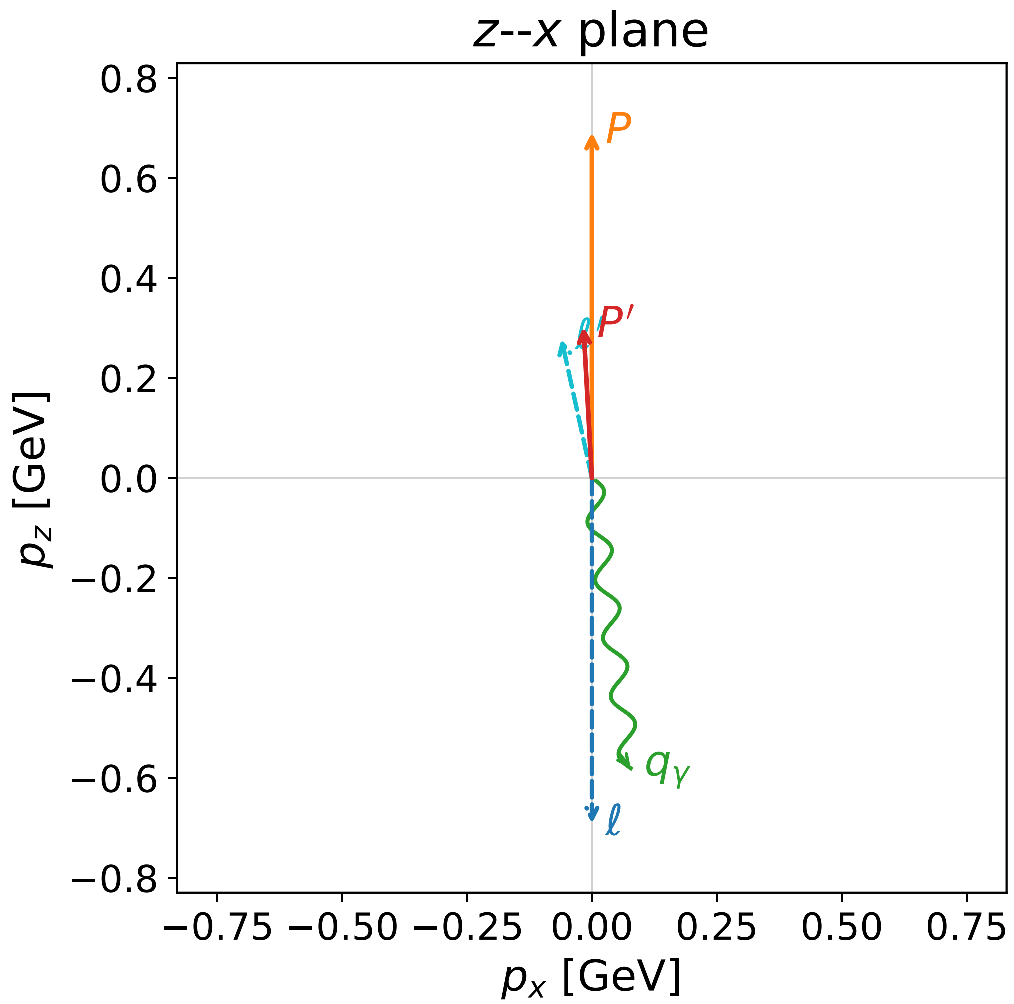
s=4.06514, $\sqrt{s}$=2.01622, $|\vec{P}|$=0.789919, $|\vec{P}'|$=0.446256
$\theta_{p'}$=0.0562449, θ_γ=2.9051, $\phi_{p'}$=3.14149, ϕ_γ=6.28313
E_γ=0.702308, $\theta_{\ell'}$=0.531494, $\phi_{\ell'}$=3.14154
Q^2=0.809472, x_B=0.280563, t=-0.0840407, W^2=2.95554, y=0.905777

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.7899 p_x= 0.0000 p_y= 0.0000 p_z= -0.7899
P E= 1.2263 p_x= 0.0000 p_y= 0.0000 p_z= 0.7899
ℓ' E= 0.2752 p_x= -0.1395 p_y= 0.0000 p_z= 0.2372
P' E= 1.0387 p_x= -0.0251 p_y= 0.0000 p_z= 0.4456
q_γ E= 0.7023 p_x= 0.1645 p_y= -0.0000 p_z= -0.6828





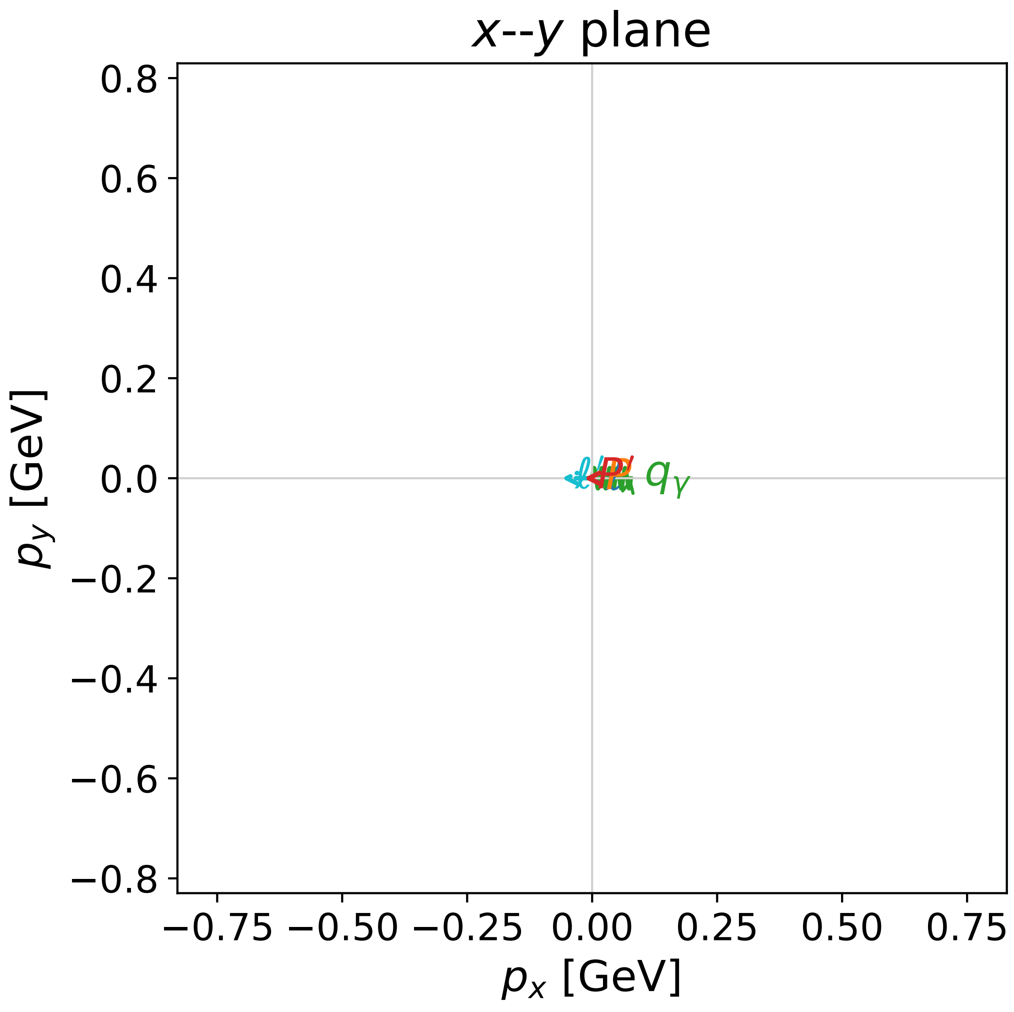
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

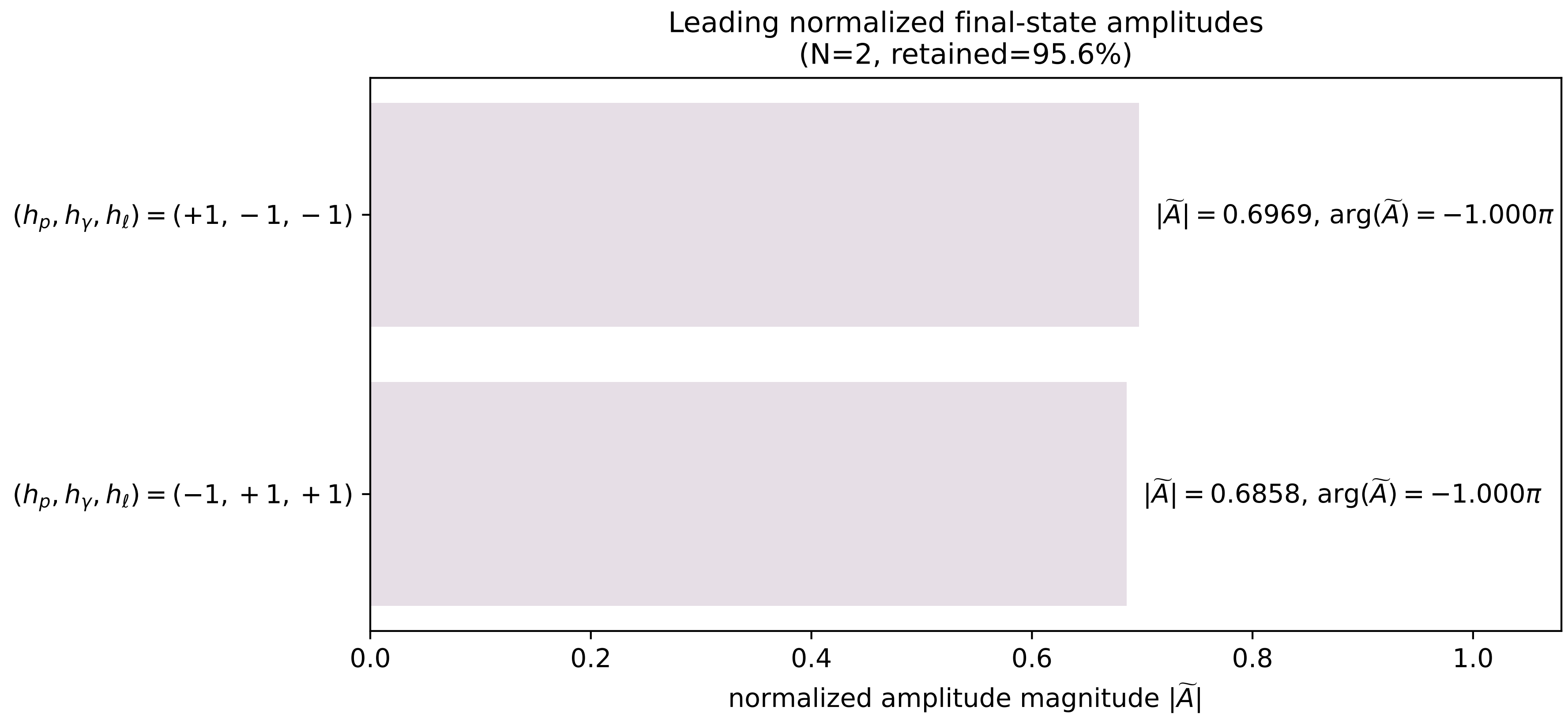



```
dghz_mixing_angles_polarization_02_local_minimum_308
d_{\rm GHZ}=0.000352127
region: polarization_cluster_02_local_minimum_308
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0308
lepton mixing angle: alpha_e=0.20616489 rad
incoming lepton: 0.978823|+> +0.204708|->
proton mixing angle: alpha_p=1.7420637 rad
incoming proton: -0.170431|+> +0.98537|->
```

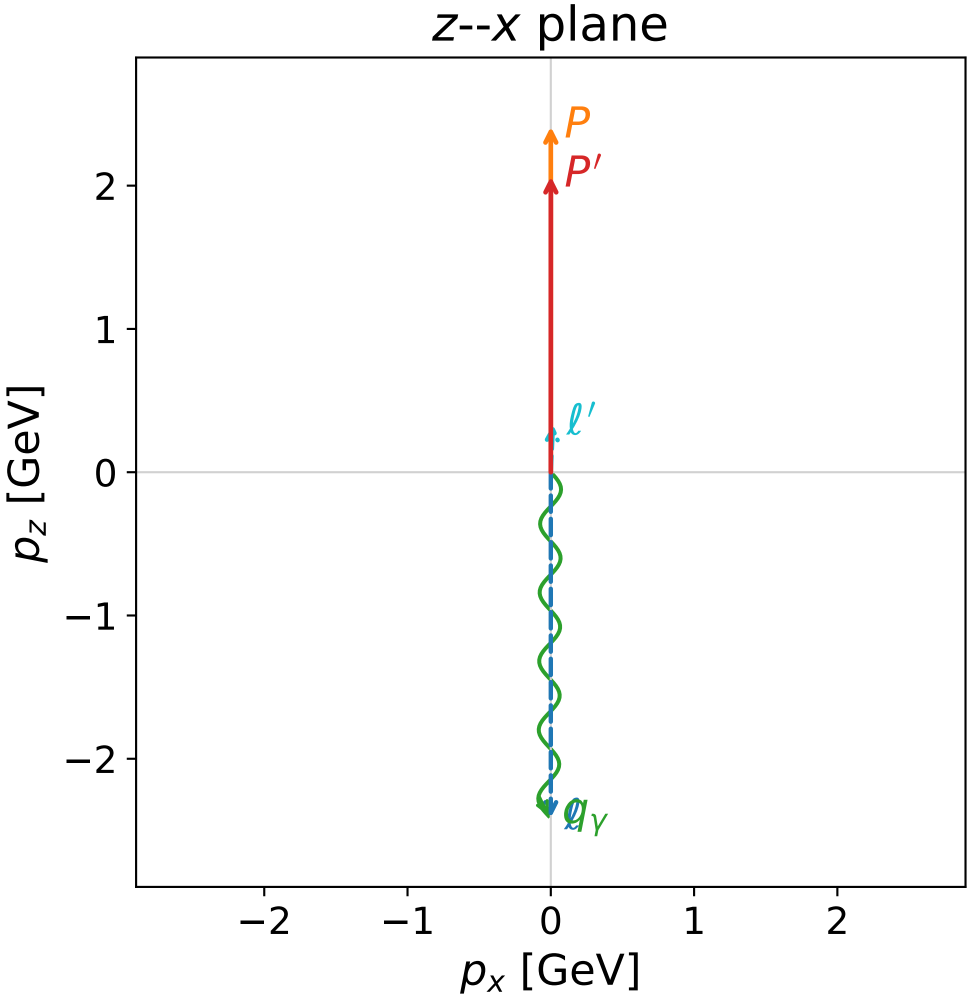
s=3.44654, $\sqrt{s}$=1.85649, $|\vec{P}|$=0.691278, $|\vec{P}'|$=0.302862
$\theta_{p'}$=0.0554148, θ_γ=3.00846, $\phi_{p'}$=3.1414, ϕ_γ=6.28312
E_γ=0.58589, $\theta_{\ell'}$=0.215754, $\phi_{\ell'}$=3.14156
Q^2=0.778686, x_B=0.340408, t=-0.11928, W^2=2.38866, y=0.891225

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.6913 p_x= 0.0000 p_y= 0.0000 p_z= -0.6913
P E= 1.1652 p_x= 0.0000 p_y= 0.0000 p_z= 0.6913
ℓ' E= 0.2849 p_x= -0.0610 p_y= 0.0000 p_z= 0.2783
P' E= 0.9857 p_x= -0.0168 p_y= 0.0000 p_z= 0.3024
q_γ E= 0.5859 p_x= 0.0778 p_y= -0.0000 p_z= -0.5807





coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

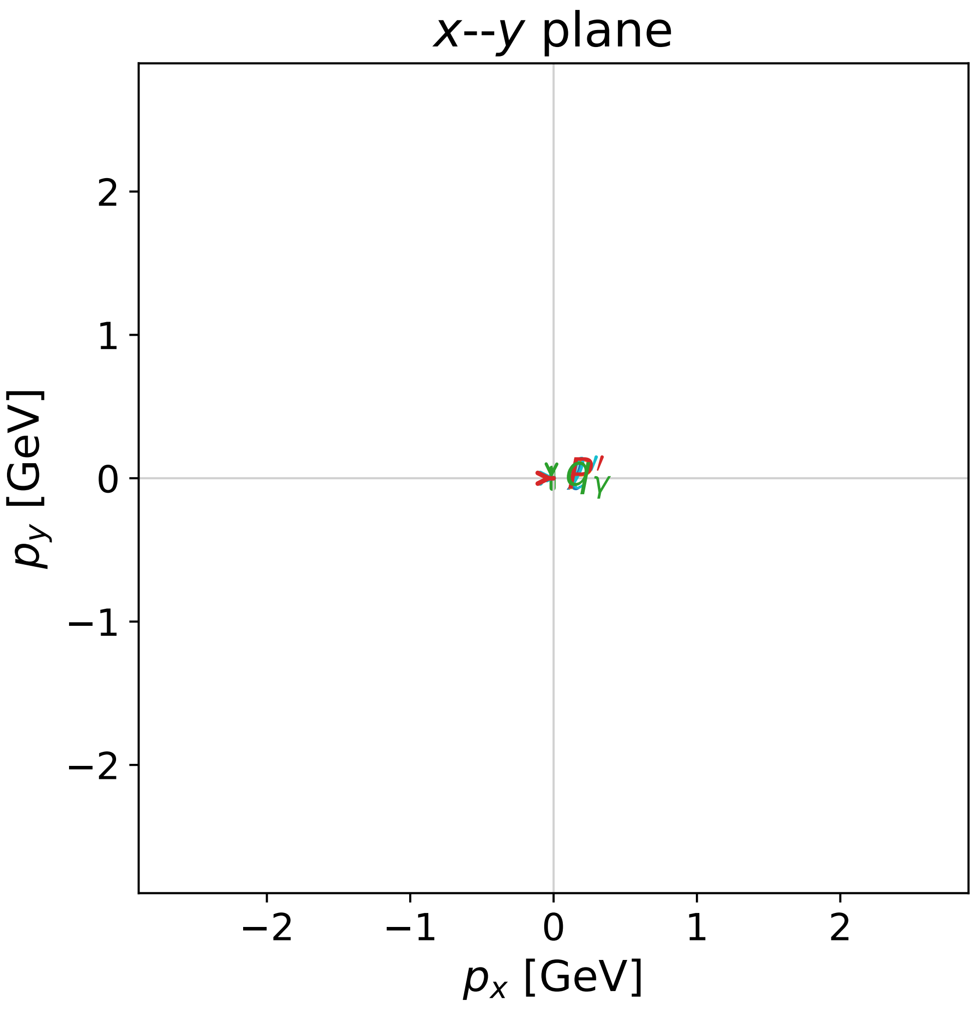


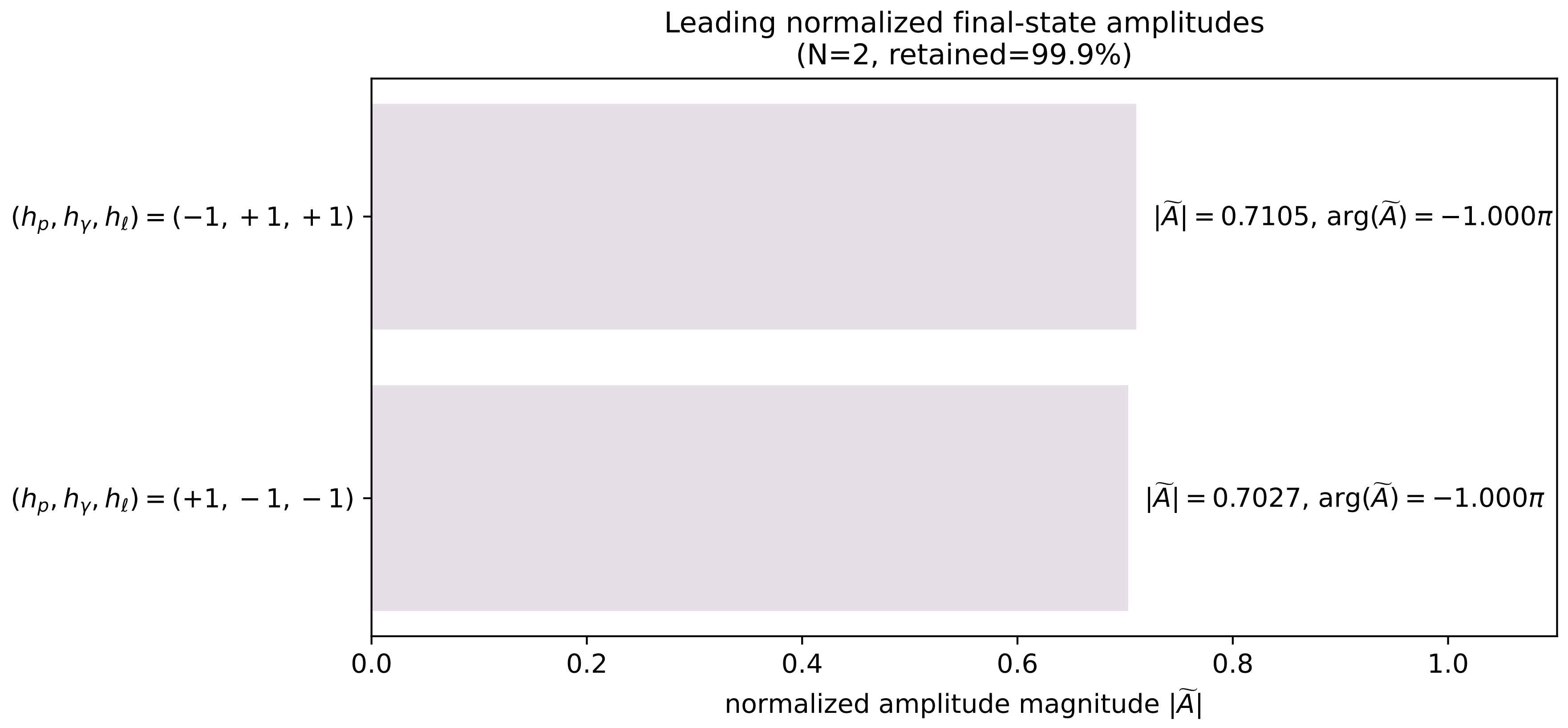

```
dghz_mixing_angles_polarization_02_local_minimum_323
d_{\rm GHZ}=0.000435487
region: polarization_cluster_02_local_minimum_323
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0323
lepton mixing angle: alpha_e=0.69364314 rad
incoming lepton: 0.768922|+> +0.639343|->
proton mixing angle: alpha_p=2.2770928 rad
incoming proton: -0.649021|+> +0.760771|->
```

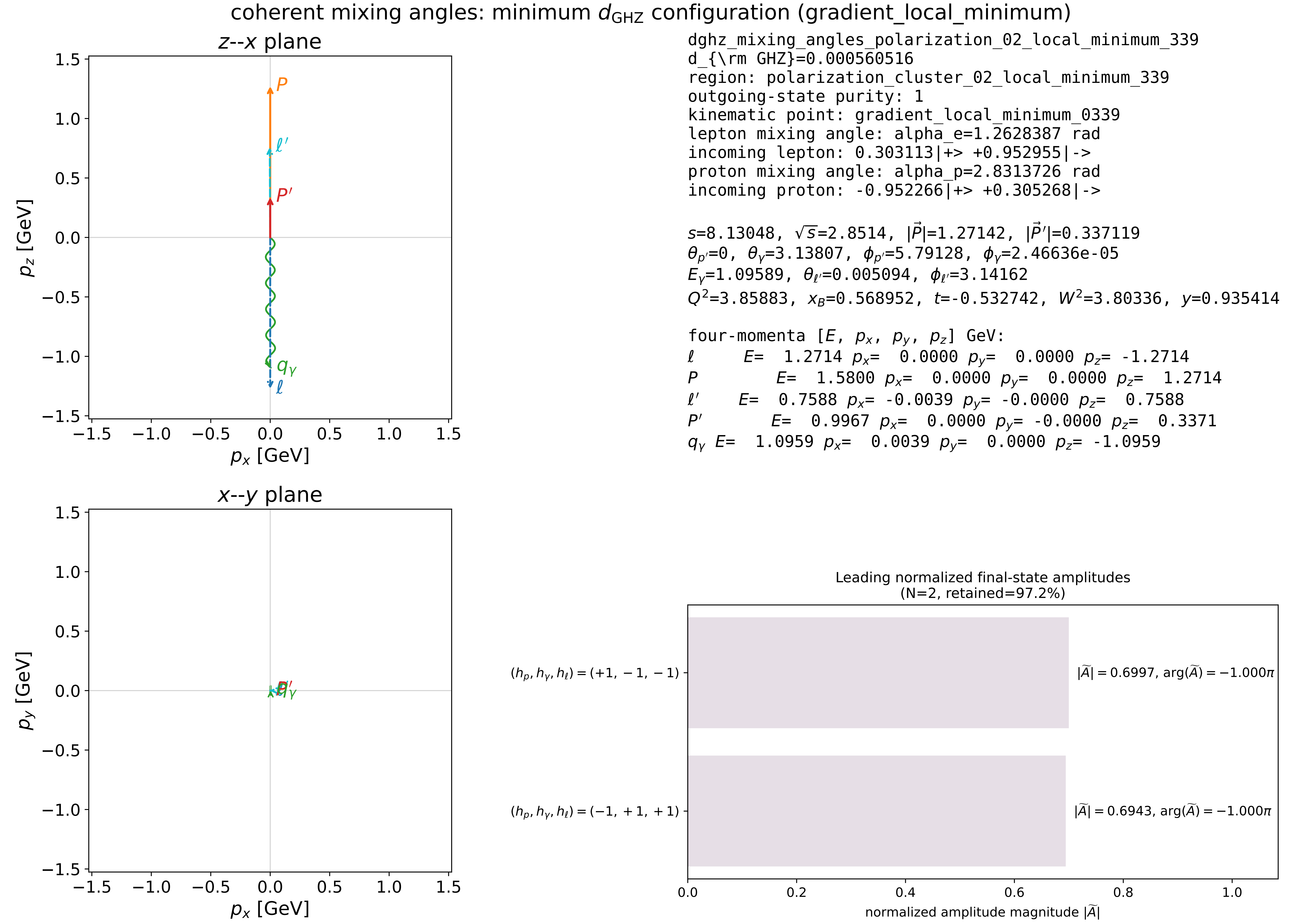
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=2.06443
$\theta_{p'}$=0.000218156, θ_γ=3.13499, $\phi_{p'}$=0.00267229, ϕ_γ=3.14155
E_γ=2.3983, $\theta_{\ell'}$=0.0460371, $\phi_{\ell'}$=6.28306
Q^2=3.22234, x_B=0.134259, t=-0.0181278, W^2=21.6583, y=0.995053

four-momenta [E, p_x, p_y, p_z] GeV:

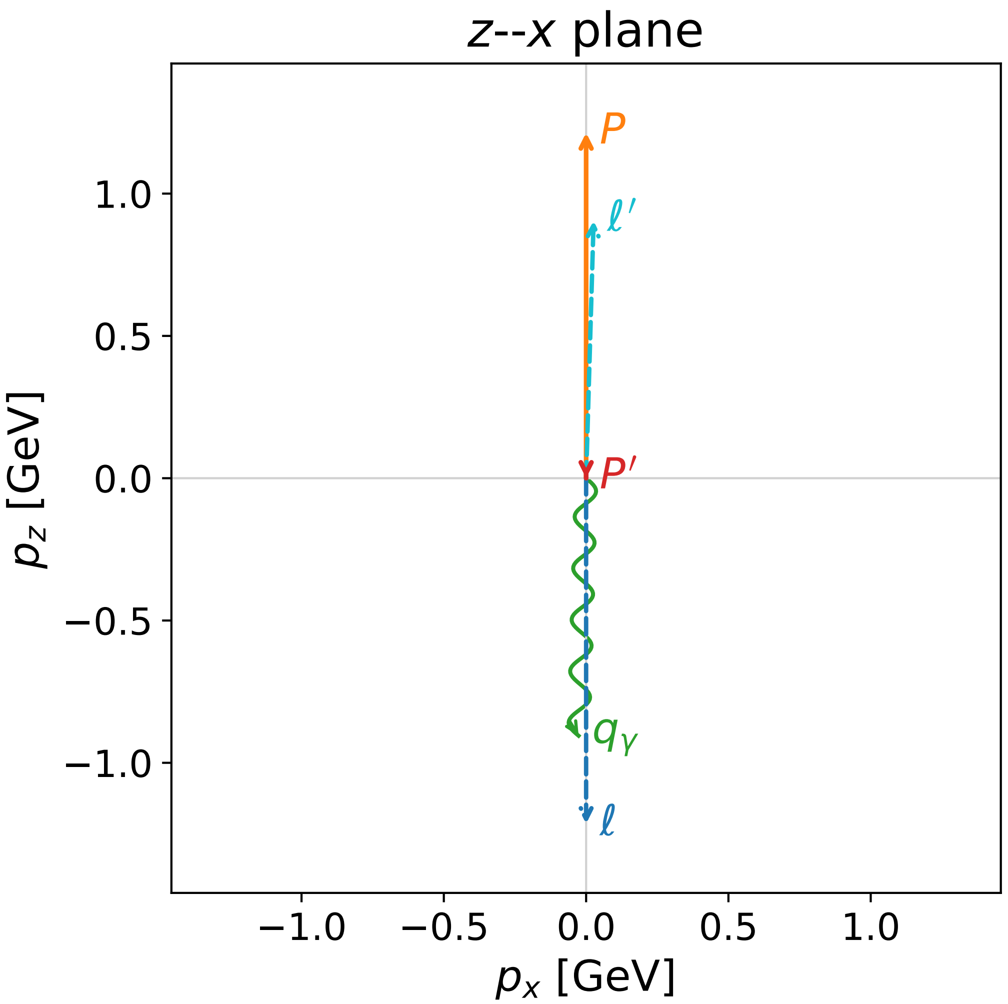
ℓ	E= 2.4120	p_x= 0.0000	p_y= 0.0000	p_z= -2.4120
P	E= 2.5880	p_x= 0.0000	p_y= 0.0000	p_z= 2.4120
ℓ'	E= 0.3342	p_x= 0.0154	p_y= -0.0000	p_z= 0.3338
P'	E= 2.2675	p_x= 0.0005	p_y= 0.0000	p_z= 2.0644
q_γ	E= 2.3983	p_x= -0.0158	p_y= 0.0000	p_z= -2.3982






coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_339
d_{\rm GHZ}=0.000560516
region: polarization_cluster_02_local_minimum_339
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0339
lepton mixing angle: alpha_e=1.2628387 rad
incoming lepton: 0.303113|+> +0.952955|->
proton mixing angle: alpha_p=2.8313726 rad
incoming proton: -0.952266|+> +0.305268|->
s=8.13048, $\sqrt{s}$=2.8514, $|\vec{P}|$=1.27142, $|\vec{P}'|$=0.337119
$\theta_{p'}$=0, θ_γ=3.13807, $\phi_{p'}$=5.79128, ϕ_γ=2.46636e-05
E_γ=1.09589, $\theta_{\ell'}$=0.005094, $\phi_{\ell'}$=3.14162
Q^2=3.85883, x_B=0.568952, t=-0.532742, W^2=3.80336, y=0.935414
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.2714 p_x= 0.0000 p_y= 0.0000 p_z= -1.2714
P E= 1.5800 p_x= 0.0000 p_y= 0.0000 p_z= 1.2714
ℓ' E= 0.7588 p_x= -0.0039 p_y= -0.0000 p_z= 0.7588
P' E= 0.9967 p_x= 0.0000 p_y= -0.0000 p_z= 0.3371
q_γ E= 1.0959 p_x= 0.0039 p_y= 0.0000 p_z= -1.0959
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=97.2%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.6997$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.6943$, $\arg(\widetilde{A}) = -1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

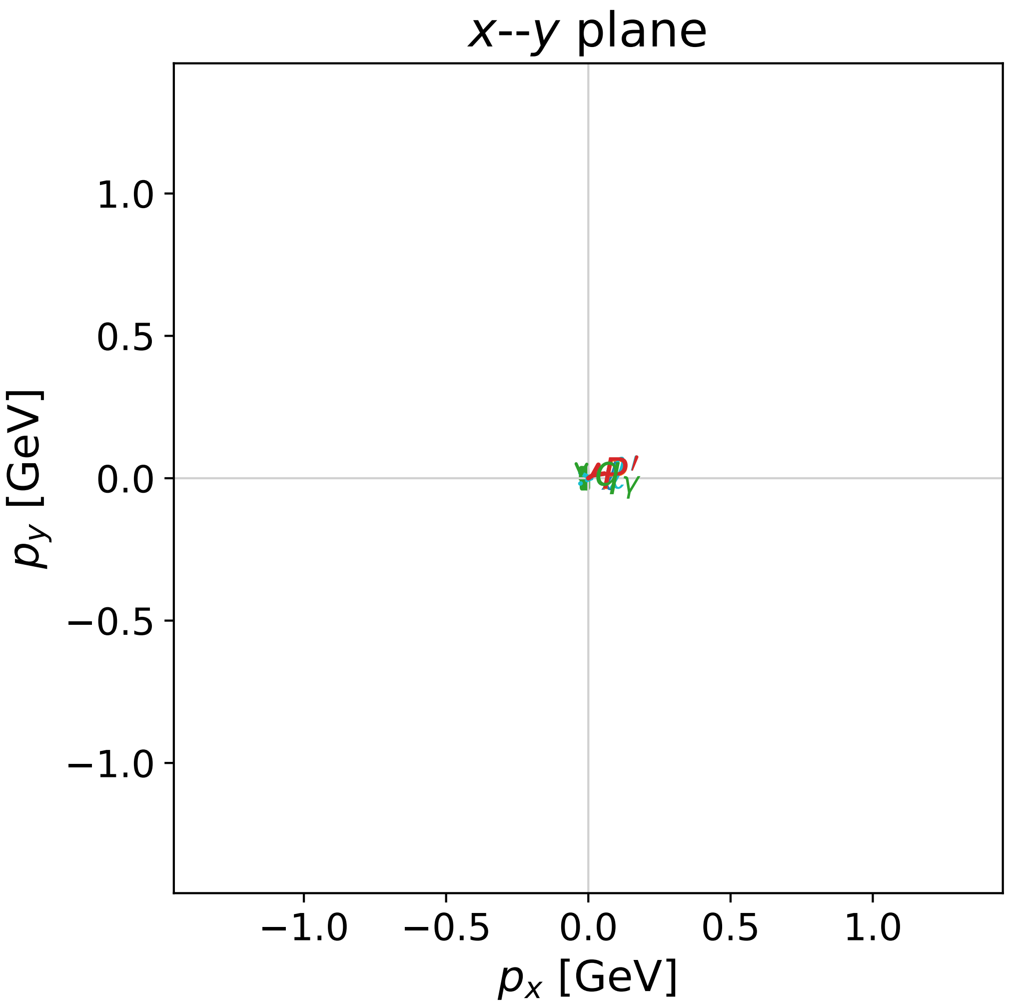
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

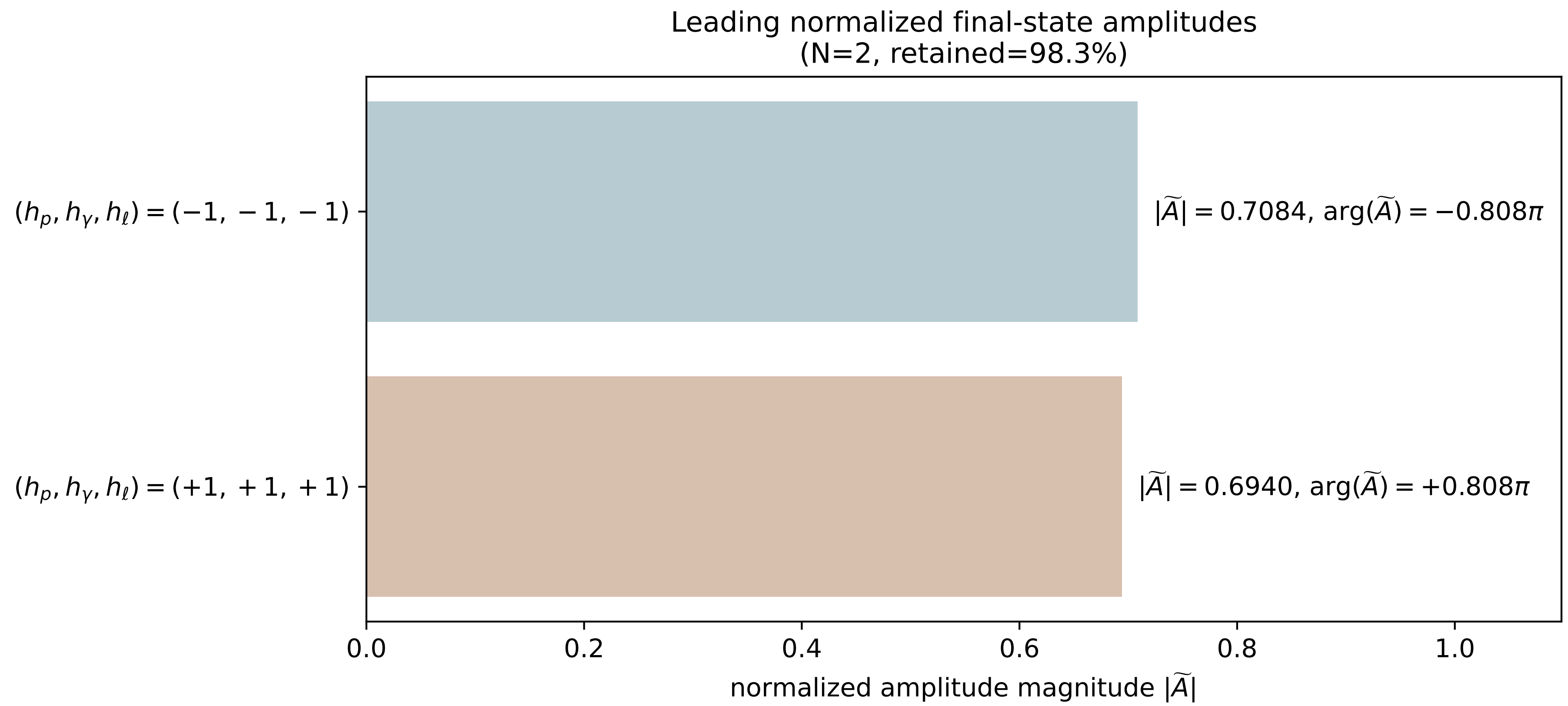


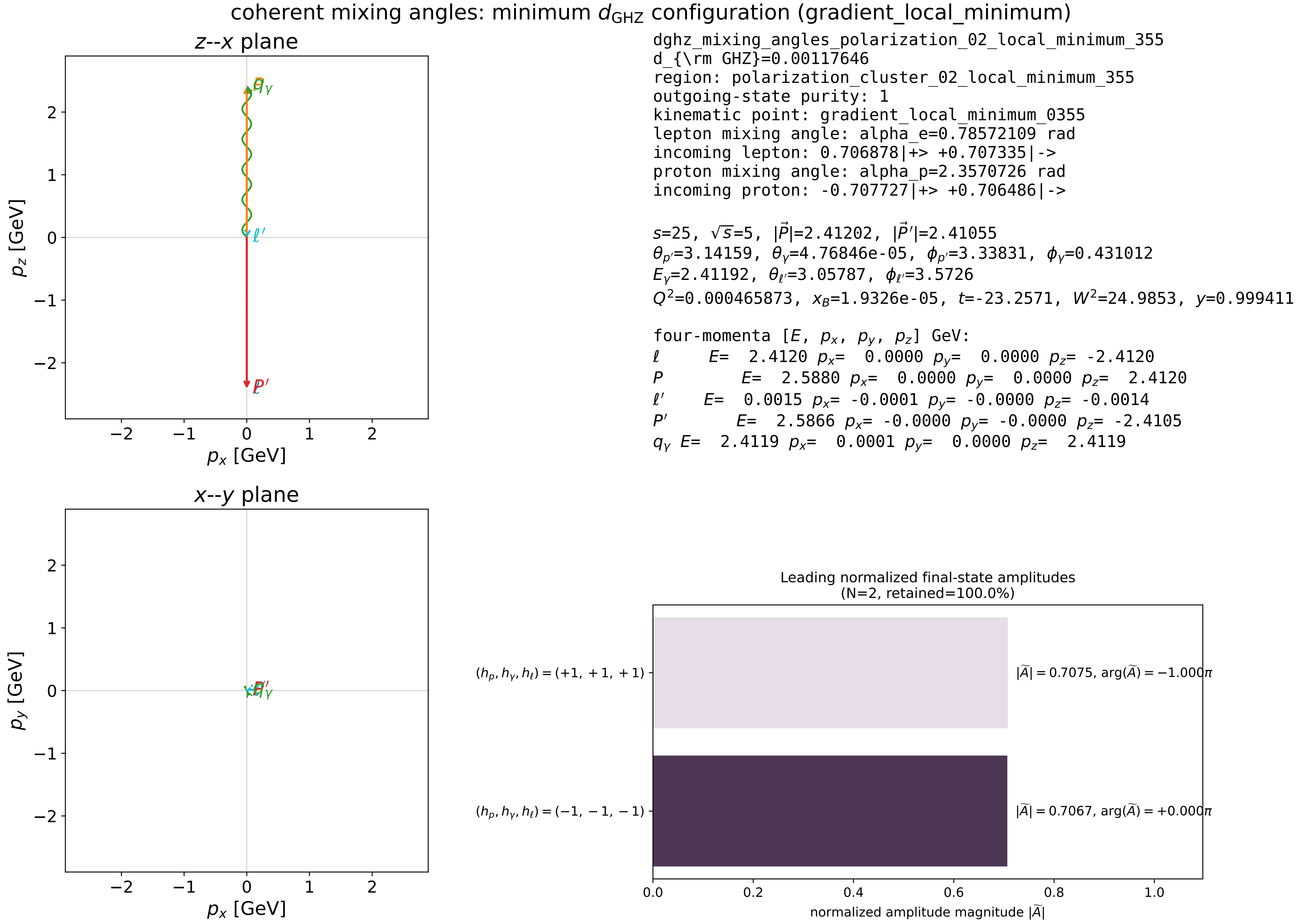

```
dghz_mixing_angles_polarization_02_local_minimum_354
d_{\rm GHZ}=0.0010861
region: polarization_cluster_02_local_minimum_354
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0354
lepton mixing angle: alpha_e=0.027969497 rad
incoming lepton: 0.999609|+> +0.0279659|->
proton mixing angle: alpha_p=1.5433927 rad
incoming proton: 0.0274002|+> +0.999625|->
```

s=7.55541, $\sqrt{s}$=2.74871, $|\vec{P}|$=1.21431, $|\vec{P}'|$=4.5944e-06
$\theta_{p'}$=3.14154, θ_γ=3.11242, $\phi_{p'}$=3.74475, ϕ_γ=3.14156
E_γ=0.905353, $\theta_{\ell'}$=0.0291745, $\phi_{\ell'}$=6.28315
Q^2=4.3966, x_B=0.721341, t=-1.11886, W^2=2.57828, y=0.913036

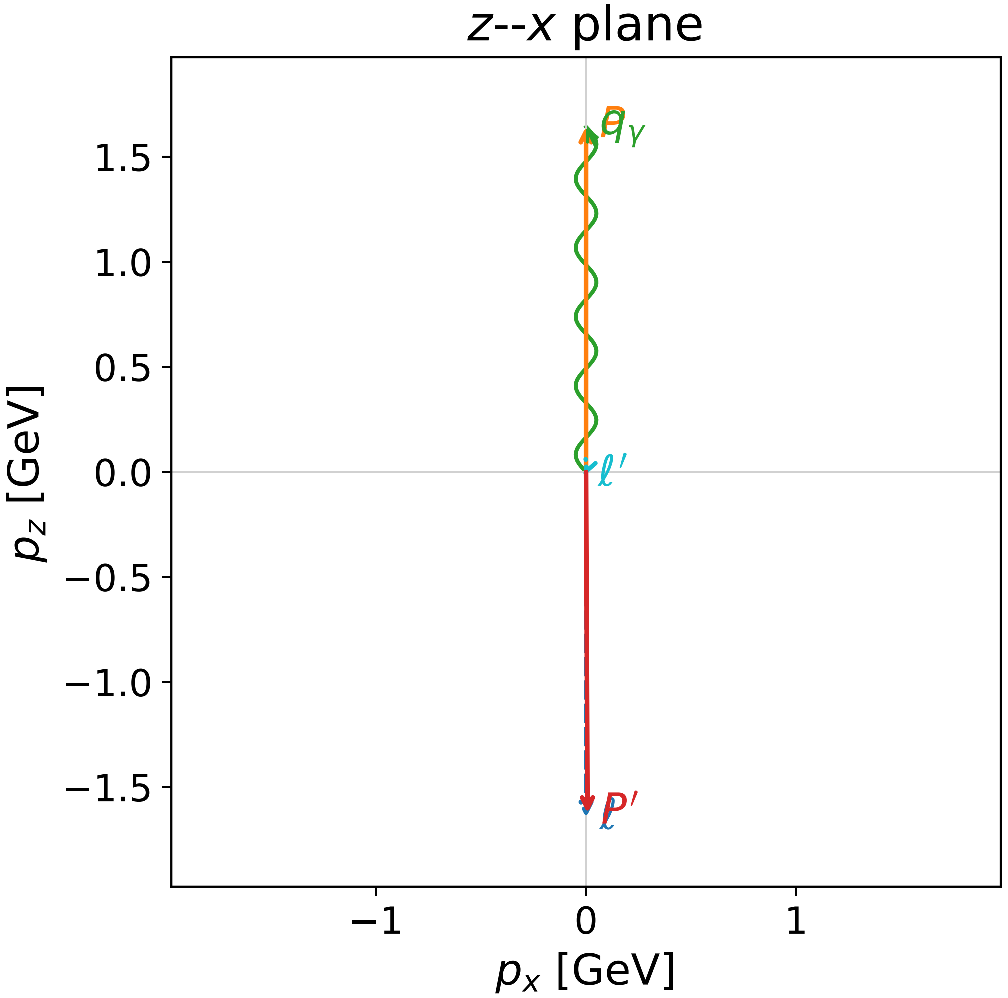
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.2143 p_x= 0.0000 p_y= 0.0000 p_z= -1.2143
P E= 1.5344 p_x= 0.0000 p_y= 0.0000 p_z= 1.2143
ℓ' E= 0.9054 p_x= 0.0264 p_y= -0.0000 p_z= 0.9050
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= -0.0000
q_γ E= 0.9054 p_x= -0.0264 p_y= 0.0000 p_z= -0.9050






coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
q_γ
ℓ'
P'
dghz_mixing_angles_polarization_02_local_minimum_355
d_{\rm GHZ}=0.00117646
region: polarization_cluster_02_local_minimum_355
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0355
lepton mixing angle: alpha_e=0.78572109 rad
incoming lepton: 0.706878|+> +0.707335|->
proton mixing angle: alpha_p=2.3570726 rad
incoming proton: -0.707727|+> +0.706486|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=2.41055
$\theta_{p'}$=3.14159, θ_γ=4.76846e-05, $\phi_{p'}$=3.33831, ϕ_γ=0.431012
E_γ=2.41192, $\theta_{\ell'}$=3.05787, $\phi_{\ell'}$=3.5726
Q^2=0.000465873, x_B=1.9326e-05, t=-23.2571, W^2=24.9853, y=0.999411
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 0.0015 p_x= -0.0001 p_y= -0.0000 p_z= -0.0014
P' E= 2.5866 p_x= -0.0000 p_y= -0.0000 p_z= -2.4105
q_γ E= 2.4119 p_x= 0.0001 p_y= 0.0000 p_z= 2.4119
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$
$|\widetilde{A}| = 0.7075$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
$|\widetilde{A}| = 0.7067$, $\arg(\widetilde{A}) = +0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

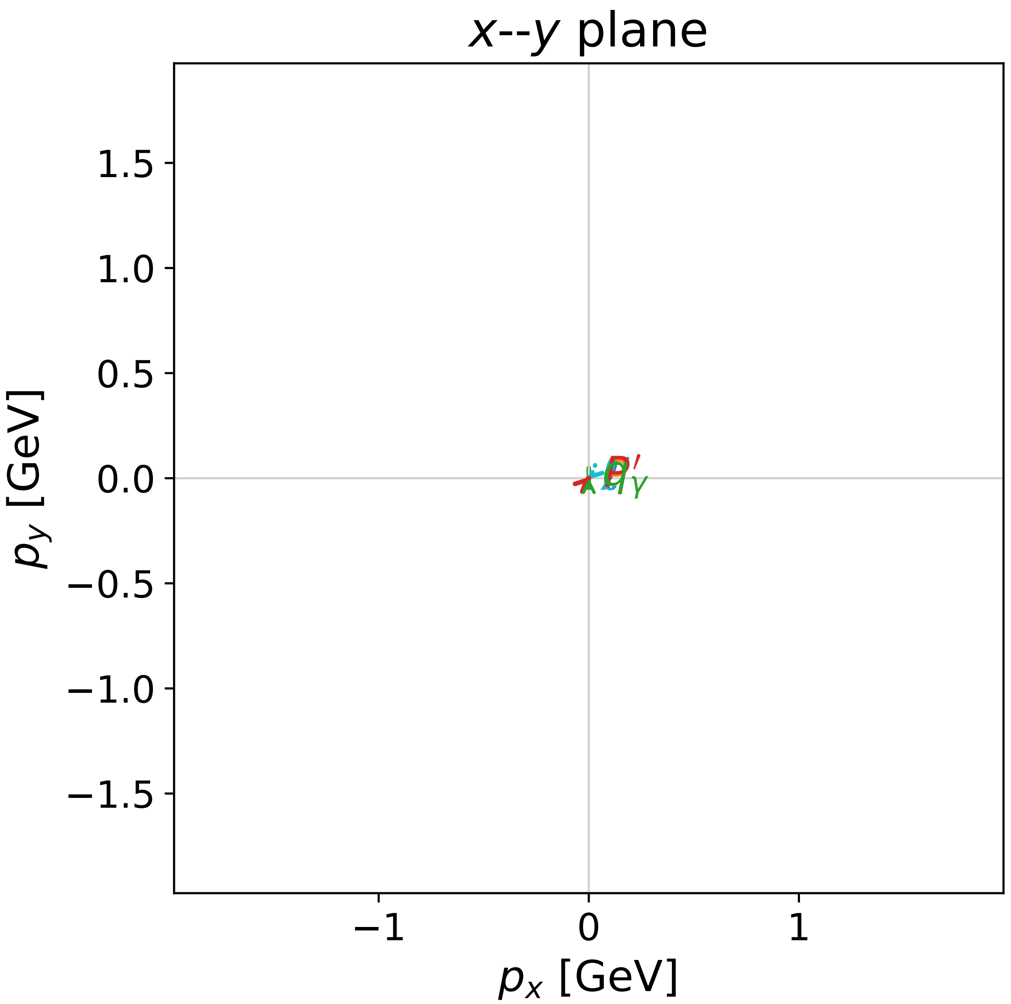
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

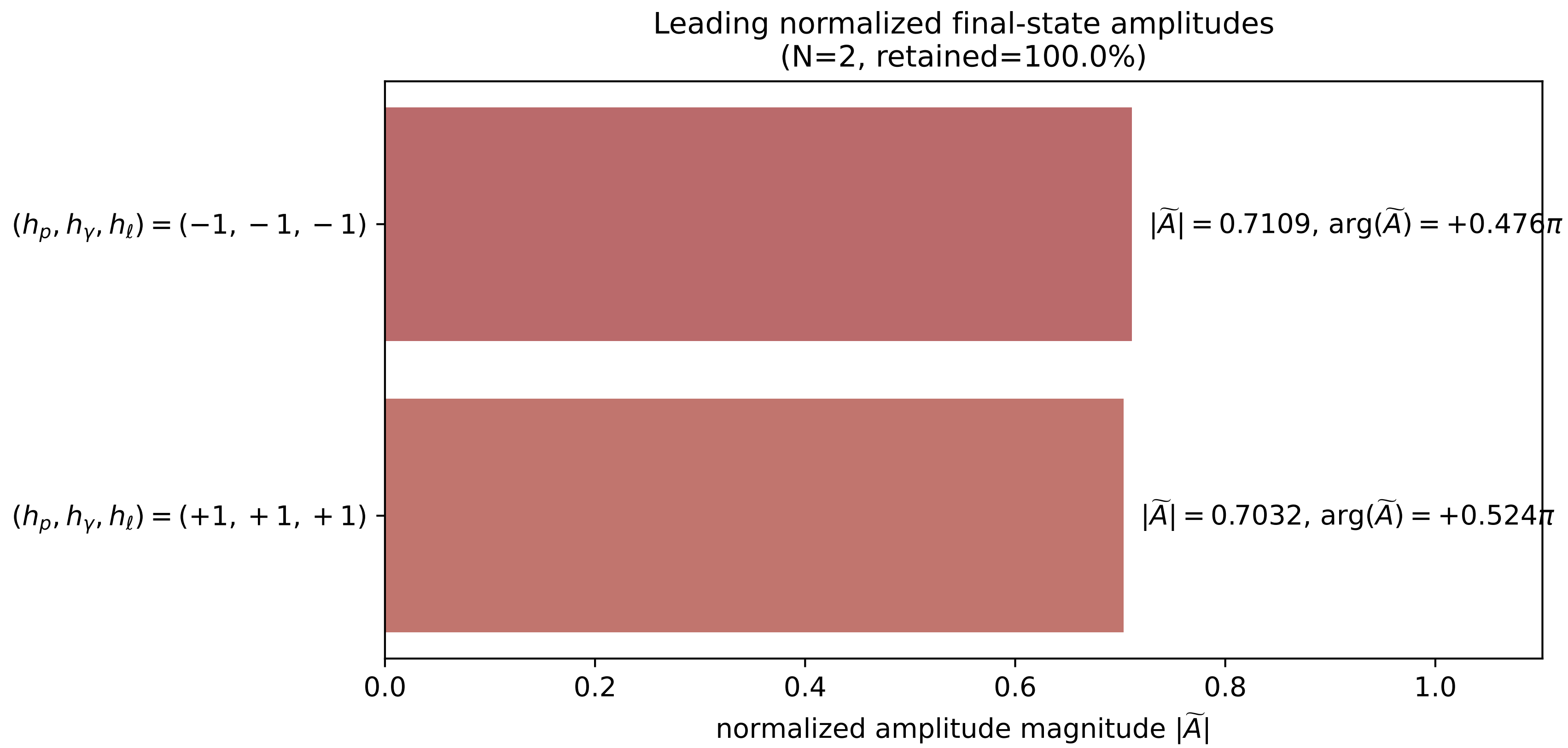



```
dghz_mixing_angles_polarization_02_local_minimum_370
d_{\rm GHZ}=0.0225057
region: polarization_cluster_02_local_minimum_370
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0370
lepton mixing angle: alpha_e=0.78532657 rad
incoming lepton: 0.707157|+> +0.707056|->
proton mixing angle: alpha_p=2.3500697 rad
incoming proton: -0.702763|+> +0.711424|->
```

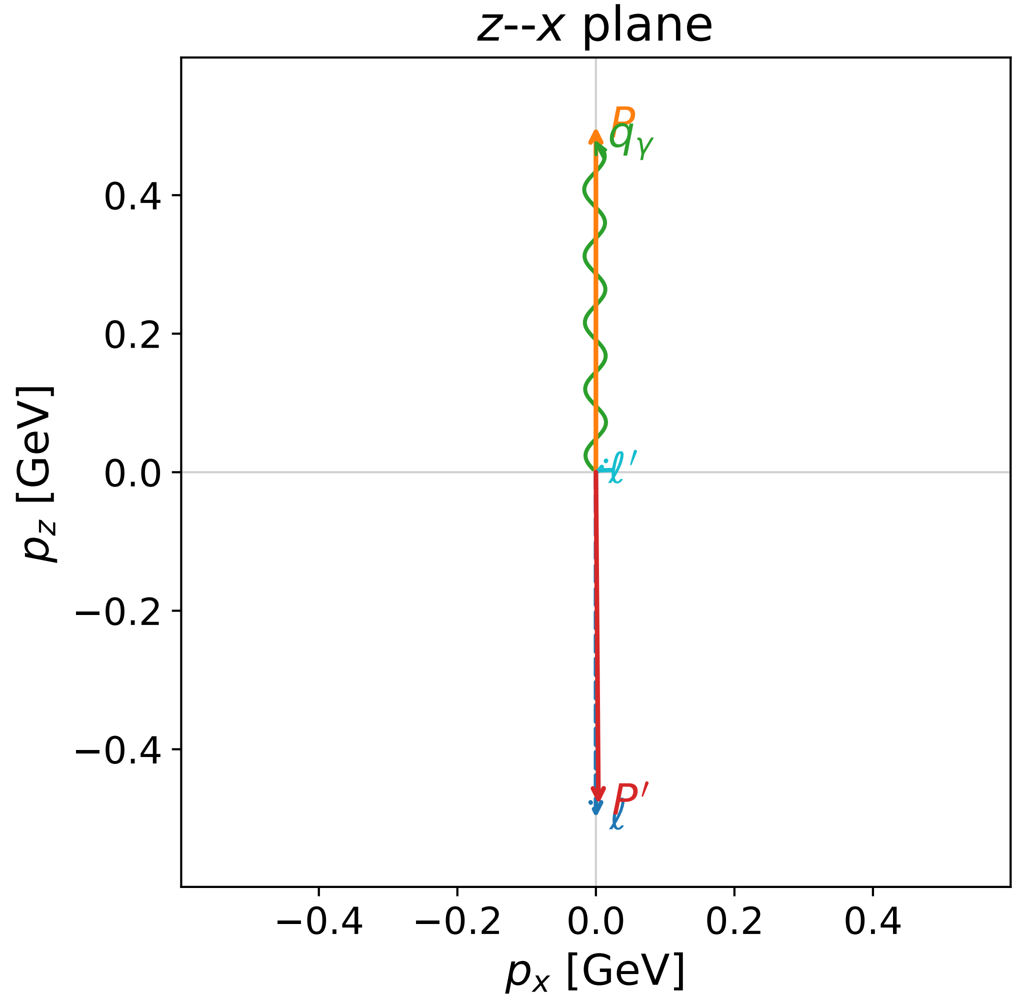
s=12.52, $\sqrt{s}$=3.53836, $|\vec{P}|$=1.64485, $|\vec{P}'|$=1.62509
$\theta_{p'}$=3.13567, θ_γ=0.000106839, $\phi_{p'}$=0.753952, ϕ_γ=0.000369628
E_γ=1.64223, $\theta_{\ell'}$=2.62489, $\phi_{\ell'}$=3.88324
Q^2=0.00850448, x_B=0.000738953, t=-10.6921, W^2=12.3802, y=0.988717

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.6449 p_x= 0.0000 p_y= 0.0000 p_z= -1.6449
P E= 1.8935 p_x= 0.0000 p_y= 0.0000 p_z= 1.6449
ℓ' E= 0.0198 p_x= -0.0072 p_y= -0.0066 p_z= -0.0172
P' E= 1.8764 p_x= 0.0070 p_y= 0.0066 p_z= -1.6251
q_γ E= 1.6422 p_x= 0.0002 p_y= 0.0000 p_z= 1.6422





coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)



dghz_mixing_angles_polarization_02_local_minimum_371
d_{\rm GHZ}=0.0256597
region: polarization_cluster_02_local_minimum_371
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0371
lepton mixing angle: alpha_e=0.78708844 rad
incoming lepton: 0.705911|+> +0.708301|->
proton mixing angle: alpha_p=2.3508161 rad
incoming proton: -0.703293|+> +0.7109|->

s=2.43798, $\sqrt{s}$=1.5614, $|\vec{P}|$=0.498955, $|\vec{P}'|$=0.496309
$\theta_{p'}$=2.87957, θ_γ=0.26554, $\phi_{p'}$=1.54217, ϕ_γ=4.69489
E_γ=0.497557, $\theta_{\ell'}$=1.86425, $\phi_{\ell'}$=2.16912
Q^2=0.0018846, x_B=0.00121447, t=-0.973645, W^2=2.42975, y=0.995924

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.4990 p_x= 0.0000 p_y= 0.0000 p_z= -0.4990
P E= 1.0624 p_x= 0.0000 p_y= 0.0000 p_z= 0.4990
ℓ' E= 0.0026 p_x= -0.0014 p_y= 0.0020 p_z= -0.0007
P' E= 1.0612 p_x= 0.0037 p_y= 0.1285 p_z= -0.4794
q_γ E= 0.4976 p_x= -0.0023 p_y= -0.1306 p_z= 0.4801